



**GREEN
CLIMATE
FUND**

Meeting of the Board
26– 28 February 2019
Songdo, Incheon, Republic of Korea
Provisional agenda item 34

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1 February 2019

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP104

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Nigeria Solar IPP Support Program” submitted by the Africa Finance Corporation (AFC);
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “Nigeria Solar IPP Support Program” submitted by the the Africa Finance Corporation is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Program Title: **Nigeria Solar IPP Support Program**

Country/Region: Nigeria

Accredited Entity: Africa Finance Corporation (AFC)

Date of Submission: 30 Jan 2019

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Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“[FP]- [Agency Short Name]-[Date]-[Serial Number]”

Acronym	Description
AFC	AFRICA FINANCE CORPORATION
AMA	ACCREDITATION MASTER AGREEMENT
BAU	BUSINESS AS USUAL
CBN	CENTRAL BANK OF NIGERIA
CCI	CERTIFICATION OF CAPITAL IMPORTATION
CESAP	CLIENT ENVIRONMENTAL AND SOCIAL ACTION PLAN
DCA	DEVELOPMENTAL CREDIT AUTHORITY
DFI	DEVELOPMENT FINANCE INSTITUTIONS
DSCR	DEBT SERVICE COVERAGE RATIO
DISCOS	ELECTRICITY DISTRIBUTION COMPANIES
EPSRA	ELECTRIC POWER SECTOR REFORM ACT
ESRS	ENVIRONMENTAL AND SOCIAL REVIEW SUMMARY
FAA	FUNDED ACTIVITY AGREEMENT
FCCC	FRAMEWORK CONVENTION ON CLIMATE CHANGE
FEC	FEDERAL EXECUTIVE COUNCIL
FGN	FEDERAL GOVERNMENT OF NIGERIA
FiT	FEED-IN TARIFF
FMEN	FEDERAL MINISTRY OF ENVIRONMENT
FNC	FIRST NATIONAL COMMUNICATION
GCF	GREEN CLIMATE FUND
GCA	GRID CONNECTION AGREEMENTS
GENCOS	GENERATION COMPANYS
GHI	GLOBAL HORIZONTAL IRRADIATION
IBRD	INTERNATIONAL BANK FOR RECONSTRCTION AND DEVELOPMENT
LC	LETTER OF CREDIT
LTA	LENDERS TECHNICAL ADVISOR
MYTO	MULTI-YEAR TARIFF ORDER
NASPA	NATIONAL ADAPTATION STRATEGY AND PLAN OF ACTION ON CLIMATE CHANGE FOR NIGERIA
NBET	NIGERIAN BULK ELECTRICITY TRADER
NDC	NATIONALLY DETERMINED CONTRIBUTATIONS
NEMSF	NIGERIA ELECTRICITY MARKET STABILIZATION FACILITY
NEPA	NATIONAL ELECTRIC POWER AUTHORITY
NERC	NERC ELECTRICITY REGULATORY COMMISSION
NESI	NIGERIAN ELECTRICITY INDUSTRY

NSE	NIGERIA STOCK EXCHANGE
PAIF	POWER AND AVIATION INTERVENTION FUND
PCOA	PUT CALL OPTION AGREEMENT
PRG	PARTIAL RISK GUARANTEE
PSRP	POWER SECTOR RECOVERY PROGRAM
SDG	SUSTAINABLE DEVELOPMENT GOALS
TCN	TRANSMISSION COMPANY OF NIGERIA
TEM	TERM ELECTRICITY MARKET
UNFCCC	UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE
WAPP	WEST AFRICA POWER POOL

A.1. Brief Project / Program Information		
A.1.1. Program title		Nigeria Solar IPP Support Program
A.1.2. Project or program		programme
A.1.3. Country (ies) / region		Nigeria
A.1.4. National designated authority (ies)		Federal Ministry of Environment
A.1.5. Accredited entity		Africa Finance Corporation
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: Africa Finance Corporation Beneficiary: Various renewable energy IPP projects in Nigeria
A.1.7. Project size category (Total investment, million US\$)		☐ Micro (≤ 10) ☐ Medium ($50 < x \leq 250$) ☐ Small ($10 < x \leq 50$) ☒ Large (> 250)
A.1.8. Mitigation / adaptation focus		☒ Mitigation ☐ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		Jan 30 th 2019
A.1.10. Project contact details	Contact person, position	Ato Gyasi, Senior Director, Investment Group Kome Ajegbo, Senior Associate, Investment Group
	Organization	Africa Finance Corporation (AFC)
	Email address	ato.gyasi@africafc.org kome.ajegbo@africafc.org
	Telephone number	+234 1279 9610 or +234 1 4480930
	Mailing address	Africa Finance Corporation, 3a Osborne Road, Ikoyi, Lagos, Nigeria

A.1.11. Results areas (mark all that apply)	
Reduced emissions from:	
☒	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☐	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Program Executive Summary (max 300 words)

Nigeria is the 6th largest oil producer in the world and with a population of over 180 million people, the availability of crude oil amidst the lack of energy supply and infrastructure, has led to an unsustainable reliance on fossil fuels for energy production across both urban and rural communities. According to the 2015 World Climate Change Vulnerability Index, Nigeria is one of the most vulnerable countries in the world and with the anticipated effects of global warming, rising temperatures will increase energy demand.

It is estimated that 98 million people, c.55% of the population, lack access to grid-connected electricity (YE 2015). Even with a gross installed capacity of c.13GW¹ available daily generation ranges between **4GW to 5GW²** as a result of gas, transmission and distribution constraints. Despite the infrastructure challenges, as the country develops, fuel sources for energy production must be diversified and include more sustainable options. In consideration of the Federal Government of Nigeria's ("FGN") commitment to reduce c.45% of its business as usual carbon emissions by 2030, in November 2015, the FGN approved its Feed-in Tariff ("FiT") regulation, facilitating the achievement of Nigeria's Nationally Determined Contributions ("NDC"). FiT is targeting a 2020 state whereby 2GW shall be generated through renewables and the electricity distribution companies ("DisCos") will be obliged to source at least 50% from renewables. Further to this, the FGN through the Nigerian Bulk Electricity Trader ("NBET"), the off-taker, signed 14 utility scale solar PPAs (the "**Nigeria Solar IPP Projects**") in July 2016 to supply c.1,125MW of power to the grid. To date, none of these projects have been able to reach financial close.

As a result, the Africa Finance Corporation ("AFC"), an Africa focused infrastructure financing institution with US\$3.9 billion in total assets and US\$4.4 billion in total disbursements, the second highest rated financial institution in Africa, is proposing to work with the Green Climate Fund ("GCF") on a Nigeria Solar IPP Support Program, which will catalyse the delivery of c.400MW of renewable power, (the "**Program**"), ensuring the successful financing, construction and operations of the first utility scale power projects in the country. Amidst the current financing challenges within the power sector i.e. 'expensive financing' from local financial institutions, the participation of local commercial banks through innovative structures, will support the further development of long-term funding solutions for the sector.

The main activity of the Program is to provide long term financing to Nigeria Solar IPP projects; senior debt financing of up to 70% of total project costs (with the balance through equity) for an estimated 3 – 5 solar projects. The senior debt financing will be comprised of a DFI Tranche (with competitive terms) a GCF tranche (with concessional terms) and potentially a commercial tranche (for local financial institutions). The financing package the Program offers will only be provided to projects that meet the selection criteria. To incentivize sponsors to optimize their financial structures, the selection criteria will incorporate a ranking that will benefit projects which require less concessional funding as a percentage of project cost.

The Program is expected to reduce or avoid 476,487t CO₂ eq on an annual basis (9,529,739 t CO₂ eq. over the life of the Program³), at least 1 million households in Northern Nigeria will be direct or indirect beneficiaries, reduce the perceived risks of investing in the Nigerian renewable energy sector and catalyse private sector investment in the sector (through a commercial tranche, on a best effort basis).

A.3. Program Milestone

Expected approval from accredited entity's Board (if applicable)	Within 120 days of GCF Board approval
Expected financial close (if applicable)	Q1 2019 – Q3 2019
Estimated implementation start and end date	Start: Q3 2019 End: Q3 2022
Program lifespan	21 years

¹ AF Mercados Overview of the Renewable Energy Sector in Nigeria

² AF Mercados Overview of the Renewable Energy Sector in Nigeria

³ If one assumes a solar PV useful life of at least 25 years, then the Program is expected to reduce or avoid 11,717,533 tCO₂ eq. over the useful life of the projects.

B.1. Description of Financial Elements of the Program

Financing Structure

The Program will consist of up to US\$300 million, with up to US\$100 million to be equally provided by AFC, AfDB and GCF. The intent is for the competitive debt to be provided alongside local financial institutions in order to support an aggregate of projects with total costs of up to US\$467 million. However, given the current exposure to the Nigerian power sector, local financial institutions have limited appetite and capacity (tenor and pricing constraints) for financing new power projects and thus the up to US\$300 million Program may only support an aggregate of projects with total costs of up to US\$430 million.

Nevertheless, following an extensive market sounding exercise with various local and regional financing institutions, there is indicative interest to participate with support from the Bank of Industry. Thus, the preferred option will be to ensure local financial institution participation for up to US\$45 million (naira equivalent). To further support local financial institution participation, the Program envisages a debt replacement tranche wherein the short tenor commercial debt tranche will be replaced by funding from the DFI's to enable the projects benefit from much needed long tenors and blended pricing. In a scenario where the terms of the local financial institutions make the Program structure unbankable, the Program will only proceed with senior debt financing provided by AFC, AfDB and GCF.

Figure 1a: Project Structure, with Local Commercial Banks

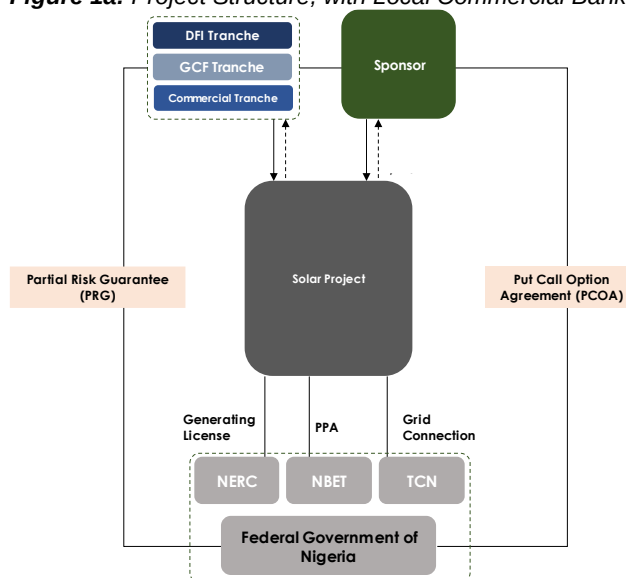
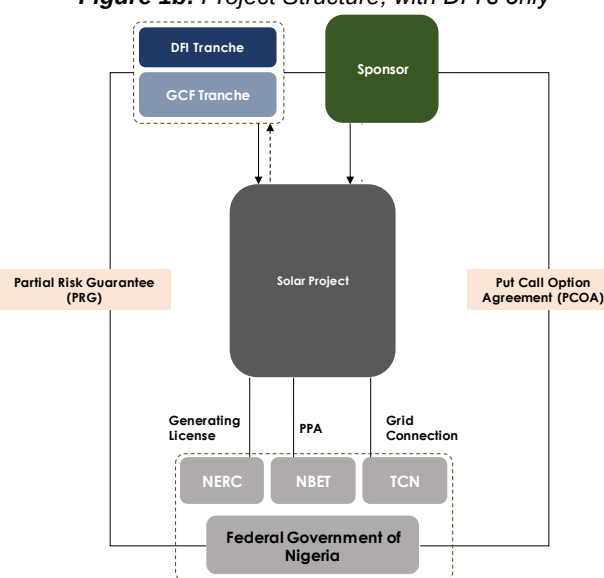


Figure 1b: Project Structure, with DFI's only



For each project, the financing structure is expected to be as follows:

Equity

The sponsors will invest a minimum of 30% equity through cash and/or shareholder loans.

Senior Debt

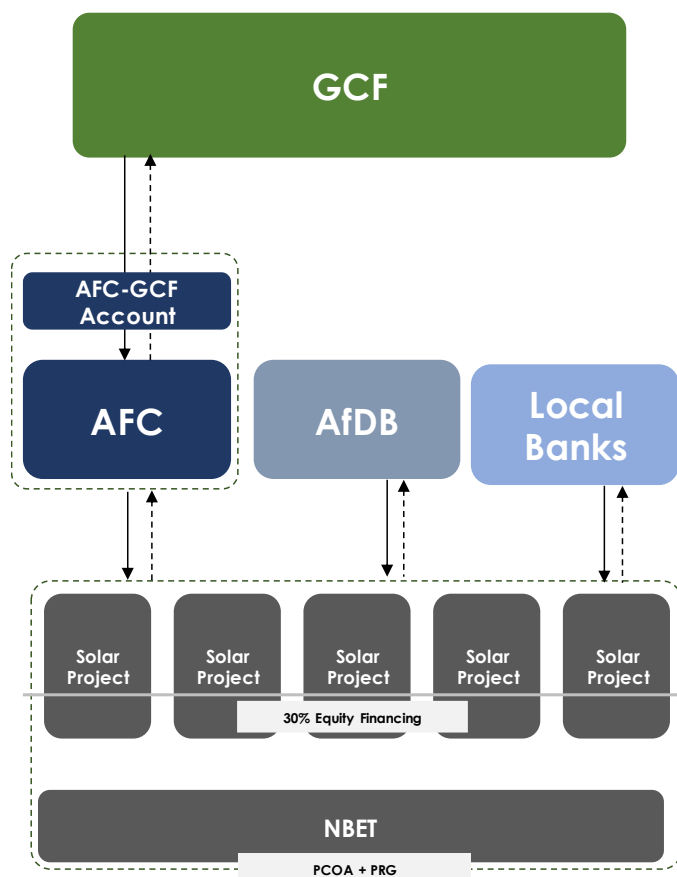
The senior debt is currently anticipated to be comprised of 2 facilities with 3 tranches:

Facility A⁴, which will consist of total debt equivalent to 70% of the project.

Facility B (Local Commercial Tranche Debt Replacement), which will be used to take out the debt outstanding of the Local Commercial Tranche in year 7 (if implemented):

Figure 2: Funds Flow, Program Overview with Local Commercial Banks

⁴ The ratio of the total financing (and per project) is intended to be such that AFC:GCF:AfDB:Local is 1:1:1:0.45. However, this is still subject to participation of the local commercial banks.



- GCF's funding amount will be transferred to the AFC as governed by the FAA.
- It is envisioned that this 'transfer' of funds will be structured as a loan between AFC and GCF, which AFC will subsequently on-lend to each of the selected projects. Thus, AFC will show as the lender of record for GCF funds provided to each of the selected projects, albeit through a separate tranche.
- As defined in the FAA, the AE will disburse to the implementing entities per an agreed schedule
- In accordance with clause 12 of the AMA between AFC and GCF, AFC shall be entitled to an AE fee for the oversight, monitoring and implementation of the funded activity, in accordance with the AMA and FAA.
- The AE will provide two tranches of financing to each of the projects, one reflecting its own investment and the other reflecting the GCF investment through AFC.
- The AE will provide monitoring / progress reports to the GCF in accordance with the AMA and FAA.
- The AE will be responsible for monitoring the activities of each of the implementing entities (the yet to be determined projects) and ensure compliance with the FAA terms in accordance with its organisational guidelines and the FAA.

Credit Enhancements

To address the liquidity risk associated with NBET, AfDB is providing a Partial Risk Guarantee ("PRG") to cover the payment risk of the offtaker for a number of projects, which will cover up to 6 months' worth of revenue. The projects also benefit from a Put Call Option Agreement ("PCOA"), which in the event of termination of the PPA due to default associated with the FGN, the PCOA will provide termination payments (in USD) guaranteed by the Ministry of Finance, to cover outstanding debt together with equity contributions and returns

B.2. Project Financing Information

	Financial Instrument	Amount	Currency	Tenor	Pricing
(a) Total project financing	(a) = (b) + (c)	up to 467	million USD (\$)		

(b) GCF financing to recipient	(i) Senior Loans		up to 100	million USD (\$) <u>Options</u> <u>Options</u> <u>Options</u> <u>Options</u> <u>Options</u>	(18) years		
	* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/program's expected performance against the investment criteria indicated in section E.						
	Total requested (i+ii+iii+iv+v+vi)		up to 100	million USD (\$)			
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Senior Loans		million USD (\$)	AFC	(18) years	TBD	pari passu
	Senior Loans		million USD (\$)	AfDB	(18) years	TBD	pari passu
	Senior Loans		million USD (\$)	Local Bank TBD	TBD	TBD	pari passu
	Senior Loans Equity		million USD (\$)				<u>Options</u>
Lead financing institution: AFC							
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.							
(d) Financial terms between GCF and AE (if applicable)	In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) or if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped. If there is a financial arrangement between the GCF and the AE, which entails a financial instrument and/or financial terms separate from the ones described in part (b), please fill out the table below to specify the proposed instrument and terms between the GCF and the AE.						
	Financial Instrument	Amount	Currency	Tenor	Pricing		
	Choose an item.		<u>Options</u>	() years	() %		
	Please provide a justification for the difference in the financial instrument and/or terms between what is provided by the AE to the recipient and what is requested from the GCF to the AE.						

B.3. Financial Markets Overview

Against the backdrop of a slowdown and rebalancing of the Chinese economy; lower commodity prices, especially sharply declining oil prices; and tightening financial conditions, with subsequent risk aversion of international investors, the Nigerian economy has gone through significant challenges. GDP growth fell from 6.3% in 2014 to 2.7% in 2015, and to -1.6% in 2016, marking Nigeria's first full-year of recession in 25 years. In 2016, global oil prices reached a 13-year low and oil production was severely constrained by vandalism and militant attacks in the Niger Delta, resulting in a significant contraction of oil GDP, although this situation seems to have been reversed subsequently.

Trading activity at the Nigeria Stock Exchange (NSE) has picked up following sharp decline during 2016-2017. The NSE all share index ended the quarter at 41,504.51, an increase of 62.7% higher than the same time in 2017. Market performance has been strong across the board, with larger gains recorded in Conglomerates, Health Care and Financial Services industries. Consequently, market capitalisation has soared reaching N 24.87 trillion (US\$ 81.36 billion) at end of first quarter of 2018, up more than 50% year-on-year. The debt market is also gaining traction, but issuances are driven largely by Treasury bills and Government bonds. Bonds market capitalisation stood at N 15.00 trillion (US\$ 49.08 billion). Yields rates in the Treasury bill primary market have declined, buoyed, by ample liquidity

and improving market sentiment. Table 1 below shows an overview of Treasury Bills in Nigeria. The yield on the 5-year tenor Government bond was 12.8% while that for the 7-year paper was recorded at 12.9%.

Table 1: Primary Treasury Bills Yields

Tenor (Primary market auction)	Stop rates 18 th April 2018 (%)	Stop rates 16 th May 2018 (%)
91-day	10.9	10.0
182-day	12.0	10.5
364-day	12.1	10.7

Source: Central Bank of Nigeria

Power Sector Financing by Local Commercial Banks

The energy sector has had significant constraints obtaining financing from local commercial banks because local banks already have significant exposure to the power sector as a result of their participation in the 2013 power privatisation process, and local banks lack confidence in the credibility of the current power sector business model.

Unique to the privatization process was the fact that participation was driven mostly by local sponsors financed by local banks. Bank exposure to the sector, either directly or indirectly through the owners, is significant. Lenders obtained corporate or personal guarantees from investors as well as a pledge on the shares of the DisCos/GenCos as security for the loans. It is understood that much of this lending was in the form of bullet financing to be repaid within a 3 to 6-year period. Owners have since faced a myriad of challenges in repaying these loans and if sustained could lead to significant loan defaults and increase the non-performing loans status of banks, negatively impacting their Capital Adequacy Ratios.

There is an estimated total exposure at c. NGN600 billion based on privatisation debt finance of c.US\$2 billion (N320 billion at 2013 exchange rates), with an additional NGN300 million provided in 2014 to support facility upgrades.

Despite these challenges, it is important to note that the Program does not attempt to fix the credibility of the current power sector business model, but rather alleviate some of the current challenges through innovative financing structures. A particular strength of the financing structure is that the PPA stipulates that payments will be calculated in USD but are payable in naira and indexed to FX as of the payment date. If there is an exchange rate loss i.e. the USD is converted to Naira at a less favorable rate than is stated on the invoice, then there is a provision for an adjustment to be made on the next invoice. Projects will also be expected to procure Certificates of Capital Importation⁵ (CCI), which prioritizes the projects ability to procure FX from the Central Bank of Nigeria.

For the banks with greatest exposure, the extent of non-performing loans is significant. Moreover, other banks have been reluctant to participate in the sector subsequently due to the poor cash flow position of the sector and the difficulty in gaining access for the limited cash received by the DisCo each month.

To support the sector, international agencies have initiated schemes to provide security to banks in sector lending. In 2014, the Development Credit Authority (DCA) of the U.S. Agency for International Development (USAID), together with GuarantCo, guaranteed a loan from 2 commercial banks with \$90 million in new capital earmarked for on-lending to the DISCOs and GENCOs. The aim of the scheme was to provide a guarantee to one commercial bank to enable them to reduce the rate at which they lend to another commercial bank, which could then on lend to the GenCos And DisCos. However, there were never any disbursements under this fund, partly due to an adverse macroeconomic environment, and later due to the need to incorporate foreign exchange hedging once the Naira depreciated significantly against the US dollar.

In 2016, a commercial bank and the French AFD signed a US\$100 million power sector credit facility to support capital expenditure by Discos. Under the agreement, a maximum of US\$50 million loan could be made available to a DisCo at single digit interest rate for a maturity of between 7 and 12 years, with a moratorium of 2 to 3.5 years, depending on the project's cash flow. However, as with the DCA/USAID scheme, no disbursements were made as the commercial bank was reluctant to lend on projects identified as cash-flow positive due to the lack of preferential access to DisCo revenues for repayment.

⁵ CCI is a certificate issued by a Nigerian bank confirming an inflow of foreign capital either in the form of cash (loan or equity) or goods. A CCI is usually issued in the name of the investor within 24- 48 hours of the inflow of the capital into Nigeria.

Case Study: Azura-Edo 450 MW IPP

Nigeria's history of utility scale renewable energy IPP projects is limited, and the only recent major IPP project that achieved financial close with international financing is the 450MW gas fired Azura Edo IPP project and this was dominated by DFI financing. Prior to this, there was the 270MW AES Barge IPP, which was completed in 2001, but is not currently operational.

The Azura-Edo IPP, the first of a new wave of project-financed greenfield IPPs currently being developed in the country, achieved financial close in December 2015. It is also the first power generation project in Nigeria to receive World Bank PRG and Multilateral Investment Guarantee Agency (MIGA) support. The 450MW open cycle gas-fired IPP project recently came on stream in 2018.

It is believed that the project played a crucial role in signaling to private investors that despite challenges facing the country, bankable power projects can be successfully developed in Nigeria. It also expected to boost power generation, (providing electricity to an estimated 14 million people) stimulate infrastructural development and economic growth, while creating in excess of 1,000 jobs during its construction and operation.

The Project will be developed in three phases, starting with a 459MW open cycle gas turbine power station and a short underground gas pipeline connecting the power plant to the Escravos-Lagos Pipeline System, and a subsequent ramp up over two phases to bring the total capacity of the plant up to 1,500MW.

As analyzed by Olaniwun Ajayi, there was no direct precedent in the market for Azura. The financing structure was rather complex with senior debt, provided by DFI's, and local and international commercial banks, inclusive of a Nigerian Power and Aviation Intervention Fund (PAIF) naira-denominated facility (equivalent to about US\$120million) provided to First City Monument Bank Limited (FCMB), for on-lending to the Project. In addition to senior debt, certain DFIs also provided a mezzanine tranche to the Project.

Total debt amounted to c.US\$687 million and total equity amounted to c.US\$190 million (including in-kind contributions from the Edo State Government of Nigeria). One lender also provided a letter of credit to backstop the payment obligations of the Borrower under the gas supply agreement (as against the more familiar structure where the LC is procured by the equity side under a collateral arrangement).

Although the successful closing of the Azura Edo transaction has demonstrated that under the right circumstances this international support for Nigerian power exists, no other utility scale IPP has achieved financial close since then.

Asides from the 14 Nigeria Solar IPP Projects, other projects at different stages of development include:

- Qua Iboe 540MW IPP Project (QIPP)
- Oma 1,080MW IPP Project
- Chevron Agura 330MW IPP Project

C.1. Strategic Context

Economic Overview

Nigeria has a population of c.180 million and a GDP per capita stands at US\$2,457 (2017). Nigeria's economy had GDP growth rates of 0.83% in 2017 and is currently classified as a Middle - Income Country⁶. In 2016, the country experienced its first recession in nearly 20 years because of the slump in oil prices and production which caused foreign exchange shortages and high inflation. GDP growth in 2017 was 0.82% recovering from recession due to higher oil production price combined with monetary and fiscal policy measures. The IMF and the World Bank are projecting GDP growth of around 2% for 2018.

Table 2: Nigeria Sovereign Debt Rating

Rating Company	Rating	Outlook	Date
Standard & Poor's	B	Stable	March 2018
Moody's	B	Stable	November 2017
Fitch	B+	Negative	January 2017

Despite being the largest economy in Africa, Nigeria is still constrained on the economic and social front (ranked 152nd among the 188 UN member states Human Development Index⁷).

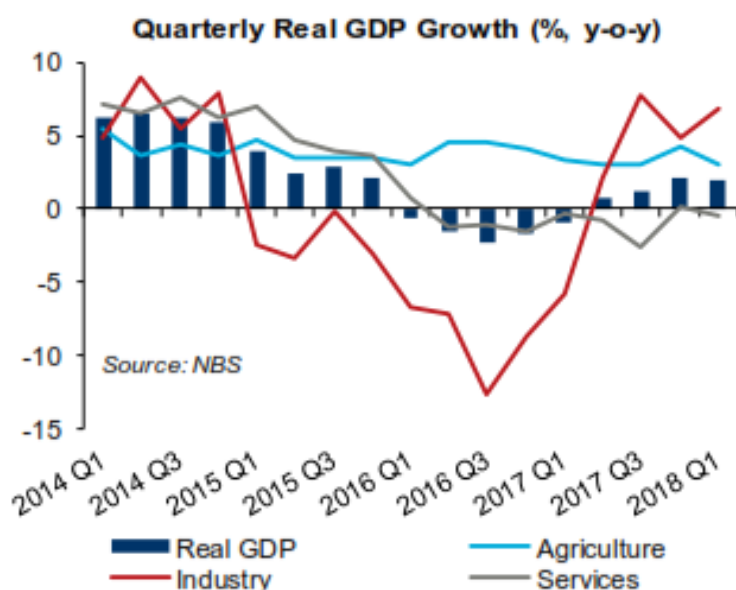
According to recent NKC African Economics reports, real GDP growth slowed to just under 2% y-o-y in the first quarter of 2018 from 2.1% y-o-y in the preceding quarter:

- Agriculture, which has been a key growth driver recently, had its growth slowing from 4.2% in Q4 2017 to 3% in Q1 2018;
- Industry continued to perform well and expanded by 6.9% in Q1 2018, driven by a 14.8% increase in hydrocarbon output;
- Manufacturing recorded a second consecutive expansion, with output growth rising from 0.1% in Q4 2017 to 3.4% in Q1 2018;
- The services sector moved back into contractionary territory (-0.5%) after recording a mild expansion (0.1%) in Q4;
- Oil output to a large extent, following the Delta attacks, is still playing a major role in driving growth.
- Non-oil activity remains subdued with GDP growth slowing from 1.5% in Q4 2017 to 0.8% in Q1 2018.

Contributing factors in this regard relate to weak private sector credit extension, fuel shortages and delays in passing the budget.

The non-oil economy's poor performance at the start of the year has dampened Nigeria's near-term growth prospects. Leading indicators also do not inspire confidence that the economy gained significantly more traction in Q2. Both the manufacturing PMI and its non-manufacturing counterpart have remained relatively stable thus far in Q2 compared to the levels witnessed during the first quarter. This, in addition to the economy's weak performance at the start of the year, prompted a downward revision in NKC Africa Economics growth forecast to 2.1% in 2018. It is expected that non-oil activity will gain more traction as the year progresses, driven by improved FX liquidity, easing price pressures and increased fiscal expenditure.

Figure 3: Quarterly Real GDP Growth



Source: NKC Nigeria Country Quarterly Update: Q2 2018

Climate Change Overview

The Program is aligned with the Nigeria Intended Nationally Determined Contributions (“NDC”) to address climate change under the United Nations Framework Convention on Climate Change (“UNFCCC”). As stated in the NDC, the climate change objective of Nigeria is reduction from Business as Usual (“BAU”) of 30% by 2030 using a range of programs specifically including solar PV. The NDC indicates that the impacts of climate change to the country vary in extent, severity and intensity and thus poses a significant threat to achievement of its development goals, especially those related to eliminating poverty and hunger and promoting environmental sustainability. Climate change brings increased variability in rainfall in Nigeria, resulting in flooding in some humid areas in the southern part of the country while a decrease in precipitation in northern savannah, putting a considerable proportion of the Nigerian population at risks of water stress.

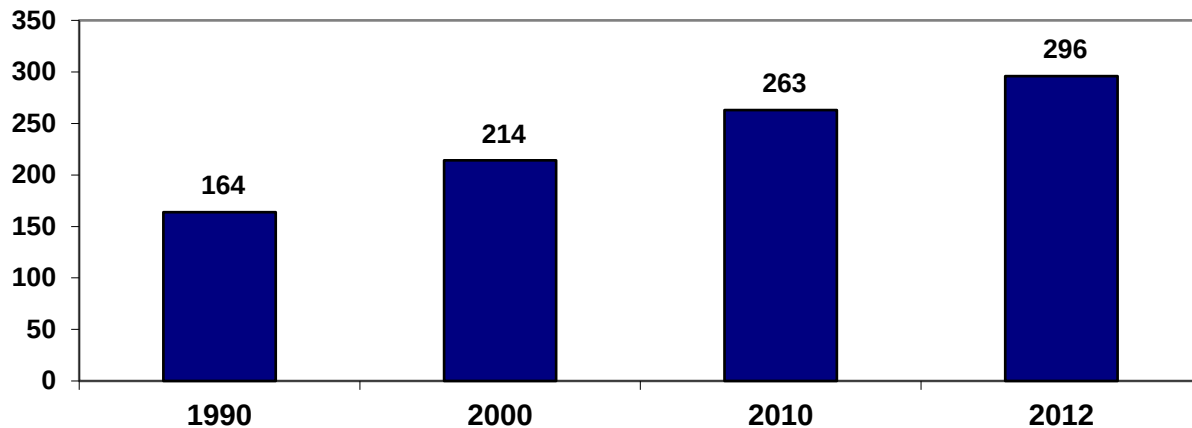
Variability in rainfall will not only impact the agriculture and food security sector, as the decline in yield in rain fed agriculture is expected to be as much as 50% but will also result in droughts and decrease in surface water resources, runoff and ground water flows in shallow aquifers which will have long-term implications for permanent and seasonable water bodies. This in turn is affecting hydroelectric power generation in Nigeria, which are for the most part located in the northern part of the country, and frequently suffers from low in-flow into the dams thereby affecting the electricity supply.

Nigeria has been actively engaged in international climate policy negotiations since it became a Party to the UN Framework Convention on Climate Change (FCCC) in 1994 and ratifying its Kyoto Protocol in 2004. Nigeria submitted its First National Communication (FNC) in 2003 and a Second National Communication in February 2014. Nigeria is host to several Clean Development Mechanism projects, as well as projects financed by the Adaptation Fund. In September 2012, the Federal Executive Council approved the Nigeria Climate Change Policy Response and Strategy.

42.5% of the Nigerian population is currently below the age of 14. The country has a high population growth rate and is expected to grow to a population of 392 million in 2050, becoming the world’s fourth most populous country. Between the years 1990 through 2010, the GHG emissions increased from 164 million tonnes (MT) CO₂eq to 263 MT CO₂eq. Population growth along with predicted economic growth is expected to drive the GHG to over 900 MT CO₂eq by 2030 (NDC, 2015). Historical emissions in Nigeria during the period 1850 – 2010 have been estimated as 2,564 MT.

Figure 4: Emission Values

Emissions in Nigeria - Million Tonnes (MT)



Sources: Nigeria's Second National Communication, Energy Commission of Nigeria, Climate Scorecard, NDC

The country has specific RE generation targets indicated in the NDC which includes working towards off-grid solar PV of 13GW, improvement of the country's electricity grid and the provision of 13GW of renewable energy to communities off the grid. The NDC clearly expresses the FGN's commitment for a transition toward renewable sources of energy (see Section C.2) and the signing of PPAs for the 14 solar IPP projects in 2016 are clear examples of the country's commitments within this space.

In terms of the Program's contribution to the GHGs emission reduction, it should be highlighted that the impact potential is twofold:

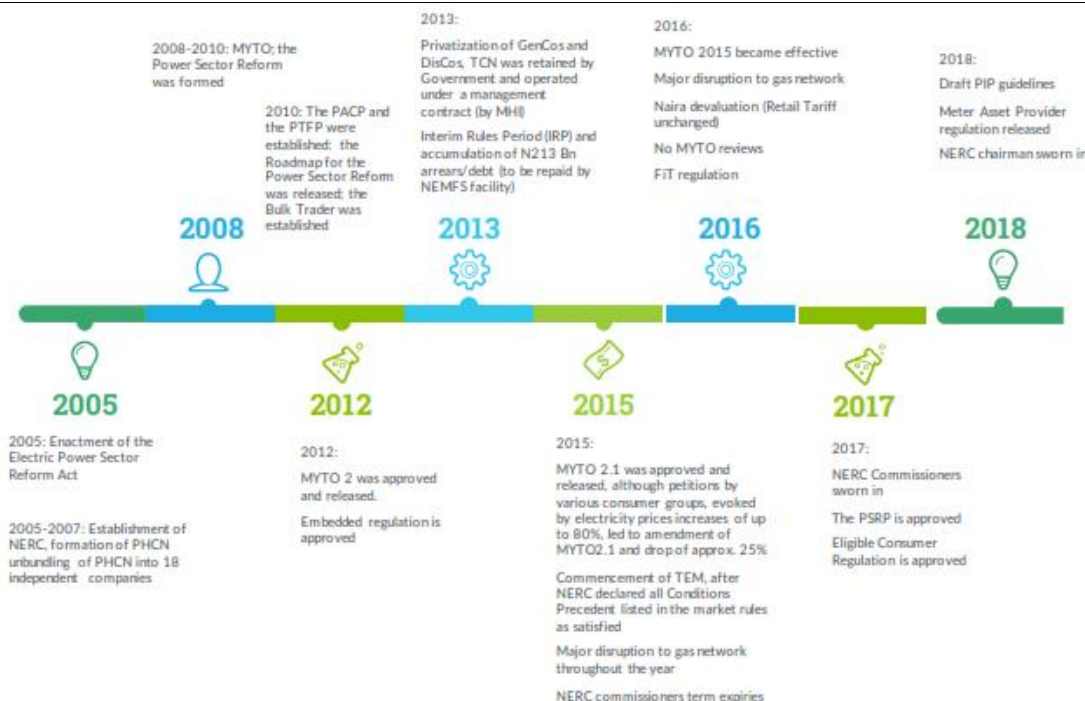
1. Increased RE generation in the energy electricity generation mix directly results in the avoiding emission of 476,487 tCO₂eq/year reduction (refer to Section E.1.2), and
2. The stable supply of electricity (especially from renewable sources) will curtail the use of fossil fuels including charcoal and firewood for cooking. This in turn reduces the potential emissions from longer term strategies that would have included more gas fired plants and diesel generators, which is a major source of GHGs emission in Nigeria (more data under Section C.2.)

Energy Sector Overview⁸

The Federal Government of Nigeria launched a far-reaching set of power reforms in 2001 that led to unbundling and subsequent privatization of electricity and distribution companies in 2013. The National Electric Power Policy in 2001 specified the reform agenda that resulted in the Electric Power Sector Reform Act 2005 (EPSRA). This Act removed the monopoly of the vertically integrated National Electric Power Authority and unbundled it into six generation companies (GenCos), 11 distribution companies (DisCos), and the Transmission Company of Nigeria (TCN).

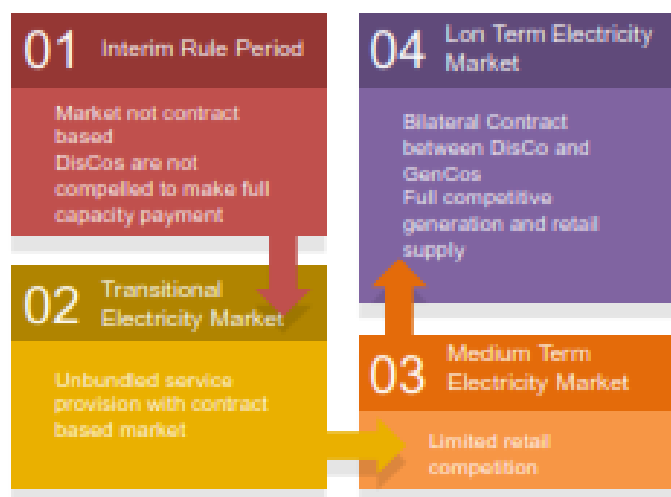
Figure 5: NESI Timeline (2005 – 2018)

⁸ Extract of the feasibility report conducted by AF Mercados for the purpose of the Program.



The 2010 Roadmap for the power sector reforms envisaged four stages of evolution after the privatization of the sector to culminate in a competitive, efficient, private sector-led power sector regulated by Nigerian Electricity Regulatory Commission ('NERC'), with the Ministry of Power providing general policy oversight. The four stages are as follows:

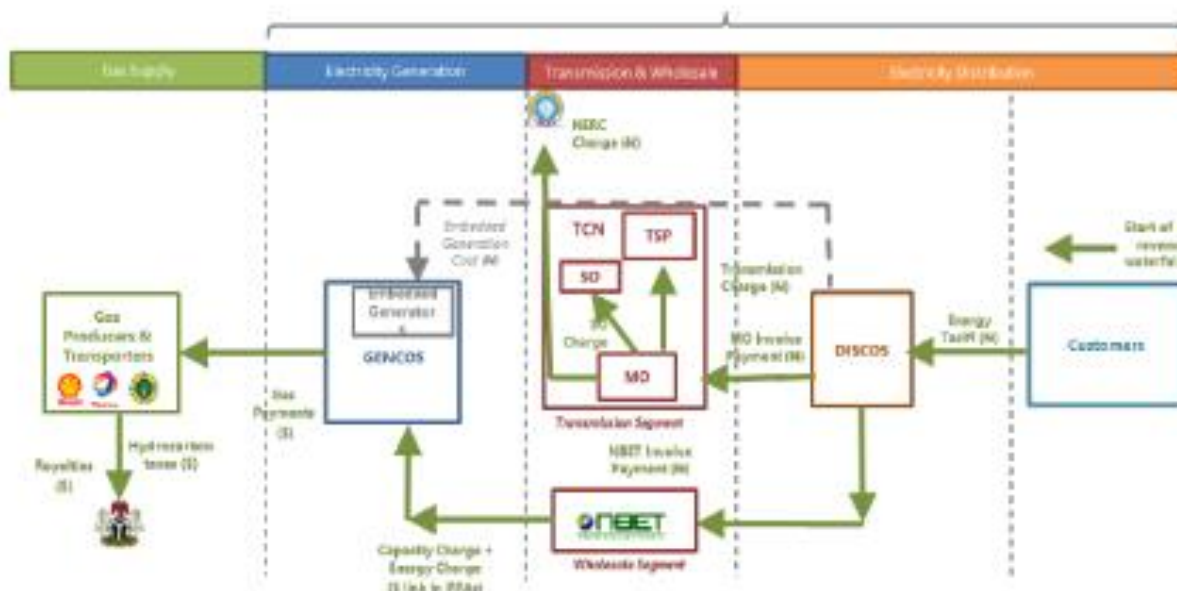
Figure 6: Market Development Stages



1. The Pre-Transition market, characterized by the unbundling and privatization of PHCN, putting in place the Market Rules and Grid Code and establishment of incentives for distribution and generation activities;
2. The Transitional Electricity Market ("TEM"), characterized by contract-based arrangements and the effectiveness of same;
3. The Medium-Term Electricity Market characterized by presence of competition and a "centrally administered balancing mechanism in the market"; and
4. The Long-Term Electricity Market characterized by bilateral contracts between electricity buyers and sellers at all levels and a central balancing mechanism through the creation of a spot electricity market. On 31 January 2015, TEM was effectively declared by an order of NERC, but without all the pre-requisites for TEM in place. The reforms are still at the TEM stage and TEM commercial operation is not yet fully implemented and unfortunately the reform program has not yet delivered substantial improvement in electricity services, as

explained in the following sections. The current Nigerian Electricity Industry (NESI) operation/agents under the TEM is summarized in the figure below.

Figure 7: Nigerian Electricity Supply Industry (NESI)



The Nigerian electricity market is currently at the Transitional Stage Electricity Market (“TEM”) phase, which came into effect on February 1, 2015. TEM was originally envisaged to be characterized by contract-based arrangements for electricity trading and the introduction of new entrants/competition into the market. This has however not materialized with contracts not fully enforceable.

The Multi-Year Tariff Order (MYTO) 2.1 established regulated prices to be paid to licensed Generation Company’s (“GenCo’s”) for electricity provided to Distribution Companies (“DisCos”) for the period between June 21st, 2012 and May 31st, 2017. However, the tariffs are negotiable if a GenCo can prove that their costs are not-in-line with the assumptions laid out in MYTO. In February 2016, the Nigeria Feed-in Tariff for renewable electricity came into effect, superseding, the MYTO order. The regulation obliges DisCos to source at least 50% of their procurement from renewable sources and prices tariffs according to the “Long Run Marginal Cost and Levelized Cost of Energy”.

Since TEM commenced, the most significant issue has been the persistent financial shortfalls in the system owing to the following:

1. Average technical commercial and collection (ATC&C) losses that exceed 50% (occasioned by electricity theft, poor collection)
2. Tariffs that are not cost reflective (below the cost of electricity)
3. Minimal remittance on the part of the DisCos, of collected revenues (currently at c.30%)
4. Significant devaluation of the Naira over the last few years, which has impacted on many of the owners of the privatized assets who procured US\$-denominated financing for the acquisitions

The above situation is exacerbated by load rejection by the DisCo’s. Nigeria also frequently struggles with attacks on gas/oil supply facilities, usually pipelines, which leads to a decline in gas supply and consequently electricity supply. Disruptions caused by attacks were a significant problem in 2016. Since then, the government has taken measures to pacify vandalizers, which has allowed much needed time for repairs to restore normal operation to the afflicted oil and gas facilities. In addition to issues with militancy, Nigeria has inadequate gas infrastructure which has precluded the processing and transporting of enough gas to satisfy the demand in the domestic market.

Because of these challenges NBET has sometimes struggled to pay for bulk supply from the GenCos. NBET is responsible for paying for energy and capacity under the power purchase agreement it signs with GenCos. The market operator is currently responsible for paying for electricity supply from legacy supplier where NBET contracts are not effective. These payments are made on a best efforts basis with the result that only a portion of capacity payments are paid to incumbent GenCos. According to NBET, outstanding payables to Gencos as at end of 2016 was NGN473 billion (US\$1.3 billion).

This trend continues to impact the attractiveness of the sector to developers and lenders evaluating investment opportunities in the power sector. Recognizing the threat these challenges pose to the survival of the sector, the Federal Government of Nigeria is working to implement many initiatives as follows:

1. Nigeria Electricity Market Stabilization Facility (“NEMSF”)

This NGN213 billion (US\$592 million) scheme was introduced by the Central Bank of Nigeria, in partnership with NERC in 2014 to cover the cash shortfalls in the sector from November 2013 to December 2014, due to the delay in declaring the commencement of the transitional electricity market. The facility was disbursed as a loan to the Discos and was also applied towards payment of the legacy shortfalls to the Gencos, post the takeover of the privatized assets.

2. NGN701 Billion Payment Assurance Guarantee Program

The NGN701 billion (US\$2 billion) payment assurance guarantee program was announced by the Central Bank of Nigeria (CBN) to restore and establish financial stability in the electricity sector given the dismal performance of the Discos in settling invoices for energy sold to them. It is a liquidity intervention program that bypasses the Discos and ensures that the gas suppliers and Gencos receive payments for their invoices. The scheme will cover payments to the gas suppliers and Gencos from January 2017 until December 2018. The introduction of the scheme has since resulted in improved gas supply and a reprieve for the Gencos from a cashflow perspective. We understand that payments through this mechanism cover c.50% of invoice payments and it is anticipated (but not confirmed) that the program will be renewed or extended.

3. Power Sector Recovery Program

The World Bank expressed its support for the Power Sector Recovery Program on April 22, 2017, a month after the Federal Executive Council of Nigeria approved it. The program focuses on supporting implementation of power sector reform, reducing losses in the DisCos, enhancing the sector's financial viability, increasing access to electricity services and mobilizing private sector investment. Specifically, the program plans to:

- Eliminate the accumulated deficit in the power sector
- Commit future funding for the energy sector from 2017 to 2021
- Ensure the performance of DisCos

To achieve this, the following actions have been approved:

- Completion of the disbursement of NEMSF to the respective gas suppliers, generation companies and distribution companies.
- Implementing NGN701 billion payment assurance program with CNB for NBET to pay future bills
- Paying the accrued shortfall accumulated in years 2015-2016 attributable to the non-cost reflective tariff, FX shock and low energy levels
- Ensuring that distribution companies that perform below the agreed loss levels at the time of the handover as approved by NERC in 2014, be held accountable for the inefficiency.

The World Bank Group has indicated its willingness to assist the plan by providing a total of US\$2.5 billion credit facility. US\$305MM of the facility will be for providing guarantees to IPPs through the group's development bank the International Bank for Reconstruction and Development (IBRD). In addition to the Group's support, IFC plans to provide US\$1.3 billion of direct investments for an additional 3.5GW of power generation and MIGA US\$1.4billion of guarantees for IPPs. The PSRP is yet to be fully implemented and is still under review by the FGN.

C.2. Program Objective against Baseline

Country Baseline: Historical emissions in Nigeria during the period 1850 – 2010 have been estimated at 2,564 million tonnes (MT) CO₂eq⁹. Between the years 1990 through 2010, the GHG emissions in Nigeria increased from 164 to 263 MT CO₂eq. Under a business-as-usual growth scenario, consistent with strong economic growth of 5% per year, Nigeria's emissions are expected to grow to around 900 MT per year in 2030, which translates to around 3.4 tonnes per person but under a high growth scenario with economic growth at 7% the emissions could go well beyond 1 billion tonnes per year. In addition, climate change will have a significant effect on the energy sector in Nigeria as rising temperatures would result in increased energy expended for air-conditioning, refrigeration and other household and industrial uses in the context of severe shortages of energy supply. Moreover, climate change results in increased variability in rainfall which will affect the water in-flow into dams. This negatively impacts hydropower generation that is predominantly located in the center and northern part of the country.

⁹ CAIT database, World Resources Institute, NNDC

Electricity system (project) baseline: At status quo, Nigeria has a gross installed capacity of approximately 13GW, composed of 74% fossil fuels and 26% hydro with a total of **25,215 GWh (25,215,000 MWh)** of power generated and delivered to the Discos in 2017¹⁰. As per NERC the 25,215 GWh is generated from about an operational capacity ranging from 4GW to 5GW depending on the month (see first para of section A.2).

Nigeria is part of the West Africa Power Pool (WAPP) as it is interconnected with Benin, Niger and Togo. The country grid emissions factor for WAPP countries was determined as part of the elaboration of the CDM standardized baseline i.e. "Grid emissions factor for West Africa Power Pool version 01.0" (ASB0034), which was approved by the CDM Executive Board on the 27th of February 2017¹¹. The baseline grid emissions factor of Nigeria was therefore determined as **0.561 t CO₂/MWh** corresponding to the combined margin emission factor of the electricity system of Nigeria. Therefore, the baseline emissions of the Nigeria Electricity system can be derived for the year 2017 as **14,145,615 tCO₂ (25,215,000 MWh x 0.561 t CO₂/MWh)** and this can be assumed as being the yearly emissions of the Nigeria electricity system in the 'business as usual' scenario or with more fossil fuel generation increasing in the electricity supply mix.

With per capita power consumption at ~151 kWh¹² per year, Nigeria lags far behind other advanced developing nations such as South Africa and Morocco, in terms of grid-based electricity consumption. Global Data forecasts electricity consumption in Nigeria to increase at a CAGR of ~7.5% between 2015-2020, fueled by economic and population growth. The lack of reliable on-grid power sources in the country, widespread and expanding use of biomass-based energy (charcoal, wood fuel) and on-site fossil-fuel based generators constitutes the major sources of energy in Nigeria, resulting in increasing GHG emissions¹³. To address electricity generation shortfall, increasing GHG emissions and make provisions for future demand requirements, the FGN has embarked on a program to increase generation capacity and to diversify its generation sources to include renewable energy. The FGN has therefore targeted the development of about 3,500 MW additional capacity of electricity through renewable energy sources as part of its vision 2020 strategy.

Anticipated climate change mitigations impact: The Program will contribute to avoid and/or reduce an average of approximately 476,487 tCO₂eq of GHG emissions annually from fossil fuels-based power generation, cumulating to approximately 9,529,739 tCO₂eq over the 20-year operational phase of the Solar PV power plants, as per the detailed calculation described in section E.1 of this document. The Solar PV power plants of this program will be constructed in the central and northern states of the country, where grid-connected households and businesses currently suffer from power shortages resulting in the constant need to resort to diesel power generators to fill energy supply gaps. The additional contribution of electricity from the Solar PV program into the Nigerian national grid will improve the reliability of grid-supplied electricity to these on-grid customers and will contribute to a reduction in the use of diesel generators thereby lowering CO₂ emissions from diesel generation. In addition, the program will increase the share of clean energy power onto the Nigeria electricity system (other than hydro) thereby diversifying the low carbon technology mix.

Anticipated gender and socio-economic impact: In Nigeria, many women and men, particularly those with low incomes living in urban and peri-urban areas, are more disadvantaged than most in terms of their ability to access electricity and modern energy. In most SSA countries, lack of access to electricity and other forms of energy adversely affects businesses, thereby reducing standards of living leading to a depression of the overall economy. Limited access to energy for the most basic needs of cooking, lighting, income generation, schooling and health centers limits human development in many spheres, including but not limited to education, nutrition and food security, water and sanitation, market-based work, health, and gender relations. [4]

Energy Use - Increased access to electricity (which will occur as a direct result of the projects) will have a positive impact on two categories of women living in grid-connected urban and peri-urban areas in two ways; by providing the urban poor, (who are at the bottom of the energy ladder in urban areas and who primarily depend on biomass sources such as cow dung, twigs, paraffin, charcoal, firewood etc.) with cleaner energy sources, and; by providing increased, stable electricity to more educated and affluent women, who will have electricity as a cheaper option for energy intensive activities such as cooking, in addition to liquefied petroleum gas.

Economic Impact - Increased access to electricity will lead to women in electrified households being more likely to find paid work outside of the home, resulting in financial and economic empowerment. Women can be found along

¹⁰ <http://www.nercng.org/index.php/library/industry-statistics/generation/541-monthly-sent-discos#data>

¹¹ https://cdm.unfccc.int/methodologies/standard_base/2015/sb102.html

¹² Powering Nigeria for the Future: The Power Sector in Nigeria July 2016 – PWC Report

¹³ Cervigni, R., Rogers, J.A., and Dvorak, I., Assessing Low-Carbon Development in Nigeria: An Analysis of Four Sectors, World Bank Study, 2013

Nigeria's energy value chain as consumers, suppliers and producers. Women entrepreneurs act as agents of change along the energy value chain, delivering benefits, and creating economic and social impact.

Education – Increased electrification, through additional input into Nigeria's national grid will result in better education outcomes for female children, as school attendance increases by releasing domestic labor constraints on women and girls. Studies have shown that countries with better electricity access have higher female literacy rates.

Health –The primary source of energy for the majority of the energy poor is biomass – which in some cases (especially in rural areas) is sourced more than 5 kilometers away on foot. This has a negative impact on the health of women and girls, because they normally carry heavy loads on their heads, thus compromising physical health. Exposure to indoor air pollution is responsible for over 4 million deaths a year globally¹⁴ – increasing electricity access will lead to decrease in biomass use and improved health.

The Program directly contributes to the implementation of the country's prioritized mitigation programs, by ensuring the take off and transition of electricity generation away from conventional energy sources. This Program will enable the financing of up to 400MW solar PV projects that will feed renewable energy to the Nigerian grid, increasing the share of renewables in the energy mix. Renewable energy and solar projects are still new in the country where this will be the first utility-scale solar PV projects. This will have a significant demonstration effect to the Nigerian market and will help place the country's energy sector onto a sustainable pathway and nurture a market for continuous renewable energy investments. Furthermore, diversification of the national energy mix with renewable energy sources will improve Nigeria's climate resilience by reducing the reliance on gas-fired power.

Overall the Program is expected to contribute to the implementation of Nigeria's Nationally Determined Contributions by bringing positive impacts to the emission profile of the country.

C.3. Program Description

Summary

To show its commitment to increasing generation and diversifying away from thermal and hydro, the FGN through the Nigerian Bulk Electricity Trading Plc ("**NBET**"), the off-taker, signed 14 utility scale solar Power Purchase Agreements (**PPAs**) in July 2016 to supply c.1,125MW of power to the Nigerian power grid. However, the **Nigeria Solar IPP Projects** that will bring significant diversification from diesel generation, stand the risk of not being able to reach financial close because of the need to decrease the tariff to a sustainable level. The current tariff of US\$11.5 cents/kWh signed under the PPAs, is considered unsustainably high by key sector players in Nigeria because it makes the solar projects amongst the most expensive sources of electricity for NBET the off-taker. It is our understanding that the weighted average cost of generation is US\$7.5 cents/kWh as at December 2017 (US\$6.6 cents/kWh as at March 2018) versus an end-user tariff in the range of US\$3.1 cents/kWh to US\$8.1 cents/kWh. Therefore, a tariff of US\$11.5 cents/kWh will exacerbate the current liquidity challenges in the Nigeria power sector. In addition, because the FGN stands behind the financial obligations of NBET under the PPA's with the Nigeria Solar IPP Projects, it also significantly increases the country's contingent liabilities beyond what is sustainable for the FGN. The Ministry of Finance is proposing a tariff of US\$7.5 cents/kWh to allow the projects to reach financial close. While a reduced tariff will strengthen the government's commitment to the solar program throughout the term of the PPA, most projects under the Program are not viable at US\$7.5 cents/kWh tariff without the GCF concessional financing, which enables a lower tariff will still covering the costs of debt.

Without the Program, it is unlikely that any of the projects will reach financial close as concessional financing is required to bring down the current tariffs of US\$11.5 cents/kWh to the target tariff of US\$7.5 cents/kWh. It is our estimation, that without concessional financing none of the projects will attain financial close because the project returns will not be acceptable i.e. too low.

In addition to enabling select solar projects achieve financial close, the Program will simultaneously support the FGN in its efforts to catalyze private investment in the renewable energy sector, thereby accelerating the achievement of its electricity generation targets and the diversification of its energy mix. Furthermore, the Program will contribute directly to the advancement of the 7th UN Sustainable Development Goal, which is "*affordable and clean energy*" by addressing the demand gap in Nigeria's electricity market and reducing Nigeria's reliance on fossil fuel through clean and renewable energy development.

¹⁴ World Health Organisation

Finally, despite the significant sector challenges, the Program will support the implementation of two critical credit enhancement structures that improve the bankability of the solar projects. Specifically, to manage the NBET liquidity risk, the off-taker is contractually obliged to provide a letter of credit (“**LC**”) covering 6 months’ worth of energy payments. To facilitate the issuance of this LC by JP Morgan, AfDB, will provide Partial Risk Guarantee (**PRG**) cover for a select number of projects¹⁵. Specifically, to cover termination risk, the off-taker has agreed to execute a Put Call Option Agreement (**PCOA**), where under all termination scenarios, senior debt will be repaid by the FGN in US Dollars.

There is currently no utility scale solar IPP PV project in Nigeria. This Program is an opportunity to spearhead the development of the first of such projects, which will have significant demonstration effect to FGN and the energy and power market in the country, with the seventh most populated country in the world (190 million) according to the world population review¹⁶.

Program Details and Project Selection Process

The main activity of the Program is to provide long term financing to Nigeria Solar IPP projects. The Program offers a financing package to selected projects meeting the selection criteria. It is expected that 3 - 5 projects will be financed under this Program. To incentivize sponsors to optimize their financial structures, the selection criteria will incorporate a ranking that will benefit projects which require less concessional funding as a percentage of project cost.

The selection criteria will have 5 phases:

Phase 1 (Confirmation of Interest): In order to be considered for the Support Program, Project Developers were required to submit expressions of interest in the Support Program.

Phase 2 (Information Packs): In order to analyze and rank projects, developers were required to provide responses to an information request pack, which covered General Project Information and 5 key selection criteria: 1) Environmental Impact 2) Social and Development Impact 3) Economic Potential 4) Project Readiness and 5) Capacity and Expertise.

Phase 3 (KYC Review): all projects will be screened for red flags. Any project with identified red flags will be engaged for clarification or enhanced due diligence if necessary. This phase is ongoing.

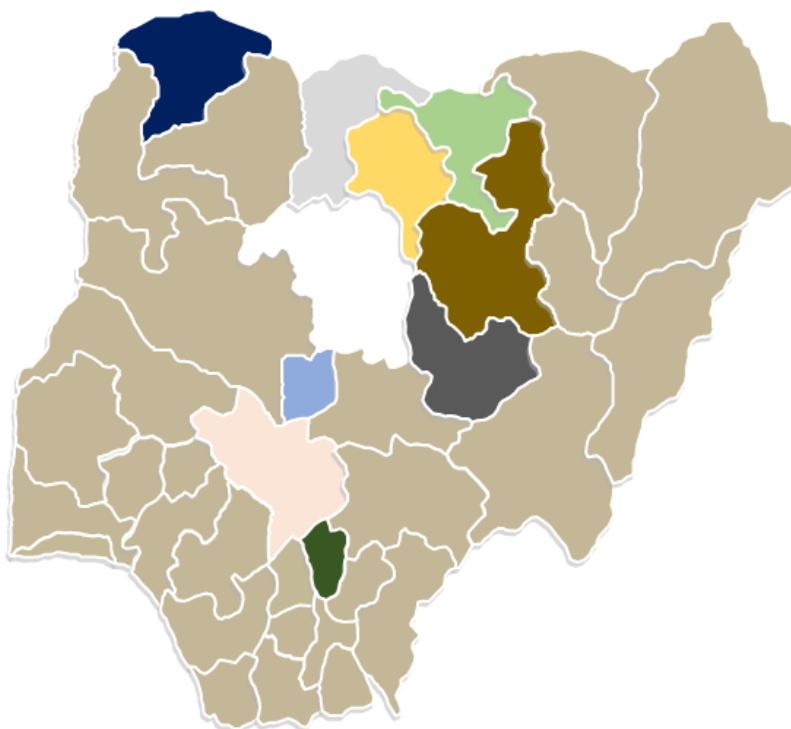
Phase 4 (Short Listing): Once the information packs have been reviewed, the projects will be shortlisted on the basis of their ability to meet each of the 5 selection criteria (i.e. no weighting of one over the other).

Phase 5 (Ranking and Negotiation): Following the analysis of the information provided and a selection of the short list based on the above selection criteria, the projects will subsequently be ranked in terms of the least need for concessionality. Based on the funding package, the top 3 – 5 projects will receive copies of the indicative term sheet for negotiation and will be engaged for further due diligence.

Figure 8: Map of Nigeria showing locations of the Nigeria Solar IPP Projects

¹⁵ It is expected that all selected projects funded by the Program will have PRG prior to first disbursement

Sokoto 1. KvK Power 55MW
Katsina 2. Pan Africa Solar 75MW 3. Nova Solar 5 Farm 100MW
Kaduna 4. Anjeed Kafanchan 100MW 5. EN Africa 50MW 6. Abiba Power 50MW
Abuja 7. LR Aaron Power 100MW
Jigawa 8. Nova Scotia Power 80MW 9. Oriental Renewable Solutions 50MW
Nassarawa 10. Afrinergia Power 50MW
Enugu 11. Motir Dusable 100MW
Bauchi 12. Nigeria Solar Capital Partners 100MW
Plateau 13. CT Cosmos 70MW
Kogi 14. Middle Band Solar One 100MW



The location of these projects is significant for a variety of reasons:

1. The bulk of them are situated in areas with high irradiation (see Figure 17)
2. Apart from the positive environmental benefit associated with solar, solar is the most feasible way of generating power in Northern Nigeria. Most of the main Northern cities are located over 1,000 km from the main thermal power stations in the Southern part of the country.
3. Solar has the potential to provide power to the underserved load centers and increase economic activity in Northern Nigeria.
4. With its location in Northern Nigeria, the Project has a key role in increasing generation capacity in power-starved areas and catalyzing economic development where the power shortage is the most severe with limited generation options.
5. The bulk of these projects are in the predominantly Derived Savannah and Guinea Savannah areas in Northern Nigeria which has high climate vulnerability, food insecurity, extensive deforestation, and with high dependency of the population on biomass energy as well as inefficient technologies which further aggravate the vulnerability of livelihoods to climate risk.
6. 50% of these projects are in the 10 poorest states in Nigeria (see Figure 18)

Table 3: Details of the 14 Solar IPP Projects in Nigeria

	Project / Project Company	State	Capacity (MW)	Financial Close?
1	Pan Africa Solar	Katsina	75	No
2	Nigerian Solar Capital Partners	Bauchi	100	No
3	Afrinergia Power Limited	Nasarawa	50	No
4	Motir Dusable Limited	Enugu	100	No
5	Nova Solar 5 Farm Limited	Katsina	100	No
6	KVK Power Limited	Sokoto	50	No
7	Middle Band Solar One	Kogi	100	No
8	LR Aaron Power	Abuja	100	No
9	Novia Scotia Power Development Company	Jigawa	80	No

10	CT Cosmos	Plateau	70	No
11	Oriental Renewable Solutions	Jigawa	50	No
12	Quaint Abiba Power	Kaduna	50	No
13	Anjeed Innova Group	Kaduna	100	No
14	EN Africa	Kaduna	50	No

Source: Infrastructure Journal 21 December 2017 & 11 January 2018, Nigeria Power Guide, Volume 4, 2018 Edition

Existing Credit Enhancement Instruments

6-month Letter of Credit from NBET

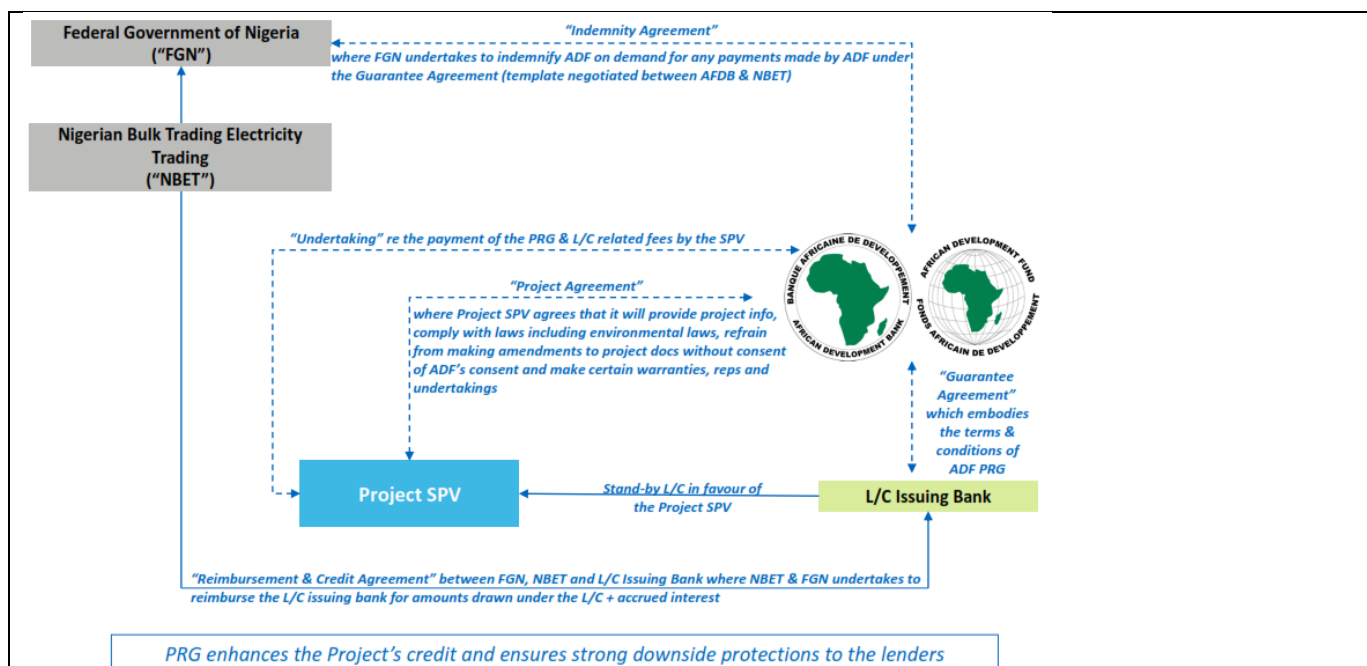
AfDB approved a UA120 million (US\$165 million equivalent)¹⁷ Partial Risk Guarantee Program in support of the Power Sector Privatization, which aims to increase electricity generation in Nigeria by catalysing private sector investment and commercial financing in Nigeria's power sector through the provision of Partial Risk Guarantees (PRGs). The PRG's are to mitigate the risk of NBET not fulfilling its payment obligations under its PPAs with IPPs (not just Solar IPP's).

The AfDB guarantee will provide liquidity support in the event of an NBET payment default. Under the LC structure:

- The PRG provides risk mitigation through a revolving standby LC
- The facility will be opened by the FGN in favor of the project company (IPP)
- The FGN would provide security under the PPA in the form of an LC, issued through a commercial bank (JP Morgan), in favor of the IPP, for 6 months
- The LC could be drawn in the event the FGN fails to make timely payments to a covered IPP under the PPA, subject to certain grace periods, for the unpaid amount
- Following a drawing, the FGN would be obligated under a Reimbursement and Credit Agreement (to be entered into between the FGN and JP Morgan) to make a repayment to the JP Morgan for the amounts drawn (plus accrued interest) within a determined period.
- If the FGN makes a payment within such period, the LC would be reinstated to the amounts repaid
- However, if the FGN fails to repay the LC bank within such period, the LC bank would have recourse to the PRG for the drawn amounts, plus any accrued interest, under a Guarantee Agreement (to be entered into between the AfDB and LC bank)
- In such case, the maximum LC amount would be reduced by the amount of payment made by the AfDB under the PRG
- Payments by the AfDB ADF under a PRG would trigger the obligation of the FGN under the Indemnity Agreement (to be entered into between the AfDB and the Government), which requires that the Government repay on demand, or as the AfDB may otherwise direct

Figure 9: PRG Schematic

¹⁷ Using UA/US\$ exchange rates as at December 2018



The typical PRG processing process once a project has been nominated is to process this in parallel with the debt for that project. This will involve undertaking the required due diligence, assessing the usual technical, financial and environmental aspects of the project as part of the approval process.

Put Call Option Agreement

The FGN provides support to the project in the event of an early termination of the PPA through the mechanism of a put and call option agreement (PCOA). The PCOA allows the project company to 'put' the plant (or its shares of the Project company) to the government in FGN default scenarios. In these circumstances, the government is obliged to pay a 'purchase price' which, at a minimum, covers the outstanding debt.

Debt Service Reserve Account (DSRA)

The DSRA, consistent with standard project finance practices, will provide an additional security to cover 6 months of principal and interest payments as a buffer.

Finally, the gender assessment (See **Annex 4**) provides information on the gender equality situation in Nigeria, and the gendered nature of energy poverty in the country. The gendered division of labor creates different energy needs, as do different perceptions of the benefits of energy and the capacity to access those benefits depending on one's gender. To meet these needs, the assessment identifies entry points (i) to increase women's participation and representation in the renewable energy sector; and (ii) to increase opportunities for women-owned businesses in the renewable energy sector. An action plan was developed to propose feasible activities to ensure equal opportunities for women and men to participate in and benefit from the project. This will aid in integrating gender equality and women economic empowerment in the projects.

Additionality / Replicability

As part of AfDB's PRG program, there is a facility in place to provide funds for capacity building in favor of key institutions involved in Nigeria's power sector reform, notably NBET and the Nigerian Electricity Regulatory Commission (NERC). This is specifically in respect to implementation and enforcement of procurement as well as environmental and social rules and regulations.

Local Commercial Bank Participation

The primary form of local financial institution participation will be through the proposed commercial tranche of the facility. While initial market soundings have indicated some interest in participation, several constraints exist:

- **Tenor:** commercial banks find it challenging to provide loans for longer than 5-7yrs.
- **Currency:** commercial banks are unable to provide significant funding in FX.
- **Pricing:** commercial banks find it challenging to meet the relatively low DFI pricing levels of sub 6% margins in USD or naira.

- **Target tariff:** the requested reduction in tariff from US\$11.5 cents/kWh to US\$7.5 cents/kWh, already stretches the capacity of DFI lenders in the funding group as well as the capacity of sponsors to continue to develop the project, due to significantly reduced IRR's.

C.4. Background Information on Program Sponsor

Africa Finance Corporation

AFC is a US\$ 4.2 billion Africa-focused infrastructure financing institution established by an agreement between sovereign states. It aims to address Africa's infrastructure development needs through financing private-sector led projects in critical sectors such as natural resources, power, heavy industries, transport and telecommunications. The company is one of the highest rated financial institutions in Africa with an international A- credit rating, assigned by Moody's Investor Service. AFC's hybrid business model combines public and private ownership, country membership and a private sector governance and decision-making structure. This allows AFC to combine a deep understanding of African governments (i.e. resources, constraints and key objectives) with a commercially driven investment process, which allows for flexibility, innovation, faster decision making and more robust project and financing structures.

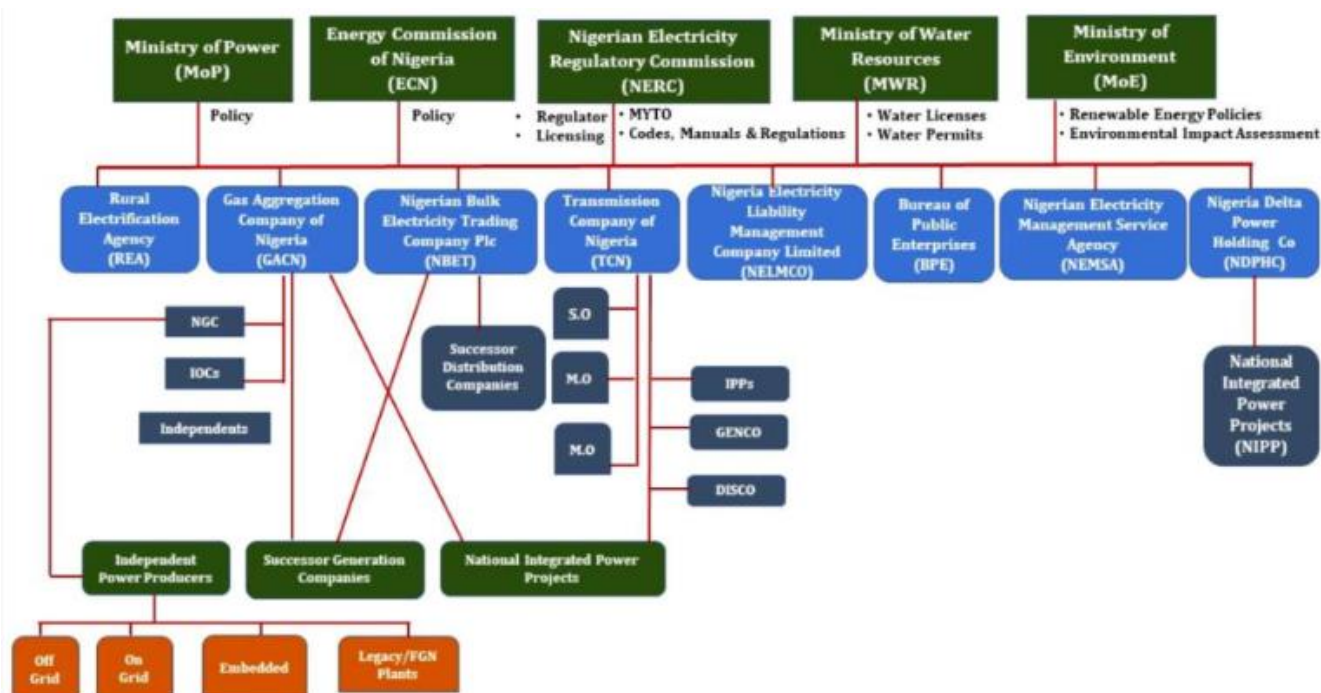
The Corporation offers project development funding, long-term debt, mezzanine debt and equity financing and has significant experience in power projects on the continent. Its track record of investments in the sector covers debt, equity and project development. To date, it has completed investments in Nigeria, Ghana, Cape Verde, Rwanda, Zambia, Ivory Coast, Benin and Ghana with a cumulative generation capacity of c. 4GW. Within Nigeria, the firm has already invested over US\$148 million on power generation and distribution projects.

Implementation teams for this Program will be constituted within the AFC and other financiers to the due diligence and execution of individual sub-projects as well as the management of **the overall Program**. The AFC team will be mobilized from a pool of investment officers (from within the power sector team), portfolio managers and risk assessment officers (inclusive of an E&S specialist), and we envision, as is the typical practice, that the same will be done by other senior lenders.

C.5. Market Overview

Regulatory Environment: In 2005, the Electricity Power Sector Reform Act ("EPSRA") came into force paving the way for unbundling of the power sector. Nigerian Electricity Regulatory Commission (NERC) was formed in the same year as industry regulator. NERC is the overarching power sector regulator with responsibility for licensing, tariffs, codes and monitoring entities engaged in generation, transmission, distribution and trading of electricity.

Figure 10: Regulatory Framework and Key players of the Nigerian Power sector



Tariff Framework: As part of the reforms, a revised Multi-Year Tariff Order (“MYTO”) was introduced in 2012 to implement cost reflective tariffs which became effective in 2015. The implementation was delayed due to elections, a change of government in 2015 and court challenges.

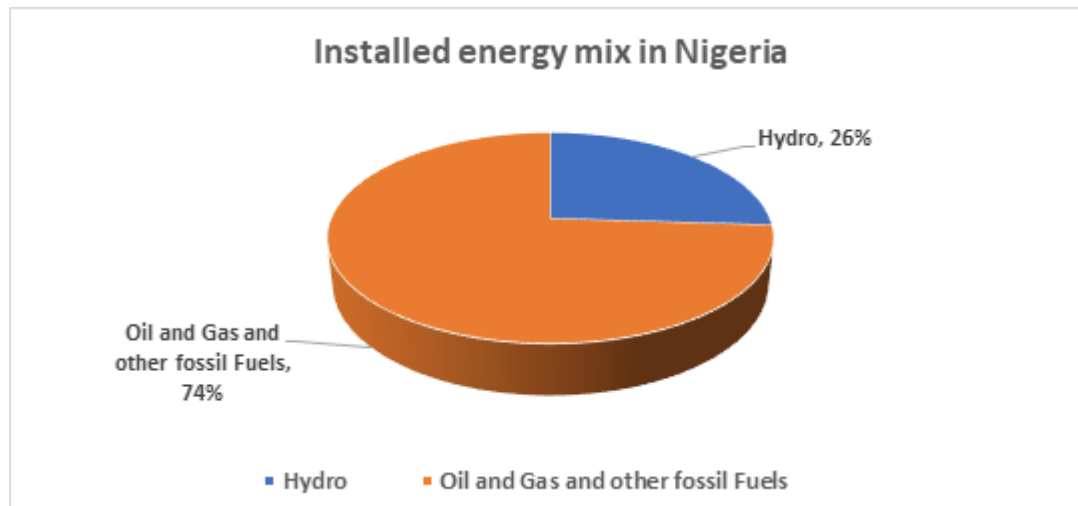
Institutional Framework: With the enactment of EPSRA, the defunct National Electric Power Authority (‘NEPA’), the sole provider of electricity, was replaced by the Power Holding Company of Nigeria (‘PHCN’), which was then unbundled into 6 Generation Companies (“GenCos”), 11 Distribution Companies (‘DisCos’) and the Transmission Company of Nigeria (‘TCN’). NBET was formed in 2010 to be the sole buyer of bulk electricity and sole seller of electricity to DisCos until such a time as GenCos and DisCos will be able to directly enter into commercial contracts. NBET was structured to act as a creditworthy intermediary between DisCos, TCN and GenCos, serving a temporary role until the sector achieves fiscal strength.

Although the power sector reform was well received, it has not yet generated the expected outcome. The electricity market is facing a liquidity crisis which does not only threaten the entire power value chain but also the collapse of key sectors in Nigeria such as the banking sector which is the main debt provider to DisCos and GenCos. The scale of the liquidity crisis is not yet clear, but it was estimated at the end of 2016 that the market revenue shortfall was approximately US\$3 billion and rising.

Transmission Network: The Nigerian electricity transmission network is managed by TCN. As part of the unbundling program, the TCN remained wholly owned by the FGN and retained its role as the market and system operator solely responsible for dispatching electricity from the GenCos and distributing power from DisCos to the ultimate consumers.

The Nigerian grid system has a capacity of up to 7G – 8GW¹⁸, which can evacuate the current available capacity, but will be insufficient once new IPPs come on stream in the future. Nevertheless, the TCN has comprehensive plans to rehabilitate and expand the existing grid to a wheeling capacity of 20GW over the next few years. This includes an estimated spend of at least US\$7.5 billion with the first major stage of the rehabilitation and expansion program coming onstream in 2020. The various works are to be funded by the FGN and DFIs.

Figure 11: Installed energy mix in Nigeria



Source: Minister of Power, Works and Housing, Business Day 17th October 2017

Electricity Demand: Electricity demand has been growing at a rate of 7% per annum over the past decade with almost no investment into generation capacity. The current unconstrained demand is estimated to be over 17 GW while actual generation is fluctuating between c. 3,000 MW - 5,000 MW reflecting severe shortages in electricity supply. With per capita power consumption at 151 kWh per year, Nigeria lags far behind other developing nations in terms of grid-based electricity consumption. Global Data forecasts electricity consumption in Nigeria to increase at a CAGR of approximately 7.5% between 2015 - 2020 fueled by economic growth and population growth. With an average GDP growth rate approaching 2.5% and an annual population growth of 2.6%, electricity consumption will have to increase about 5% more than current consumption to support economic growth.

¹⁸ AF Mercados Overview of the Renewable Energy Sector in Nigeria

Electricity Supply (Generation): Nigeria has a total gross installed electricity generation capacity of about 13GW (As at November 2016) of power with peak energy generation of **4,885 MW** (as at 2015)¹⁹²⁰ due to various constraints including insufficient gas supply due to low production, insufficient infrastructure and vandalism, poor water management, line constraints due to inadequate transmission infrastructure. On-grid electricity generation by type is 74% oil & gas and 26% hydro, however, 80% of the total population still rely on traditional solid biomass as their primary source of energy due to the low levels of grid-based energy.

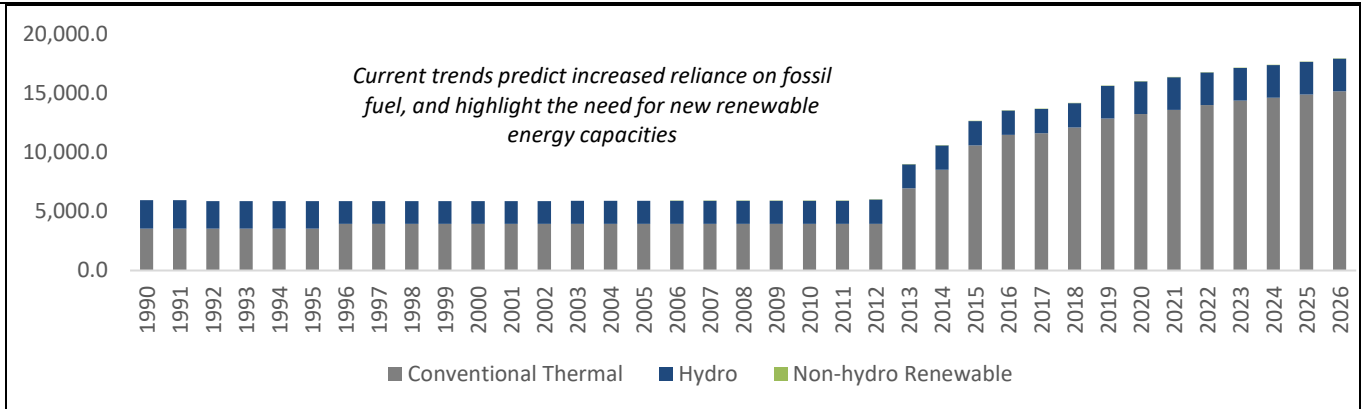
Table 4: Major power plants installed in Nigeria

Generation Plant / Power station	Fuel Type	Installed Capacity (MW)	% Share Installed Capacity
Aba power station	Gas	140	0.9%
AES barge	Gas	270	1.6%
Afam I-V power station	Gas	977	6.0%
Afam VI power station	Gas	642	3.9%
Alaoji power station NIPP	Gas	1,074	6.6%
ASCO Power plant GT1	Gas	55	0.3%
Azura -Edo IPP	Gas	450	2.7%
Calabar power station (NIPP)	Gas	561	3.4%
Delta-Ughelli Thermal Power Plant	Gas	921	5.6%
Egbema power station (NIPP)	Gas	338	2.1%
Egbin thermal power station	Steam	1,320	8.1%
Gbarain power station (NIPP)	Gas	225	1.4%
Geregu I power station – NIPP	Gas	414	2.5%
Geregu ii power station – Gas	Gas	434	2.7%
Ibom power station	Gas	190	1.2%
Ihovbor power station NIPP	Gas	450	2.7%
Jebba hydro power station reservoir	Hydro	577	3.5%
Kainji power station reservoir	Hydro	760	4.6%
Kwale okpai power station (IPP)	Gas	480	2.9%
Odukpani NIPP	Gas	561	3.4%
Olorunsogo (Papalanto) I power station Gas	Gas	335	2.0%
Olorunsogo II power station NIPP	Gas	675	4.1%
Omoku power station	Gas	375	2.3%
Omosho I power station	Gas	335	2.0%
Omosho II power station NIPP	Gas	450	2.7%
Rivers IPP	Gas	180	1.1%
Sapele power plant (NIPP)	Steam	450	2.7%
Sapele power station	Steam	1,020	6.2%
Shiroro power station reservoir	Hydro	600	3.7%
Trans-amadi power station	Gas	136	0.8%
Transcorp – ughelli power station	Gas	972	5.9%

Figure 12: Trend in electricity generation

¹⁹ AF Mercados Overview of the Renewable Energy Sector in Nigeria

²⁰ Nigeria Power Guide, Volume 4, 2018 Edition



Source: BMI Research

Nigeria is yet to have a utility-scale solar power plants on-grid. As the country stands at the critical juncture of introducing diversified RE technologies into the electricity market, the Program will add significant value and enable the transition toward a sustainable energy system.

Integrity of the Grid

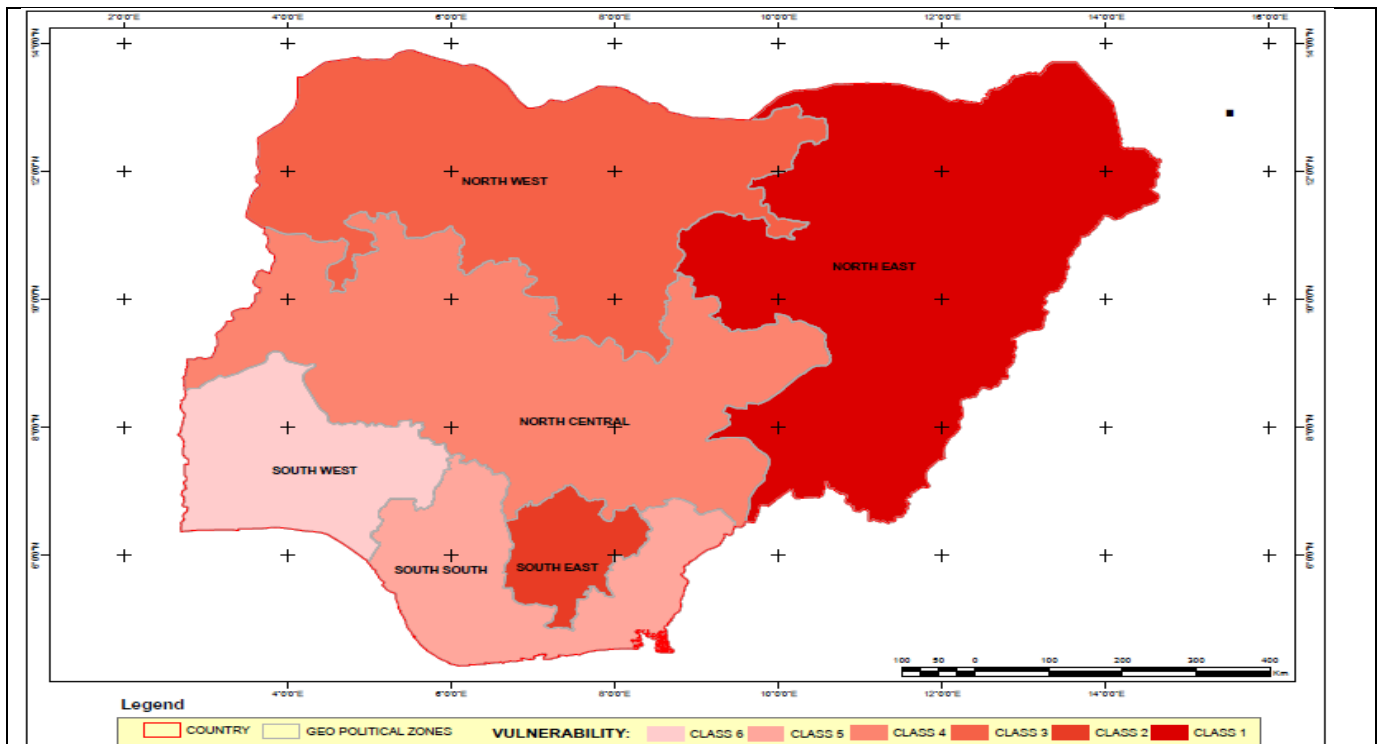
The Nigerian grid system has a capacity of up to 8GW which can evacuate the current available capacity and the solar IPP projects but will be insufficient once new IPPs come on stream in the future. Similarly, the transmission capacity of TCN is constrained and needs significant CAPEX to keep up with the expanding generation capacity. To help address transmission challenges, TCN launched a US\$ 7.5 billion Network Expansion Plan to double its wheeling capacity by 2020, which is expected to be funded by FGN, World Bank and other DFIs and is part of the wider sector and structural reform initiatives which form the Power Sector Recovery Program (PSRP), a FGN and World Bank collaboration, details shown under Figure 19 - PSRP illustrated. Individual projects will have Grid Connection Agreements (GCAs) in place with the Transmission Company of Nigeria (TCN) and corresponding power evacuation approvals. The GCAs will have confirmation (and necessary approvals) that individual projects can evacuate power to the grid. In addition, GCAs will form part of the selection criteria.

Climate Change Adaptation and location of the Nigeria Solar IPP Projects

The 2014 World Climate Change Vulnerability Index, published by the global risk analytics company Verisk Maplecroft, classifies Nigeria as one of the 10 most vulnerable countries in the world. A recent government study determined vulnerability across Nigeria's geographical regions, focusing on the 3 principal determinants of vulnerability: adaptive capacity, sensitivity and exposure. The relative vulnerability of the 6 geopolitical zones of Nigeria is shown in figure 16. There is a general south-north divide. The 3 northern zones show higher vulnerability than those in the south. This reflects the higher rainfall and socio-economic development of the south. The south-south shows highest relative variability among the three southern zones, reflecting the challenges of coastal flooding and erosion, as well as the impact of petroleum exploration and exploitation in that part of the country.

The southwest is least vulnerable, the northeast, on the other hand, is most vulnerable. Understanding these spatial vulnerabilities is crucial to shaping climate-resilient development in Nigeria. Notably the Nigeria Solar IPP Projects are in the most vulnerable areas (see figure 16) which will assist in addressing the vulnerability impacts.

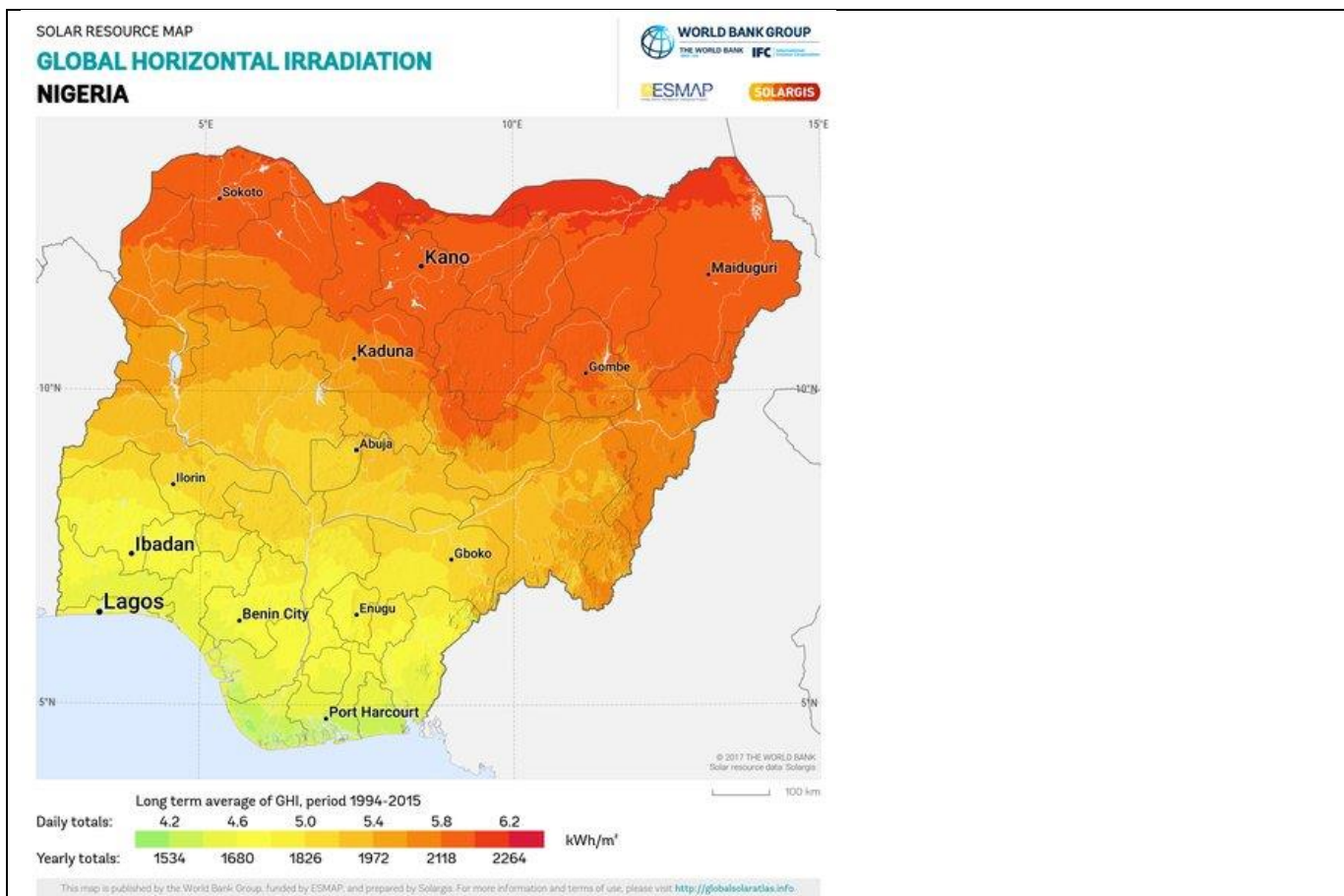
Figure 13: Spatial variation in relative climate change vulnerability



Source: NDC 2015

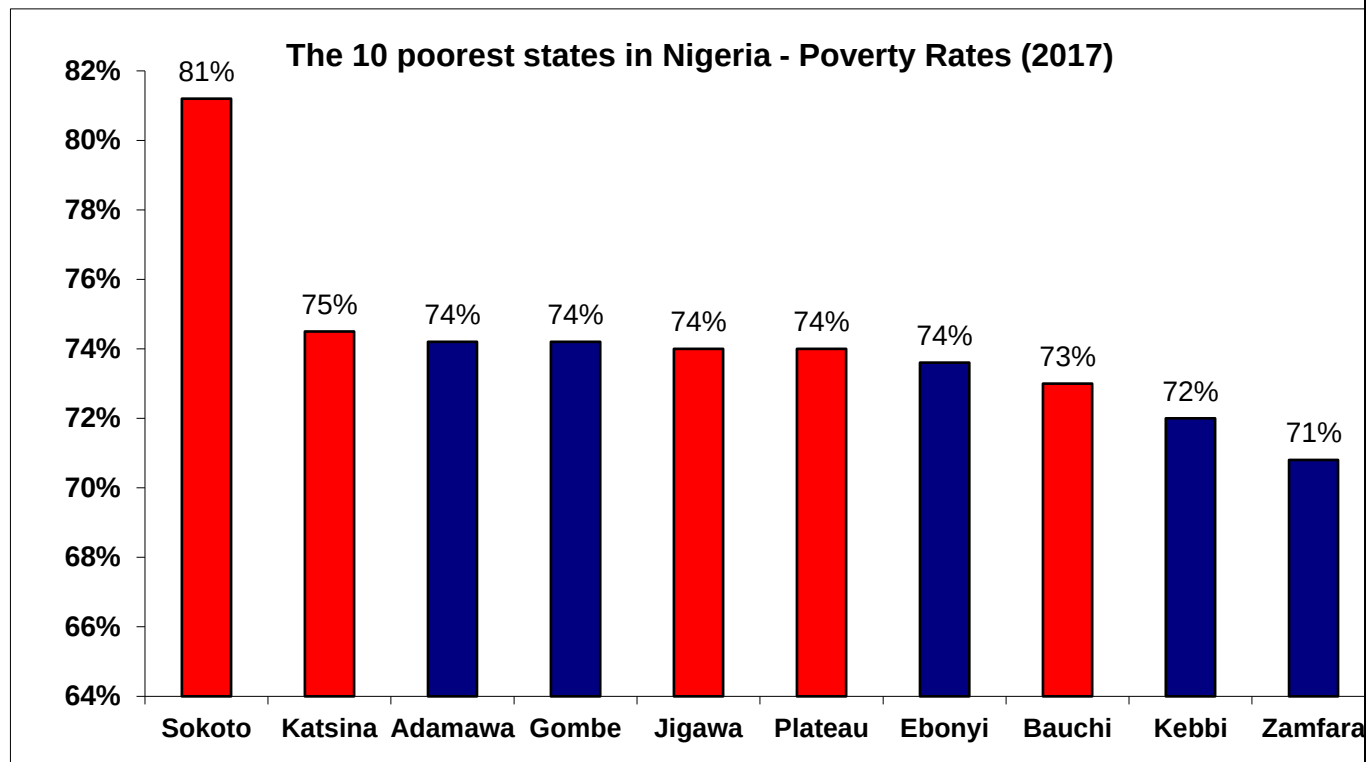
Nigeria has excellent solar resources with a GHI (Global Horizontal Irradiation) ranging from 1,800 - 2,200 kWh/m², 485 million MWh/day of solar energy and an average of 6.2 hours of daily sunshine. According to estimates, the designation of only 5% of suitable land in central and northern Nigeria for solar thermal would provide a theoretical generation capacity of 42,700 MW²¹.

Figure 14: Global Horizontal Solar Irradiation in Nigeria (kWh/m²)



Source: The World Bank, Solar resource data: Solargis

Figure 15: The 10 poorest states in Nigeria, red states are where some projects are located



Source: National Bureau of Statistics

Market Opportunity: Investment in Nigeria power sector is necessary due to the growth opportunities in the Nigerian electricity market where demand far outstrips current supply. The historic gap between the demand for power in Nigeria and the electricity available from the grid has led to widespread self-generation of power in the commercial, industrial and residential sectors; many individuals and businesses own generators, which is more expensive than grid-based electricity to compensate for lack of access to and supply of energy.

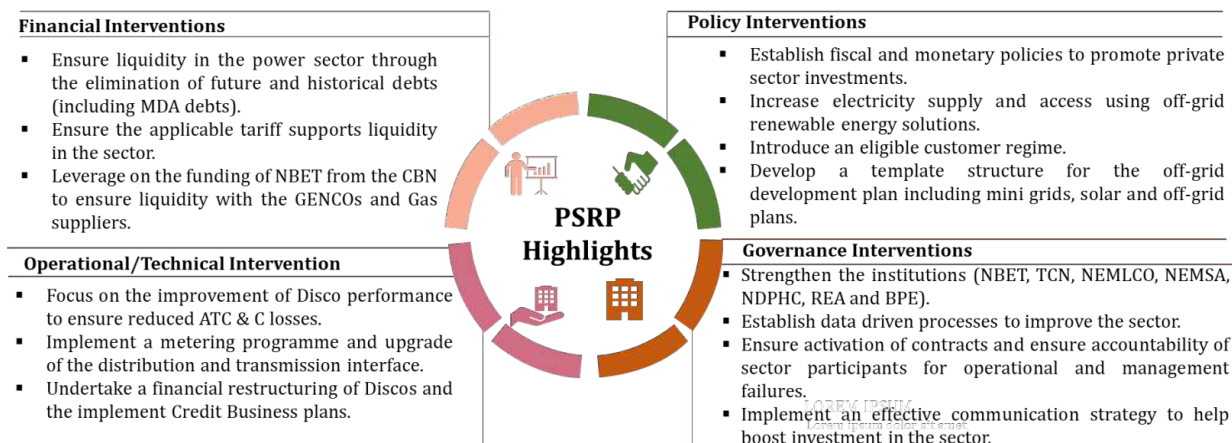
Competition: Total existing operating plants have an installed capacity of around 13GW with peak generating capacity of **4,885 MW** and average generating capacity of **3,850 MW** (as of December 2017) Some IPP projects currently under implementation and development in Nigeria includes non-solar sources of energy such as Quantum power 350 MW, Proton 500 MW, Qua Iboe (QIPP) 540 MW, Oma 1,080 MW and Chevron Agura 330 MW. These projects when aggregated will still not meet the huge unconstrained demand in Nigeria, therefore there is more than enough room for more IPPs especially from renewable energy sources.

Electricity tariffs: As part of the reforms, a revised Multi-Year Tariff Order ("MYTO") was introduced in 2012 to implement cost reflective tariffs which became effective in 2015.

Sector and Structural Reform Initiatives: There are two significant initiatives that are expected to be at the forefront of structural reforms in the sector to help tackle the energy sector challenges are: (a) the Power Sector Recovery Program ("PSRP") by the FGN in collaboration with the World Bank and (b) the granting of a NGN701 billion (US\$2 billion) Payment Assurance Guarantee ("PAG") for the power sector to be provided by the Central Bank of Nigeria ("CBN").

The PSRP is a comprehensive program of policy, legal, regulatory, operational and financial interventions aimed at restoring service efficiency and long-term power sector viability. These interventions to be implemented by the FGN over the next 5 years aims primarily to achieve a restoration of the financial viability of Nigeria's power sector, improvement of power supply reliability, strengthening of the power sector's institutional framework and increase transparency, and encourage investor confidence in the sector. To achieve the above objectives, the PSRP comprises 4 broad components covering Financial, Operational, Policy and Governance interventions.

Figure 16: PSRP illustrated



Source: Nigeria Power Guide Volume 4, 2018

Additionally, the Federal Executive Council ("FEC") of Nigeria in March 2017 approved a two-year N 701 billion PAG through the CBN to increase the liquidity position of the GenCos and ultimately increase revenue certainty and confidence in the power sector. The objectives of PAG include assisting in stabilizing power generation to the grid, enabling NBET to meet its payment obligations under the PPAs with GenCos and providing assurance of payment to GenCos.

C.6. Regulation, Taxation and Insurance (if applicable)

One of the key government licenses will be generation licenses that will be issued by Nigerian Electricity Regulatory Commission ('NERC') for the individual IPP projects. These licenses will be for the generation of power for the respective IPPs and specifies the amount of energy which the project can evacuate. All the 14 solar IPP's have been issued generation licenses. As part of the due diligence, prior to the program implementation, the AE will re-confirm the status of the license and confirm that ownership is in line with the project company.

The various Projects SPV will be incorporated under Nigerian law and will be subject to the local regulations regarding taxation. However, the Projects will benefit from the same tax incentive scheme as other IPPs in Nigeria, one of which is a 5-year tax holiday given to companies named “Pioneer Status” and 7-year tax holiday given for industries located in economically disadvantaged local government areas of the Federation, announced and implemented in 2014.

The Nigerian Naira is not pegged to any other currency. The projects to be funded by the Program will require hard currency financing in US\$ as the capital expenditures of IPPs typically involve importing equipment and services paid in hard currency. The PPAs have been signed with NBET as the off-taker in US\$, payable in Naira but pegged to the US\$ (hedging out FX rate risk). In this regard, exchange rate risk will be mitigated through a reconciliation mechanism. Once the Naira payment is received by the Project, the Project will seek to convert the proceeds into US\$. If the proceeds received in US\$ are less than what was invoiced in US\$, the PPA allows for adjustment by the Project on subsequent invoice as exchange rate loss adjustment. Similarly, exchange rate gains will require adjustments in favor of NBET in subsequent invoices.

Privileges and Immunities

the Africa Finance Corporation enjoys certain immunities, privileges and exemptions under its Establishment Agreement and Charter in its Member Countries. The assets of the Corporation are exempt from attachment unless pursuant to a final judgment or final arbitral award. GCF Funds disbursed to the Corporation for the Corporation's implementation as lender of record or equity provider will be treated as funds disbursed by the Corporation and enjoy the same privileges and exemptions as the Corporation.

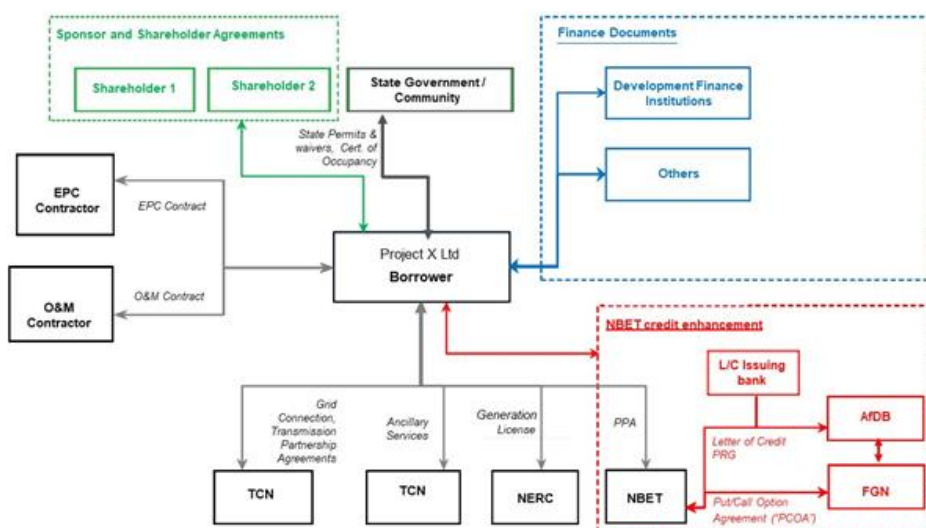
Insurance

The AFC as lender of record, and as part of the detailed due diligence process required when evaluating projects to lend to, will ensure that selected projects are adequately insured as per standard industry practices. The PPA requires sellers to ensure that the insurances for the construction (Construction and Erection All Risk) and operations phases (Commercial / General Liability), are affected and maintained in full force under the PPA Clause 15 – Insurance and Schedule 7 - Insurances.

C.7. Institutional / Implementation Arrangements

The AFC will be responsible for the overall oversight of the Program implementation and will report to GCF per the terms agreed under the Accreditation Master Agreement (AMA) and to be agreed the Funded Activity Agreement (FAA). For managing the GCF resources, the AFC will maintain a separate account for the GCF funds and independently administer the funds. It is envisaged that the AFC will be the direct lender to the projects under its capacity as an AE, and under one Facility Agreement to each respective project. It is expected that AFC will receive GCF funds under the terms of a loan agreement to be detailed in the FAA. Figure 21 below shows the overall implementation structure of the Program.

Figure 17: Project-level contractual structure



A summary of the key contractual agreements is shown below.

Table 5: Summary of key Project contractual agreements

Key Agreements	Counterparty	Term	Description
PPA	NBET	20 years from Commercial Operations. Date (COD)	Power purchase agreement with NBET
EPC	Project SPV and EPC contractor	Contract signing to end of warranty period	Contract with respective contractor for engineering, procurement, and construction of the solar project
O&M	Project SPV and O&M contractor	From O&M mobilization period	Contract with respective O&M contractor operation and maintenance of the solar project
Land Lease	State Government + Community (where applicable)	Varying number of years and start dates according to the project	Rights conveyed to the Project.
PCOA	Federal Government of Nigerian represented by the Ministry of Finance	Financial Close through to the end of the term of the PPA	Contract with the Federal Government of Nigeria that covers FGN related default
PRG	NBET and AfDB	Corresponding to the tenor of the debt facility	Set of agreements that aim to provide a guarantee backing NBET's Letter of Credit provider per the terms of the PPA.
Grid Connection Agreement	Transmission Company of Nigeria (TCN)		This allows the project to connect its plant to the national grid
Generation License	Nigerian Electricity Regulatory Commission (NERC)	[10] years starting from specified periods	Specifies the amount of energy which the project can evacuate.
Financing agreement	TBD		

Detailed structural mitigants to cover the counterparty risks are illustrated in [section G](#) .

C.8. Timetable of Project/Program Implementation

Please provide a project/program implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

	2018				2019				2020				2021				2022				2023				- 2039				2040			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Task: Senior debt financing for Nigeria Solar IPPs																																
Approvals																																
GCF Board Approval																																
AFC Board Approval																																
Program Implementation																																
Project Selection Process																																
Due Diligence																																
Financial Close																																
Project Construction																																
Project Operations																																
Program Reporting																																
APRs																																
Interim Evaluation																																
Final Evaluation																																
Self Assesment																																
AE Reports																																

Notes

1. Project Construction: ranges from 6 - 12 months. Start date for each selected project will vary
2. Project Operations: commercial operations of the power plant, estimated to be 18 – 20 years i.e. PPA term less construction
3. Program Reporting: as defined in the AMA and as may be amended in the FAA
4. AE Reports: any relevant reports conducted for the AE above and beyond what has been agreed with the GCF.

D.1. Value Added for GCF Involvement

The result of the recent economic downturn, especially as it relates to the significant downward trend in commodity prices, particularly the oil prices slump has taken a toll on the Nigerian economy and specifically its ability to attract foreign investors. Despite the provision of a Partial Risk Guarantee (PRG) from the AfDB, local commercial banks are unable to provide:

1. The type of tenors (15 – 18 years) required to finance these projects in a sustainable way, as they are typically limited to 5 years on average and 7 years on the long end;
2. The preferred currency of funding (usually US\$) because of recent prolonged and severe FX convertibility and availability shortages;
3. The typical quantum of funding (US\$ 20 million and upwards) because of the already existing significant exposure to the power sector through the significant involvement of banks in the 2013 / 2014 privatization of the generation and distribution companies; and
4. Competitive financing structures given their lack of appetite and interest in a novel and niche sub sector.

Without the proposed GCF investment size and structure, the Nigerian Solar IPP Projects will be unable to achieve the successful and bankable closure of any of the 14 projects under development. The already executed PPA's are set at US\$ 11.5 cents/kWh. However, since none of the 14 projects could reach financial close within the already extended long stop date of January 21st, 2018, the Nigerian Ministry of Finance has indicated its strong intent to review tariffs down to US\$ 7.5 cents/kWh which is more sustainable for FGN. At US\$ 7.5 cents/kWh the projects require concessional financing from GCF to make it bankable and reach financial close.

The objective of this proposal is to support the GHG mitigation efforts of the FGN. The proposed Program is a mitigation, large size, international project. It consists of the development, financing, construction and operation of up to 400MW of solar power in Nigeria, consisting of 3 - 5 projects as part of a wider 1 GW solar program:

In line with the NDC implementation activities, Nigeria is encouraging energy generation from renewable energy sources. Increased energy access will build the resilience of Nigerian communities by providing reliable energy infrastructure for productive activities. This Program will therefore support the strategic implementation of the NDC in Nigeria, and the involvement from the GCF will be essential to realize the expected outcomes that align with the Paris Agreement and with Nigeria's Vision 2020 in which solar power is a critical component of the power policy of the FGN who has reaffirmed its commitment to increasing renewable energy capacity.

Barriers to RE investment: RE projects usually have a long-term investment horizon of over 10 years and as such require long-term financing. The RE investment scene in Nigeria and other sub-Saharan African countries have been dominated by the Development Finance Institutions (DFIs) with little space of participation from commercial and local financial players. Commercial banks are not able to provide long-term credit for RE projects partly because they depend largely on short-term deposits for loan transactions. This mismatch of using short-term deposits for long-term loans and the financing constraints it presents to the sponsors/developers of RE projects will persist without availability of long-term credit facility. Further, the cost of financing from local commercial banks is high thus they cannot compete with other sources of financing from sponsor's point of view. This crowds out private investment into the RE sector. Still, maintaining the status quo cannot bring a transformative change that is required for dramatically increasing access to energy in Nigeria with stable and sustainable energy sources. Nigeria needs to tap into an innovative approach to crowd-in diverse sources of financing for RE project investment, both on-grid and off-grid.

Value added for GCF involvement

1) RE project loans: Involvement of the GCF will enable a financing scheme that will offer long-term and reasonably priced loans to upcoming RE projects. Currently the Nigeria Solar IPPP projects consists of a total of ~1,125 MW RE projects at advance development stages and these projects need a financial package that can fill the volume gap at affordable price. Without the GCF participation, it is difficult to realize the proposed scheme that accommodates private sector participation in the RE market, which in fact is pivotal to the implementation of NDC related RE activities in Nigeria. The intervention by the AFC, AfDB and GCF will give confidence to both the various developers of these projects as well as the FGN to proceed with this Program. Needs-based concessional financing from the GCF will compensate the elevated cost from the participation of DFIs, making sure that the all-in price offered to the projects is affordable and that the desired Tariff levels are achieved.

As this is Nigeria's first wave of utility scale solar IPP projects the GCF involvement will add value by unlocking the potential of private-led RE financing with strong local participation, which will come with high demonstration effect and possible replication in other countries and regions.

D.2. Exit Strategy

Qualifying projects under the Program will be fully evaluated using AFC's internal credit approval process to ensure the soundness of the projects in the long term. The evaluation will include the financial assessments, strategic fit, regulatory compliance, environmental, social and gender considerations all aimed at ensuring that approval for the projects to benefit from the Program's funding is based on a comprehensive due diligence exercise. Legal documentations covering the loan structure and covenants will ensure the repayments of the loans are made on time and according to schedule. Adequate fall back (security) package will be put in place to cover the resources allocated to the Program in the event of default of any project under the Program.

Once the funds have been disbursed, the projects will be monitored by AFC's portfolio management department to ensure that project is being executed as planned and the plants are being managed in a way that does not jeopardize the investments of the Program (this will occur alongside other portfolio management processes by the AfDB and any other lender to the project. Such periodic monitoring activities allow the Banks to intervene and offer support as and when required to ensure the objectives are achieved.

Also, the first few grid-connected utility scale solar PV projects in Nigeria are expected to require concessional funding support due to high transaction costs, high risks, uncertainty, and lack of country experience. Over time, however, the need for concessional funds will likely diminish. The perception of risk will be expected to decrease, attracting greater interest from domestic and international financial communities. Equipment costs are expected to also continue to fall, allowing for prevailing market tariffs to become enough to deliver desired rates of return to investors. Transaction costs will also fall, the government will start seeing early benefits of scaling-up solar PV projects, and ultimately, market perception of the regulatory environment will improve.

Thus, the development efforts, persistence, and high costs encountered by the early movers in the sector, will ease the development and implementation process and lower entry costs for future project developers. These demonstration efforts will also improve the capacity of solar PV technology service providers (equipment supply, engineering, advisors etc.), and prove the technical and economic realities of solar PV. Through these mechanisms, the Program expects to promote the sustainability of utility-scale solar PV projects, thereby accelerating the development of the sector.

In terms of replicability, this Program proposal seeks to support the growth and market maturation of Solar PV projects in Nigeria. The objective is to promote private sector participation in developing first grid-connected utility-scale solar PV projects in Nigeria. The sub-projects under the Utility-scale Solar PV Program will: (i) pave the way for developers, investors and lenders to follow with solar investments; (ii) demonstrate to other investors the potential of technical and economic aspects of grid-connected solar technologies in Nigeria; and (iii) build local expertise, enabling adaptation and replication for similar grid-connected projects. In addition, as the first IPP solar project(s) in Nigeria connected to the national grid, the Program success will serve as a demonstration project to evaluate technical, institutional and economic viability in Nigeria and to build local expertise to foster rapid adoption and replication for similar grid-connected projects.

- **Low Cost Financing:** Project tariffs have already been set and to attain acceptable returns for the equity investors, long term, low cost financing is critical for project implementation. Local commercial banks are typically only able to provide financing for tenors of up to 7 years with high double-digit local currency financing. To attain acceptable returns, projects require financing with tenors of at least 18 years and concessional pricing, based on initial financial modelling to determine affordable tariff levels. Pricing GCF's participation in the initial funding of the project will enable risk to be shared between AfDB, AFC and other DFI's that are considering investing in multiple projects within the country and thus have limited resources to deploy in any one project.
- **Facility Management:** The selected projects will be built by technically qualified EPC contractors who have experience in solar plants in the region. Reputable O&M contractors will also be employed and on this basis the projects will be self-sustainable and financially viable in the long run even if the GCF is fully repaid. The contractor will ensure uninterrupted operation of the project and optimum use of the plant. Maintenance work

will be performed in such a manner that the work has as minimal impact on the guaranteed performance ratio levels and plant availability.

For individual sub-projects, the loans will be paid back according to the individual project's repayment schedule. For the entire sector, accelerating private sector involvement in RE delivery pioneered by this Program will enable market transformation that is required for the paradigm shift.

E.1. Impact Potential

Potential of the project/program to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation impact potential

The AFC, AfDB and GCF joint intervention in the energy sector of Nigeria is expected to deliver up to 400MW of renewable energy through (3-5) utility scale solar PV IPPs, therefore reducing over concentration on the limited available national electricity grid and reduce the usage of fossil fuel-based power generation by factories and households. Deployment of additional generation capacity will in turn contribute to the increased energy access by people excluded from the grid. The diversification of the energy mix will improve the resilience of the energy system in Nigeria against climate variability. In the Nigeria market where the use of solar and other RE technologies is still new (no utility-scale solar plants installed yet), and where the concept of private-led energy project financing is yet emerging, this Program has an important role to play to unlock the full potential by demonstrating a viable model and this will be seen as 'pathfinder' projects in the country.

The Program will also complement GCF's existing efforts to support the Nigeria power sector through its approval of US\$100 million for the FMO, Climate Investor One project and the AFD, Transforming Financial Systems for Climate project. While these two projects support Nigeria, none of them are dedicated solely to Nigeria as they are both multi-country projects. The NSISP is a dedicated Nigeria Program.

Nigeria is one of the highest GHGs emitter in Sub-Saharan Africa (492 MtCO₂eq. in 2014²²) and as such this intervention is timely and the expected climate change mitigation impact is high. The Program will result in the reduction of carbon emission in Nigeria up to 9,529,739 t CO₂ eq over its 20 years²³ operational lifetime. Grid-connected customers (businesses and households) in urban and peri-urban areas that are currently using diesel-powered generators to fill power supply shortages will consume the electricity provided by the Solar PV power plants under this program. The electricity provided will contribute to reducing the use of diesel generators, which have a greater emissions factor than the grid (0.8 - 1.2 tCO₂/MWh compared to 0.561 tCO₂/MWh). This also provides a positive environmental impact, as diesel exhaust contains more than 40 toxic air contaminants, including many known or suspected cancer-causing substances such as benzene, arsenic and formaldehyde. Exposure to diesel exhaust fumes has been linked to lung cancer in occupational settings. In Nigeria, indirect evidence of the impact of diesel exhaust on lung cancer is indicated by its rising incidence among urban-based nonsmokers less than 60 years old, most of whom use diesel generators daily. Regional surveys suggest that Nigeria's asthma prevalence among adults increased from 5.1–7.5% in 2003 to 13.1–14.2% in 2006, with Nigeria currently having the highest prevalence of asthma in Africa, after South Africa.²⁴

Through the envisaged demonstration effect and performance track-record, the sector would be able to gradually rely on local financial institutions to play a major role in harnessing sustainable energy opportunities in the country. The success of this Program will demonstrate investment opportunities in the sector and contribute to the national development agenda of Nigeria.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided Expected total number of direct and	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided	Annual	476,487 CO ₂ eq
		Lifetime	9,529,739 t CO ₂ eq
	Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience);	Total	1,000,000

²² CAIT Climate Data Explorer

²³ If one assumes a solar PV useful life of at least 25 years, then the Program is expected to reduce or avoid 11,717,533 tCO₂ eq. over the useful life of the projects.

²⁴ Generator Diesel Exhaust: a Major Hazard to Health and the Environment in Nigeria." American Journal of Respiratory and Critical Care Medicine, 183(10), p. 1437

<i>indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience);</i>		Percentage (%)	51% Men 49% Women ²⁵
Proportion of low-emission power supply in a jurisdiction or market: 0 non-hydro RE in the national power generation mix	Proportion of low-emission power supply in a jurisdiction or market: 0 non-hydro RE in the national power generation mix		

GHG Accounting Methodology

The methodology used to estimate the CO₂ emissions of this program is the CDM methodology ACM0002, "Consolidated baseline methodology for grid-connected electricity from renewable sources" (version 19.1)²⁶. The ACM0002 methodology is appropriate because it applies to project activities that include the construction and operation of a Greenfield grid connected renewable energy power plants, which is the case for the Program, providing the first utility scale solar power plants in Nigeria.

To demonstrate additionality, the ACM0002 methodology indicates that any specific technology in the positive list is defined as automatically additional if at the time of submission any of the following conditions are met:

- (a) The percentage share of total installed capacity of the specific technology in the total installed grid connected power generation capacity in the host country is equal to or less than two per cent; or
- (b) The total installed capacity of the technology in the host country is less than or equal to 50MW

All the power plants of the Nigeria Solar Program will be using Solar PV technology, which is in the positive list. In addition, currently there are no grid connected Solar PV power plants in the Nigeria electricity system therefore both, conditions (a) and (b) above are met. The Nigeria Solar Program is therefore automatically additional.

Emission Reduction Calculation Formula

Per the ACM0002, emissions reduction are calculated as follows: Emission reductions (ER) = Baseline Emissions (BE) - Project Emissions (PE).

Since project activity uses solar energy, PE= zero, therefore (ER) = BE.

To calculate the Baseline emissions, the methodology assumes that existing grid-connected power plants and the addition of new grid-connected power plants would have generated all project electricity generation above baseline levels. The baseline emissions are to be calculated as follows

$$BE = EG \times EF$$

Where

BE = Baseline emissions per year (tCO₂/y)

EG = Quantity of net electricity generation that is produced and fed into the grid because of the implementation of the project activity per year (MWh/y)

EF = Combined Margin CO₂ emissions factor for grid connected power generation per year using the latest version of the "Tool to calculate the emission factor for an electricity system (tCO₂/y)"

EF = 0.561tCO₂/MWh, this corresponds to the Combined Margin CO₂ emissions factor that is derived from the CDM standardized baseline "Grid emissions factor for West Africa Power Pool version 01.0" (ASB0034). This

²⁵ 5,000,000 individuals split as per Nigeria Statistics Office

²⁶ https://cdm.unfccc.int/filestorage/5/8/1/58IAGB7SZUDEO2VN6LYM30K41HFPRO/EB100_repan06_ACM0002.pdf?t=cUI8cGZiMiRlFDCynBqe5oboUOM2CC08xwut

standardized baseline was approved by the CDM Executive Board on the 27th of February 2017²⁷ and is based on the latest version of the tool ("Tool to calculate the emission factor for an electricity system")

EG = The net electricity generation was derived from the technical and financial model of the Nigeria Solar Program which yielded an average annual net electricity generation per year of **849,353 MWh** as per the following assumptions:

- Installed DC Capacity : 502.44MWp
- Specific Yield at P90 : 1,846kWh/kWp
- Capacity Factor : 21%

ER = BE = EG x EF = **849,353 MWh x 0.561 (t CO₂ eq /MWh) = 476,487 t CO₂ eq**

- Lifetime of the sub-projects is 20 years (PPA period)

Total emission reduction (ER) over the lifetime of the projects: 476,487 t CO₂ eq x 20 years = 9,529,739 t CO₂ eq

The 'high' estimated cost per tCO₂ is due to the fact that the project is quite capital intensive while the grid emission factor (or the baseline emissions) of Nigeria is a mix of hydro and thermal (mainly natural gas) which are clean when compared to HFO, LFO and coal generation.

Expected total number of beneficiaries:

As the solar PV power plants will provide grid-connected electricity, the beneficiaries of the power are to be households and businesses (SMEs and Industries) who are either lacking access to grid-connected power or experience shortages.

Since electricity consumption per capita in Nigeria is estimated at 151 kWh and an average household size of 5 persons the provisions of 849,353MWh power will benefit a total number of 1,000,000 households in the country.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/program investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

The proposed Program has the potential to play a transformational role in Nigeria's RE sector by supporting some of the first grid-connected utility-scale solar PV IPP projects in the country and establishing an initial track record that demonstrates the viability of financing utility-scale solar PV IPP projects by the private sector. By helping to address the fundraising challenge associated with being the first solar power projects in Nigeria, this Program is designed to help give private financing institutions confidence about the viability of investing in the sector. Risk return perception of future projects is expected to improve and attract more private sector players as well as international and regional DFI's to the solar PV market in Nigeria. In the long term, the need for concessionality for solar PV investments is expected to diminish in Nigeria for several reasons, including:

- i. the pipeline of first-mover projects will establish track record, demonstrating capacity of the government to process needed contracts in satisfactory and bankable manner and reducing uncertainties and costs associated with the solar PV project preparation and execution process;
- ii. there will be a reduction in investor risk perception, lowering the cost of capital, and enabling future projects to achieve reasonable returns and proceed with less or no subsidy; and
- iii. there will be an increase in the overall market experience with grid-connected utility-scale solar PV plants in Nigeria, including establishing supply chains, potentially local manufacturing of components, specialized service and labor, etc.

E.2.2. Potential for knowledge and learning

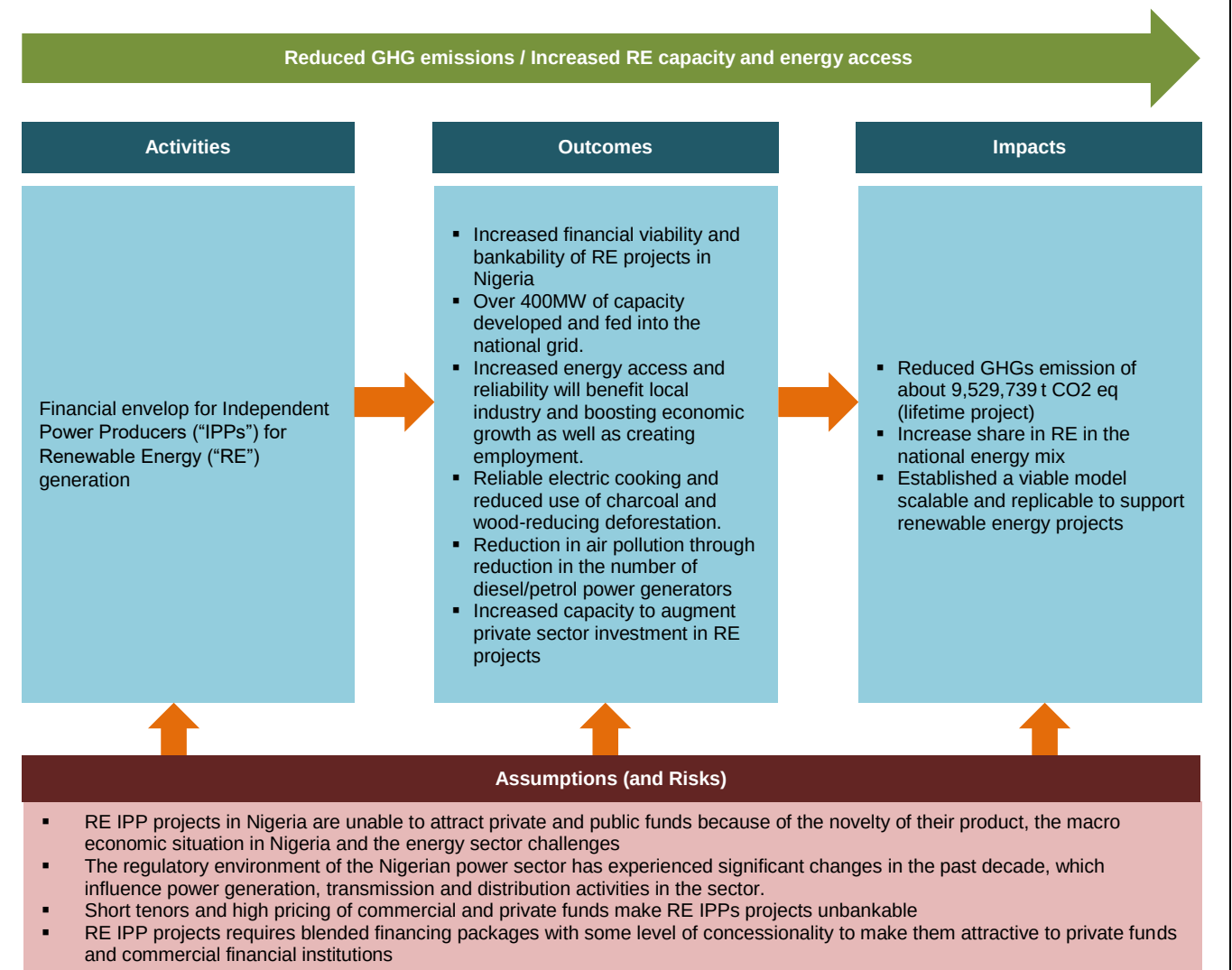
The Program will generate knowledge that will inform the design and implementation of similar RE programs across the country and in the region. The IPP financing structure will leverage international financial institutions and will have a high demonstration effect and set the bar for replication. Lessons learned from the IPP financing will be distilled and shared systematically through the AfDB, AFC and GCF knowledge network. Primarily this will involve knowledge sharing within Nigeria, inviting all the relevant stakeholders in the energy, financial and climate

²⁷ https://cdm.unfccc.int/methodologies/standard_base/2015/sb102.html

change sector to learn from the success of the program and pave way for replication in other part of the country and other African countries with similar climate pattern for example Mali, Burkina Faso, Chad and Niger republic.

During the implementation of the Program, knowledge will be generated and embedded within the institutions that are responsible or have critical roles to play for scaling-up sustainable energy generation and distribution in Nigeria. The experience in Nigeria will demonstrate what works and what does not work – the knowledge that scarcely exists in the Sub-Saharan African context.

Figure 18: Theory of Change



E.2.3. Contribution to the creation of an enabling environment

Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program. Describe how the proposal contributes to innovation, market development and transformation.

Examples include:

- *Introducing and demonstrating a new market or a new technology in a country or a region*
- *Using innovative funding scheme such as initial public offerings and/or bond markets for projects/program*

The proposed commercial tranche will facilitate the participation of local commercial financial institutions that are interested in financing and participating in the renewable energy sector. In addition, the financing of the first scale

utility solar power projects is a critical milestone for the FGN who hopes to have 50% of its power distributors procuring their energy from renewable sources by 2020.

E.2.4. Contribution to regulatory framework and policies

National Renewable Energy and Energy Efficiency Policy (NREEEP), 2015 This policy is regarded as an umbrella document consolidating the various climate change policies and strategies in one document. It is the national renewable energy action plan and a national energy efficiency action plan. The overall focus of the policy is on optimal utilization of the nation's energy resources for sustainable development:

- To guarantee adequate, reliable, affordable, equitable and sustainable supply of renewable energy at cost-reflective and appropriate costs and in an environmentally friendly manner, to the various sectors of the economy, for national development.
- To accelerate the process of acquisition and diffusion of technology, managerial expertise and indigenous participation in the renewable energy and energy efficiency sector industries, for stability and self-reliance and guarantee efficient, location-specific and cost-effective consumption pattern of renewable energy resources and improved energy efficiency which will promote and increased investments and development of the renewable energy and energy efficiency sector, with substantial private sector participation and ensure a comprehensive, integrated and well informed renewable energy and energy efficiency sector, with plans and programs for effective development.

In addition to the **National Adaptation Strategy and Plan of Action on Climate Change for Nigeria (NASPA-CCN) 2011** which seeks to minimize risks, improve local and national adaptive capacity and resilience, leverage new opportunities, and facilitate collaboration with the global community, all with a view to reducing Nigeria's vulnerability to the negative impacts of climate change through various method including renewable energy mix in the national grid which will give energy security to the country.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The Program is expected to deliver wider environmental and socio-economic benefits on top of adding RE generation capacity to the grid and diversifying the energy mix in the country. Such wide-ranging benefits will make sure that the Program contributes to the accomplishment of the National Development Plan as well as the United Nations Sustainable Development Goals (SDGs).

- Nigeria's concentration on gas and hydro plants in its current energy mix will be reduced with solar PV, which will mitigate social and environmental impacts from large hydro construction. Nigeria has large hydropower and gas plants concentration (provided in **C.1**), which generate and distribute electricity to mainly northern Nigeria where this Program will be implemented. Although the potential of hydropower will still have to be harnessed in short-term to medium-term to meet the country's rising demand, scaling-up of solar power can fast-track the transition.
- Job creation during the construction and operations: Among other development impacts, this Program will contribute to the economy by creating about 500 green jobs (per project) during the construction phase ranging from highly skilled (field technicians, operators, corporate staff, etc.) to unskilled labor work (support and service), and additional ~50 jobs for long term operations and maintenance. This will be a starting point of building the national ecosystem of technicians and skilled experts in RE technologies.
- Reduced reliance on fossil-fuel based generation, under emergency situation, will lead to the cleaner air quality by reducing the emission of pollutants including GHGs. This will have additional positive health impact to the population.
- Stable supply of electricity in both urban and rural areas will reduce the dependency on charcoal and firewood for cooking, contributing to the reduction of deforestation and associated carbon emissions. In addition, better access to clean energy will impact positively on women and children who disproportionately hurts most and who are forced to spend up to 20 hours a week gathering biomass (wood, dung, crop waste) and drinking water, instead of going to school, learning, and bettering their situation.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

N/A

E.4.2. Financial, economic, social and institutional needs

Economic and social development level of the country and the affected population

In Nigeria, lack of access to reliable energy is the most pressing social and economic development issue and is one of the reasons for Nigeria's poor outcomes in health, nutrition, food security, education, insecurity, lack of business competitiveness and high unemployment. Given that most of the solar projects are in the northern part of Nigeria, power is brought closer to many people who would otherwise depend on power from the grid which would be difficult to receive in the short to medium term because of their remoteness. Many of the projects have got transmission infrastructure which will contribute to the transmission and distribution infrastructure in the remote areas. Nigeria has the second largest population without access to electricity in the world (after India which has >240 million people without access). A significant proportion of this population without access reside in rural areas. This amounts to approximately 120 million people in Nigeria living without access to reliable and affordable power supply. Nigeria's energy sector has struggled with low investment, inadequate and ailing infrastructure, which resulted in shortages and consequent rise in costs, access and affordability to the vast majority of people, in particular low-income earners most of who are women, the vulnerable, society's marginalized and poor men.²⁸

Absence of alternative sources of financing (e.g. fiscal or balance of payment gap that prevents from addressing the needs of the country; and lack of depth and history in the local capital market)

Nigeria is heavily dependent of crude oil to fund government expenditure and other government spending. The Export prices for Nigerian crude oil correlate with the international market trends. The price has climbed consistently over time, bar the signals seen in crisis years. The US free on board (FOB) price for Nigerian crude oil has risen since 1973 from US\$ 7.81 to an initial peak of US\$ 38.10 in the 1980's, before steadily climbing since 2001, when the price was US\$ 24.85 upward to as much as US\$ 114.51 in 2012 [IEA; 2013]. However, between July 2014 and January 2015 the oil price dropped massively from US\$ 115 to US\$ 45 per barrel, which resulted in a ~28% drop of Nigeria's revenue (based on Nigeria's gross receipts). This has contributed to putting the country in a low growth cycle and low government spending on infrastructure including in the energy sector. This makes crowding-in commercial financing for the IPP projects challenging under this circumstance. Considering this background, sources of financing are limited for Nigeria electricity market with only a few DFIs having an appetite to make investments. This Program is designed to address this specific bottleneck and will aim to attract alternative sources of financing to the RE market once more confidence returns in the energy market after this demonstration effect.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or program

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

The Program is being designed to support Nigeria's plans to increase its available electricity capacity and diversify its energy mix. It is also aligned with Nigeria's Nationally Determined Contribution ("NDC") with the objective of GHG reduction from Business as Usual (BAU) by using the following measures:

- i. work towards ending gas flaring by 2030
- ii. work towards off-grid solar PV of 13 GW and provision of up to 13GW of renewable energy to communities currently off the grid.
- iii. increase the use of efficient gas generators
- iv. 2% per year energy efficiency (30% by 2030)
- v. improve electricity grid and
- vi. climate smart agriculture and reforestation.

The NDC also aspires to reduce CO₂ emission per US\$ (real) GDP from 0.873kg (in 2015) to 0.491kg by 2030. By supporting the deployment of renewable energy. National Policy on Climate Change Nigeria 2013 enhances the country ownership of this financing Program with a strategic goal to foster low-carbon, high growth

economic development and build a climate resilient society through the attainment of the following objectives among others:

- i. Implement mitigation measures that will promote low carbon as well as sustainable and high economic growth;
- ii. Significantly increase public awareness and involve private sector participation in addressing the challenges of climate change; and
- iii. Strengthen national institutions and mechanisms (policy, legislative and economic) to establish a suitable and functional framework for climate change governance.

The country ownership is ensured by close collaboration with the NDA and key sector stakeholders in Nigeria. Consultations will be organized in due course to fine-tune the implementation arrangement and other necessary details of the Program, under the leadership of the NDA and the line Ministries. In addition, country ownership is augmented by the design of the Program where local institutions will be capacitated to play a key role in the RE IPP development.

E.5.2. Capacity of accredited entities and executing entities to deliver

AFC has developed and financed (debt and equity) more than 3GW of power projects across Africa. Of this portfolio, more than 30% are from renewable sources. AFC also has more than 1 GW of power projects in development in Africa. A selection of some of AFC's projects under development/ construction are noted below:

1. Cenpower, 340 MW combined cycle power plant under construction in Ghana
2. WAPP, 2 x 450 MW combined cycle power plants under development for ECOWAS in Benin and Ghana
3. Singrobo, 44 MW hydro power plant under development in Cote d'Ivoire
4. Djibouti, [60 MW] wind power plant under development in Djibouti

AFC has also supported the Federal Government of Nigeria through its technical adviser role to the Central Bank of Nigeria's NGN300 billion Power and Aviation Intervention Fund (PAIF). The PAIF was established to in 2007 to i) fast-track the development of electric power projects, especially in the identified industrial clusters in the country; ii) fast-track the development of the aviation sector of the Nigerian economy by improving the terms of credit to Airlines; and iii) improve power supply, generate employment, and enhance the living standard of the citizens through consistent power supply; iv. provide leverage for additional private sector investments in the power and aviation sectors.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

The AE is working with the Ministry of Energy in Nigeria to engage the National Designated Authority for national stakeholder consultation and community participation in the design of the implementation arrangements and how to ensure the community benefits from the Program. In addition, community engagement and participation during the design and implementation is part of the of the environmental and social safeguards procedures. In this regard, the E&S will ensure that the consultation is inclusive and will ensure the establishment of an accountable grievance mechanism.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/program

E.6.1. Cost-effectiveness and efficiency

The Program will ensure cost-effectiveness and efficiency by introducing an adequate financial structure that will provide a blended pricing at a reasonable cost and a long tenor to ensure that the tariff is acceptable. While operating under the principle of minimum concessionality, the GCF's concessional financing in terms of pricing is critical for the entire structure to be financially viable. The GCF's offer will be a key enabler of the market

transformation, and as local financiers learn from the experience, the RE financing market will gain efficiency and move toward an adequate price equilibrium point.

The analysis shows that the Program overall will be able to produce 849,353MWh/year of renewable electricity and is estimated to contribute to approximately 476,487 tCO₂eq of annual avoided emissions, which results in around 9,529,739t CO₂ eq of total emission reduction over the lifetime of the projects (20 years).

In addition, establishment of a commercial tranche and the participation of local financial institutions with Bol's support will send a strong signal for increased participation of the private sector (both local and international financiers) in the financing of future stages of the renewable (and more generally, other elements of the power) sector.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

The Program will support the development of RE projects totaling up to US\$467 million (i.e. estimated total project cost under this Program) including 3-5 solar PV sub-projects. The projects are expected to have equity of 30% and be financed with a leverage of 70% (inclusive of a debt replacement facility), corresponding to total debt financing needs of US\$327 million.

GCF Co-Financing Ratio: Total Co-Financing / GCF Investment = 385 / 100 = 3.85x.

E.6.3. Financial viability

N/A

E.6.4. Application of best practices

AFC will ensure that the best available technologies are applied. The due-diligence will ensure the use of the most suitable industry technologies. In such way, the implementation of the Program will bring the best international practices as well as well-adjusted technologies to the RE sector in Nigeria. These best practices include the assessments of financially and technically feasible technologies as well as the high-standard procurement principles which will also bring desirable environmental and social benefits. A thorough assessment process will enable a comparison between the proposed projects and good international practices.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
	(a) Total project financing	US\$467 million
	(b) Requested GCF amount	US\$100 million
	(c) Expected lifetime emission reductions overtime	9,529,739 t CO ₂ eq
	(d) Estimated cost per tCO₂eq (d = a / c)	US\$49 / tCO₂eq
	(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	US\$10 / tCO₂eq
	Describe the detailed methodology used for calculating the indicators (d) and (e) above.	
	Assumptions:	
	- 400MW capacity (all solar PV)	
	- [927,504 MWh/year] delivered capacity at P90	
	- 20 years operational lifetime	
	- Nigeria Grid emission factor: 0.561 tCO ₂ /Mwh (standardized baseline for West Africa Power pool)	
	Detailed calculation can be found under Section E.1.2. and in Annex_NSISP_CO2 Emissions Reduction Calculation Sheet	

Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/program)	N/A
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** The information can be drawn from the project/program appraisal document.*

F.1. Economic and Financial Analysis

The original projects were developed with the assumptions that tariff is US\$ 11.5 cents/kWh and debt terms were being provided by AFC, AfDB and other lenders at market pricing. All the projects were bankable and provided adequate returns to the sponsors and did not require concessional financing to do so. Due to concerns of the sustainability of the tariff of US\$ 11.5 cents/kWh, this was reduced to US\$ 7.5 cents/kWh and this has changed the dynamics of the project's economics and financials.

With reduced tariffs, when the debt portions of the capital structures of the projects (assumed at 70% of project costs) is funded entirely by DFIs such as AFC and AfDB, the projects will not be bankable. This is because the cash flows from the projects are not able to provide adequate debt cover (1.16x) for the periodic obligations under the debt facilities. With the introduction of concessional financing by GCF of up to 23% of the project cost, the project bankability improves.

Lower financing cost contributes to lower total costs of projects and consequently end users will benefit from relatively lower tariffs. The ability to intervene through the provision of concessional financing for up to 5 of the 14 projects will ensure the successful implementation of the first utility scale renewable Program in Nigeria and most importantly, the provision of up to 400MW of power across the country and most especially in the most deprived and lower income areas.

F.2. Technical Evaluation

The projects are solar PV generation plants in the range of 55 MW to 100 MW capacity. This is to tap into vast solar resources available in the northern part of the country, which makes it a more cost-efficient option than other RE technologies. Technical quality and compliance with the technical specifications will be part of the evaluation criteria during the selection and due diligence process, with detailed technical assessment to be conducted for each selected sub-project as part of the overall due diligence process.

F.3. Environmental, Social Assessment, including Gender Considerations

AFC's E&S Risk Management Policy is based on Good International Industry Practice reference frameworks, including the IFC Performance Standards on Environmental and Social Sustainability and the World Bank Environmental, Health and Safety Guidelines. AFC's E&S Risk Management Process is detailed in Fig 21 below.

AFC's ESDD process is premised on the categorization allocated to each project. Environmental and Social risk categories (as detailed in Fig 21 below) are allocated by imputing project details into an automated tool.

For Category C projects, no extensive ESDD is necessary. Compliance with E&S laws is checked, and a reputational risk screening is conducted.

For Category B projects, the AFC E&S Due Diligence Questionnaire and E&S sector- specific Questionnaire is used to assess the E&S impacts and risks. If any red flag issues become apparent, further investigations are initiated and external consultants' involvement may be necessary. If considered necessary by the E&S Risk Officer or external consultant, a Client Environmental and Social Action Plan (CESAP) and mitigatory activities, leading the client to eventual compliance with AFC's E&S requirements must be developed.

The ESDD for Category B projects involves the following steps:

- Requiring the client to fill in AFC's E&S Due Diligence Questionnaire and the E&S sector-specific Questionnaire;
- Collecting and studying original Environmental and Social Impact Assessment ("ESIA") documents or other relevant documents (audit reports, etc.)
- Carrying out research on potential risks / potential NGO interest;
- Optionally - visit the site;
- Optionally - preparing a CESAP if the AFC's E&S requirements for Category B projects are not met; and
- Optionally - assistance of an external expert.

Based on the ESDD, the ESRO will prepare an Environmental and Social Review Summary report ("ESRS"), containing the various mitigation measures.

For Category A projects, assistance of an external expert is mandatory. The most important element of the ESDD is to determine the gaps between the current E&S performance and the benchmark performance/reference standards (IFC Performance Standards, World Bank EHS Sector Guidelines and the AFC's E&S requirements). The gap analysis with regard to these requirements is considered crucial to ensure that the client is managing E&S risks in accordance with local or internationally recognized E&S risk management standards and laws (whichever is more stringent).

The gap analysis must be provided by a qualified external expert. It must lead to the formulation of a reasonable action plan, which describes all the actions necessary to achieve AFC's E&S requirements in a given time frame after factoring in the criticality of the E&S risks.

The ESDD for Category A projects involves the following steps:

- Requiring the client to fill in AFC's E&S Questionnaire;
- Handing over to the client the relevant, sector-specific Sector Guidelines
- Requiring the client to prepare a gap analysis between client's performance and the performance defined in the IFC Performance Standards and World Bank EHS Sector Guidelines with the help of an external expert;
- Collecting and studying original ESIA documents or other relevant documents;
- Carrying out research on potential risks/non-governmental organizations' ("NGO") interest;
- Site visit by an external E&S consultant; and
- Preparing a CESAP if the AFC's E&S requirements are not met (based on the gap analysis's findings).

AFC has a Gender Policy which seeks to integrate gender considerations in its investment activities. AFC's approach and commitment to its gender policy as it relates to its investment activities, implies that AFC will:

- work towards conducting gender analysis to ensure that gender issues are addressed as part of its investment process;
- encourage investment entities to educate their supply chain in managing gender equity in accordance with global best practice;
- integrate gender equity assessment into the AFC Investment process;
- not establish business relationships with organizations that condone, support or otherwise participate in trafficking, including labor or sexual exploitation;
- include sex disaggregated data in environmental and social action plans.

Working in collaboration with the AfDB, a co-financier, all the projects under the Program will have ESIA's in line with AfDB Integrated Safeguards System which also include gender considerations and IFC Environmental and Social Standards. As part of the selection criteria, the projects will have to meet the AfDB Integrated Safeguards and Systems (ISS). AfDB ISS is designed to promote the sustainability of project outcomes by protecting the environment and people from the potentially adverse impacts of projects. This requires that all the projects will comply with these safeguards' requirements of the ISS during project preparation and implementation.

F.4. Financial Management and Procurement

Due Diligence

AFC in collaboration with AfDB as co-financier, be responsible for the deal selection and implementation of projects under this Program. AFC will also conduct its own standard KYC and due diligence process including anti-money laundering and other evaluations of sponsors according to each respective institution's internal policy requirements. It is important to note that AFC as AE will not be delegating any of its responsibilities through the selection and implementation of projects, to the AfDB.

Financial Management

AFC's treasury team will be responsible for the financial management of all funds AFC will use for the Program. The eventual application of the funds to each of the projects, will be based on the guidelines of the investments team and the implementation of the financial operations team.

Disbursement

All disbursements will be consistent with a pre-agreed mechanism to be set out in the financing agreements. These will be subject to the fulfillment of all conditions precedent as agreed by the AFC Investments team, risk management team and financial operations team.

Supervision and Portfolio Management

The AFC has dedicated portfolio management and risk assessment teams, both of whom are primarily responsible for the monitoring and supervision of portfolio projects. There are a set of stipulated quarterly, semi-annual and annual reporting requirements, part of which also require site visits on a case by case basis. AFC will work with each of the projects to ensure they are compliant with all the monitoring and reporting requirements as stipulated in the AMA and to be agreed in the Funding Agreement.

Procurement

AFC has an internal procurement process that is consistent with international procurement standards. The procurement manual outlines the processes and procedures for each procurement method, as well as their applicable thresholds. The manual addresses i) the procurement of goods and services for general use ii) the procurement of specialized services that support investment operations and project development activities, the management and oversight of procurement through a data bank of suppliers, contractors, consultants and standards of conduct.

The AFC Vendor Data Base is constantly refreshed and therefore not cast stone. AFC currently shortlists pre-registered vendors based on their core Product or Service, relative to the Corporation's individual business requirements. Potential vendors/consultants are required to register on the AFC vendor Management System, via the following link: http://procure.africafc.org:8080/afc_web/frontend/addvendor.php. Thereafter, AFC will undertake a systematic pre-qualification to assess the suitability of the firm, for inclusion in the Corporation's database of consultants and service providers – This process can be expedited on grounds of urgency.

Audit

The framework will be subject to AFC's normal internal audit policies, which means it will be audited annually by external auditors. AFC's internal audit team monitors transaction activities relating to disbursements of funds to the project as well as repayments.

G.1. Risk Assessment Summary

Several sector-wide risks pose modest risks to the implementation of the Program; however, this will be managed through preventive measures.

Identified risks include:

1. Off-taker (NBET) financial stability
2. Grid absorptive capacity and connection risk
3. Environmental and social risk (site-specific)
4. Construction risk
5. Operation and technology risk
6. Foreign exchange risk

Note: terrorism risk is not included because terrorism is not a nationwide predicament. Nevertheless, while the majority of the projects are located in the northern part of the country, which is where terrorism activities have occurred, the activities have primarily been refined to the north eastern part of the country where there is limited commercial activity. It is also important to note that sponsors would not develop projects in regions that are deemed to be high risk for terrorist activity as it will deem their projects unbankable. Finally, as projects are selected, it is expected that several (perhaps all), as is typical of most power generation businesses, will procure applicable insurance products, which though not inclusive of terrorism, will include cover for business disruption due to force majeure etc.

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/program objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Off-taker (NBET) financial stability	Financial	High (>20% of project value)	High

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The Nigerian Bulk Electricity Trading Plc (NBET) is the off-taker of the power to be produced under the solar Program. It has a 20-year, US\$ linked, take-or-pay Power Purchase Agreement (PPA) with the solar IPPs. As the off-taker, NBET credit risk depends on the proceeds from power sold to Distribution Companies (DisCos). These receivables from the DisCos have not been regular and have created a shortfall in the revenue collected by NBET. The shortfall is being addressed as part of a country wide energy sector recovery plan. AfDB is also looking to assist the FGN by collaborating with the World Bank to bring the AfDB full toolkit of instruments to assist the power sector in line with the energy sector recovery plan.

Additionally, the following mitigation measures are in place:

- For every selected project, there will be Debt Service Reserve Accounts (DSRA) of up to 6 months built into the structure to reduce the risk from an unexpected volatility in cash flows.
- The PPAs for some of the Projects will be supported by AfDB's Partial Risk Guarantee (PRG) whereby a Letter of Credit (LC) covering up to 6 months of NBET's payment obligations under the PPA will be put in place. This LC will be backed by the Bank's PRG in favor of the relevant Project(s). The intention is that the project can draw on the LC if NBET's payment obligation under the PPA is not met. The LC with the backing of the PRG will basically provide liquidity for up to 6 months' worth of revenue in event that the off-taker defaults on its payment obligations and will be provided prior to Financial Close.

Put/Call Option Agreement (PCOA) between NBET, the Federal Government of Nigeria (FGN) and the Project Company is expected to be put in place. In the event of default by the off-taker or early termination as defined by the PPA, the FGN is obligated to purchase the PV facility at a pre-determined purchase price. In effect the PCOA provides for the FGN to backstop NBET obligation on termination through a put option to the benefit of the Project and subsequently protect the debt portion.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Grid absorptive capacity and connection risk	Technical and operational	Medium (5.1-20% of project value) Medium (5.1-20% of project value) Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The status of these conditions will be determined as part due diligence for the selected projects. The focus will be on distance between project sites and individual substations, as well as relevant Grid Connection Agreements with the Transmission Company of Nigeria (TCN) as well as verification of generation licenses. Additionally, solar yield studies and likely performance ratios will be verified and advised by the Lenders Technical Advisor ('LTA') regarding the effectiveness of the technology in meeting the project yields associated with the project cash flows in the Project's financial model.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
Environmental and social risk (site-specific)	Social and environmental	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

AFC will apply its highest standards in the screening and managing of E&S related issues. The sites are multi-location, which will reduce the overall E&S risk. Each project will be examined on a case by case basis to ensure compliance of the individual Banks's E&S policies and ensure that Environmental Social Impact Assessments ("ESIA") have been approved by the Federal Ministry of Environment of Nigeria ("FMEN").

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Construction risk	Technical and operational	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Project-specific construction risk (completion and cost overrun) will be mitigated by a thorough due diligence process led by the AFC technical and financial experts, and by ensuring robust lump sum, turnkey, date certain, EPC contracts that will ensure that construction risks will be passed on to the EPC contractors through a combination of Liquidated Damages and penalties for late delivery of the plant and performance below contractual levels. The EPC and O&M contract will be assessed for bankability by AFC and other lenders together with the LTA. There are also certain levels of construction contingencies built into the project's costs.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Operation and technology risk	Technical and operational	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Project-specific operation risk will be mitigated by a thorough due diligence process led by the AFC technical and financial experts and by ensuring robust O&M contracts. Resource risk (e.g. insufficient irradiation for solar PV projects) will be mitigated by ensuring high-standard technological studies in place and comfort from the LTA.

Selected Risk Factor 6

Description	Risk category	Level of impact	Probability of risk occurring
Irradiation Risk	Technical and operational	Low (<5% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Project-specific irradiation risks refer to the potential that the Project site is not exposed to enough sunshine for the duration of the PPA. This will be mitigated by examining data on the irradiation of the project site's surface, which will date back several years, and this will be incorporated into solar yield studies. Additionally, AFC will structure Projects based on P90 and at appropriate minimum DSCRs.

Selected Risk Factor 7

Description	Risk category	Level of impact	Probability of risk occurring
Foreign exchange risk	Social and environmental	Low (<5% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Exchange rate risks are mitigated through the “true-up” mechanism that indexes payments in Naira to the US\$ under the PPA. The PPA also provides for adjustment to allow for the depreciation in Naira from the point that it is paid to the Project and the Project converts it into US\$. Convertibility risk for termination payment is removed because the current drafts of the PCOAs provide that termination payments will be settled in US\$. Convertibility risk remotely foreseen in the capacity and availability payments by NBET and this will be mitigated by NBET’s selection of a bank with a USD balance sheet.

Other Potential Risks in the Horizon

Please describe other potential issues which will be monitored as “emerging risks” during the life of the projects (i.e., issues that have not yet raised to the level of “risk factor” but which will need monitoring). This could include issues related to external stakeholders such as project beneficiaries or the pool of potential contractors.

Final approval of financing for individual projects is subject to both AFC and AfDB’s project-specific risk screening. AFC will put in place an implementation team for the Program, to oversee due diligence and execution of individual projects. The team will mobilize staff and energy financing experts from their respective offices in Cote d’Ivoire and Nigeria. This mechanism will ensure a speedy and seamless execution of the Program.

** Please expand this sub-section when needed to address all potential material and relevant risks*

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level²⁹

Paradigm shift objectives

<i>Shift to low-emission sustainable development pathways</i>	Please elaborate on the paradigm shift objectives to which the project/program contributes. The Program is primarily aimed at contributing to Nigeria's strategic and government-driven goal of contributing to the reduction of global carbon emissions. Nigeria is one of the highest GHGs emitters in Sub-Saharan Africa and as such, this intervention is timely, and the expected climate change mitigation impact is high. Specifically, the Program has the following paradigm shifting objectives: • The Program aims to deliver up to 400MW of power on to the national grid, thereby reducing the power supply deficit in Nigeria by partially displacing fossil fuel-based generation currently used to bridge the gap, with renewable and sustainable energy sources in the form of solar power. • The Program is aimed at increasing the resilience of the energy system in Nigeria against climate variability. • In the Nigerian market, where the use of solar and other RE technologies is still new and where the concept of private-led energy project financing is emerging, this Program plays an important role by demonstrating a viable model for RE IPP financing. • Given the ambitious RE generation targets of Nigeria as specified in the NDC and the country's Vision 2020, the Program is expected to make a crucial contribution in launching Nigeria's transition toward a low-emission sustainable development pathway. • Furthermore, the Program is aimed at facilitating a stable supply of electricity, which will curtail the use of charcoal and firewood for cooking, reducing emissions from deforestation and land use change, which is a major source of GHGs emissions in Nigeria. The Nigerian Government recognizes the need for increased private sector involvement in the country's power sector, in order to facilitate the achievement of the dual objectives of incremental power supply and decarbonization. This Program will provide an important setting for the Government to realize these objectives.
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Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	

Fund-level impacts

M1.0 Reduced emissions through increased low-emission energy access and power generation	<i>Tonnes of carbon dioxide equivalent (t CO₂eq) reduced as a result of Fund-funded projects/ programs</i>	Program report (mid-term and final) Regular reporting from AFC to GCF as agreed in the AMA and FAA	0	476,487 mtCO ₂ eq	9,529,739mtCO ₂ eq	Monitoring of this performance indicator will be based on an average annual CO ₂ reduction figure of 476,487tCO ₂ eq given the following assumptions: • Full disbursement of the approved financing for each Project • Successful completion of power plant(s) construction for each Project • Successful installation of project MW amounts • Existence of extraneous factors
	<i>Cost per t CO₂eq decreased for all Fund-funded mitigation projects/ programs</i>		0		US\$49 / tCO ₂ eq	
	Volume of finance leveraged by Fund funding		0		Co-financing leverage ratio: 3.85x Private finance leverage ratio: 0.45x Private Finance share: 10%	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Program level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/program Outcomes	Outcomes that contribute to Fund-level impacts					
M6.0 Increased number of small, medium and large low-emission power supplies	M6.1 - Increased proportion of low-emission power supply in a jurisdiction or market	1. Annual report of Nigeria Electricity Regulatory Commission (NERC) and of the Nigerian Bulk Electricity Trading (NBET). 2.Regular report from AFC to GCF as agreed in AMA. 3.Reports received from individual projects	0 non-hydro RE in the national electricity generation mix		3.5%No n-hydro RE in the national electricity generation mix (400MW)	Assuming a current installed capacity of 11.1GW and full commercial operation of the 400MW
	M6.2 - Number of households, and individuals (males and females) with improved access to low-emission energy sources;		0	-----	1,000,000 households and 5,000,000 individuals	Per capita electricity consumption in Nigeria is estimated at 151 kWh Average yearly production of the program estimated at 849,353MWh Average household size of 5 persons per household ³⁰
	M6.3 - MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support		0		400MW	Full disbursement of the approved financing for each Project Successful completion of power plant(s) construction for each Project
Project/program outputs	Outputs that contribute to outcomes					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	

³⁰ According to the Nigerian National Bureau of Statistics the total population is 140,431,790 for 28,197,085 total regular household. This gives an average of 4.98 people per households: <http://nigeria.opendataforafrica.org/xspplpb/nigeria-census>. This indicator is based on new additional households with access to electricity.

1. 1. Solar IPP projects financed, and plants installed and operational	Number of solar IPP projects financed	Regular reporting provided by project companies to AFC	0		3-5 solar IPPs	It is assumed that the NBET will extend PPA for IPPs shortlisted by the Program.
	Amount of solar power fed into the grid		0		400MW	It is assumed that the Transmission Company of Nigeria has the capacity to evacuate and transmit power generated.
2. Direct IPP financing (senior debt) Due diligence	Successful due diligence on shortlisted projects	AFC Due Diligence Process	0		3-5 projects successfully diligenced	Adequate Due Diligence Elements: • Successful KYC • Legal • E&S • Bankability • Financial strength
3. Direct IPP financing (senior debt) Financial closure	Senior debt for 3 – 5 (between 50MW to 100MW) Solar IPP projects.	AFC, AfDB, and GCF Board approvals AFC, AfDB, and GCF: senior debt	0		3-5 projects financed	The ultimate outcome of the Due Diligence Process will be Commercial Close. All IPP projects will successfully meet Conditions Precedent
4. Other economic indicators of the program	Number of Direct Jobs created during Construction and Operation phase	Reports received from individual projects	0		200 permanent jobs to be created in the Operation and Maintenance phases	It is assumed that the required skill set will be available locally and there will be no need for importation of skills Assuming 10 job-year per MW installed in the construction phase
			0		4,000 direct and indirect jobs (construction phase)	

	Number of new women-operated businesses in determined project area		0		50 new businesses per sub-project	
	Increased number of households, and individuals (males and females) with improved access to low-emission energy sources;		TBD for each sub-project		A minimum increase of 15% of households (in each sub-project area) with increased access to low energy sources	Assuming that each sub-project will establish a baseline of households with access to high and low emission sources
Activities	Description		Inputs		Description	
Component 1. Project Financing	Up to US\$327 million of senior debt financing to construct c.5 solar pv on-grid projects in Nigeria.		Loans		The activities and processes will follow the AE's standards for appraising and investing in power projects.	
Project Selection Process	Review of all 14 projects, selection of qualifying projects and ranking of shortlist.		-		-	
Due Diligence	Environmental, social, financial, and technical due diligence of the shortlisted projects.		-		-	
Financial Close	Signing of the facility agreements between the AE and the selected projects.		-		-	
Project Construction	Construction of the selected projects.		-		-	
Project Operations	Operations of the selected projects i.e. generation of solar power and distribution to the national grid.		-		-	

H.2. Arrangements for Monitoring, Reporting and Evaluation

Monitoring

All projects financed under the proposed Program will be monitored by the AE's Portfolio Management team in line with internal relevant internal policies and procedures. AFC will be responsible for direct monitoring of implementation conditions and reporting periodically to the GCF under the terms to be agreed in the FAA between the AFC and GCF, and the respective project facility agreements between AFC, AfDB and any other lender. All IPP projects financed under the proposed Program will comply with AFC and other lender appraisal, approval, monitoring and supervision standards and procedures involving representatives or all relevant teams (engineers,

lawyers, project finance specialists, procurement experts, E&S specialists, climate finance officers, financial management officers, supervision and monitoring specialists).

All projects will be monitored in accordance with the existing AFC portfolio monitoring process as referenced above in section F.

Reporting and Evaluation

All project reporting will be in accordance with the existing AFC portfolio monitoring process as referenced above in section C8 and F3 as well as in line with the terms of the AMA. Specifically, the following reports will be produced:

Milestones/Reports	Expected Timing
Start of Programme implementation	Upon effectiveness of the FAA
Inception Report	Upon effectiveness of the FAA
Interim independent evaluation report	Within six (6) months from the mid-point of Programme implementation
End of Programme implementation	Three (3) years from the start of Programme implementation
Programme Completion Report (Final APR)	Within three (3) months after the end of Programme implementation
Final independent evaluation report	Within six (6) months after the End of Programme implementation
End of Programme Financial and Performance Reporting Period	Upon full repayment (or write off) of the Programme Loans and/or Programme Replacement Loans

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☐ Feasibility Study
- ☐ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☐ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☐ Project/Program Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – *see the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☒ Map indicating the location of the project/program
- ☒ Timetable of project/program implementation

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



FEDERAL MINISTRY OF ENVIRONMENT

HEADQUARTERS, MABUSHI, ABUJA.

Ref No FMENV/DCC/GCF/013/I/95

Date 15 November, 2018

The Green Climate Fund ("GCF")
Songdo International Business District,
175, Art Center-daero,
Yeonsu-gu, Incheon 406-840
Republic of Korea.

Dear Sir,

Re: Funding Proposal for the GCF by the Africa Finance Corporation regarding the "Nigeria Solar IPP Support Programme"

We refer to the programme, "Nigeria Solar IPP Support Programme" in Nigeria as included in the funding proposal submitted by the African Finance Corporation to us on July 31, 2018.

The undersigned is the duly authorized representative of the National Designated Authority/Focal Point of Nigeria.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate Nigeria's no-objection to the programme "Nigeria Solar IPP Support Programme" as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The government of Nigeria has no-objection to the programme as included in the funding proposal;
- b) The programme as included in the funding proposal is in conformity with Nigeria's national priorities, strategies and plans;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Dr. Yerima Peter Tarfa

GCF Focal Person Nigeria/Director, Department of Climate Change
Federal Ministry of Environment, Nigeria

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Nigeria Solar IPP Support Program
Accredited entity	Africa Finance Corporation (AFC)
Environmental and social safeguards (ESS) category	Category B
Specific project location or target country/ies of the programme, as the case may be	Nigeria Approximately five locations will be selected from the following list: Katsina, Bauchi, Nasarawa, Enugu, Sokoto, Kogi, Abuja, Jigawa, Plateau and/or Kaduna
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-25
Language(s) of disclosure	English
Link to disclosure	https://www.africafc.org/What-We-Do/Environmental-and-Social-Risk-Management.aspx An ESIA consistent with the requirements for a category B project is contained in the environmental and social management framework titled "E&S Management Framework (ESMF)".
Other link(s)	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-25
Language(s) of disclosure	English
Link to disclosure	https://www.africafc.org/What-We-Do/Environmental-and-Social-Risk-Management.aspx An ESMP consistent with the requirements for a category B project is contained in the environmental and social management framework titled "E&S Management Framework (ESMF)".
Other link(s)	N/A
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	N/A
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Link to disclosure	N/A
Other link(s)	N/A

Disclosure in locations convenient to affected peoples	
Date	2019-01-25
Place	Africa Finance Corporation Headquarters. Please refer to the link for further information: https://www.africafc.org/Contact-Us.aspx

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's review of FP104

Proposal name:	Nigeria Solar IPP Support Program
Accredited entity:	Africa Finance Corporation (AFC)
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: Medium

- According to the information provided in the feasibility study and funding proposal, Nigeria's renewable energy resources are significant, and, in particular, the country has great potential for solar energy industry development. Nigeria lies just north of the equator, in a region where there is an abundance of sunshine all year round. Annually, Nigeria's territory receives an average of 2,000 kilowatt hours per metre squared (kWh/m²) of irradiation. Studies say that the best potential for large-scale solar power plants lies in the northern part of the country, which is the least developed and poorest region of Nigeria. Horizontal irradiation in the northern part of the country, where the project solar power plants are planned to be constructed, is estimated as 7 kWh/m²/day, while in the south it is about 4 kWh/m²/day because this part of the country is often cloudy and has a longer rainy season.¹ This region receives an average of six hours of sunshine per day.²
- In Nigeria's nationally determined contribution (NDC), submitted by the Federal Government of Nigeria (FGN) in 2015 to the United Nations Framework Convention on Climate Change prior to the Paris Agreement, the country committed to reducing its business-as-usual baseline emissions by 45 per cent (equal to 1.4 tonnes of carbon dioxide equivalent (tCO₂eq) per capita) by 2030 as a conditional commitment and by 20 per cent (equal to 0.6 tCO₂eq per capita) as an unconditional commitment. Reduction of transmission losses in power and gas industries and increases of the share of renewables in the energy mix are highlighted in the NDC document as key measures along with the economy-wise energy efficiency plans for low-emissions development. In order to facilitate fulfilment of the NDC commitments in the power supply sector, FGN with the support of the World Bank, developed and started implementation of the Power Sector Recovery Programme (PSRP) oriented on policy, regulatory, operational, governance and financial interventions to be implemented by FGN in 2017–2021. The programme aims to restore the financial viability of Nigeria's power sector, improve transparency and service delivery, resolve consumer complaints, reduce losses and energy theft, and reset the Nigerian Electricity Supply Industry (NESI) for future growth. This programme should increase the national grid supply capacity and support the development of on-grid solar energy potential existing in country.
- As the first step in implementing the NDC commitments, FGN approved its feed-in tariff (FIT) regulation for renewable energy in November 2015. The FIT targets a 2020 state where 2 gigawatts (GW) will be generated through renewables and the electricity distribution companies will be obliged to source at least 50 per cent from renewables. In February 2016, the FIT policy came into effect but later it was changed (see next paragraph).

¹ Feasibility study, page 44.

² Available at <<https://www.africa-eu-renewables.org/market-information/nigeria/renewable-energy-potential/>>.

4. In addition, within the framework of the Nigeria solar independent power producer (IPP) projects, FGN (through the Nigerian Bulk Electricity Trader (NBET), which is the power off-taker) signed 14 utility-scale solar power purchase agreements (PPAs) in 2016 to supply 1,125 megawatts (MW) of renewable power to the grid. So far, none of these projects have been able to reach financial close because of financial constraints and high risk. In particular, the reduction of the FIT tariff by NBET from USD 11.5 cents/kWh (offered by FIT) to USD 7.5 cents/kWh stretches the capacity of development financial institution lenders in the funding group as well as the capacity of sponsors to continue to develop the project due to significantly reduced internal rates of return. Hence, there is currently no utility-scale solar on-grid project in Nigeria.

5. Based on the recommendations from the Ministry of Finance of Nigeria, NBET is proposing a reduced tariff of USD 7.5 cents/kWh to allow the projects to reach financial close. Meanwhile, FGN committed to the solar programme throughout the term of the PPA. Still, most projects under the programme are not financially viable without concessional financing, which enables a lower tariff while still covering the costs of debt.

6. In order to remove the mentioned financial barriers to the local private sector and utility-scale solar power producers, the Africa Finance Corporation (AFC) jointly with the African Development Bank (AfDB) has developed this funding proposal aimed at reducing the financial risks of these IPPs through concessional loans. This funding proposal mainly targets existing financial barriers in parallel to other ongoing programmes implemented by FGN to reduce existing technical and regulatory risks. This is an opportunity for the country to implement the first utility-scale, on-grid solar power project, which could have a significant demonstration effect.

7. The funding proposal falls in the GCF results area for energy access and power generation, and its main activity is to provide long-term financing to the solar IPP projects. The funding proposal plans to financially support a maximum 3–5 selected projects, totalling 400 MW. Among the project's selection criteria, the key criteria supporting the GCF investment policy are: the potential to reduce greenhouse (GHG) emissions and lower concessional funding requests as a percentage of project cost in order to incentivize sponsors to optimize their financial structures.

8. The total project cost is up to USD 467 million.³ The financial contribution from GCF is up to USD 100 million (USD 94 million as a senior loan for concessional lending and a USD 6 million standby facility for replacement of the commercial tranche)⁴ through AFC. AFC and AfDB also contribute up to USD 100 million each (USD 94 million for a commercial loan below the market rate and USD 6 million for a standby facility for replacement of the commercial tranche). USD 140 million is 30 per cent equity from sponsors and USD 45 million potential contribution from local banks. The senior debt financing will comprise a development financial institution tranche (with competitive terms), a GCF tranche (with concessional terms) and potentially a commercial tranche (from local financial institutions). Support will be provided in the form of senior debt financing of up to 70 per cent of total project costs with the balance through equity up to 30 per cent.

9. Despite the significant technical and regulatory challenges existing in the sector, the programme will support the implementation of two critical credit enhancement structures that improve the bankability of the solar projects:

³ A standby facility (USD 18 million) is not included.

⁴ In the event that commercial financial institutions do not participate in providing debt, the standby facility will be revoked and incorporated into the senior debt facility under senior debt terms.

- (a) To manage the NBET liquidity risk, the off-taker is contractually obliged to provide a letter of credit covering six months' worth of energy payments. To facilitate the issuance of this letter of credit by JP Morgan, AfDB will provide partial risk guarantee cover for a select number of projects; and
 - (b) To cover termination risk, the off-taker has agreed to execute a put-call option agreement, where, under all termination scenarios, senior debt will be repaid by the FGN in USD.
10. The programme is expected to avoid 476,487 tCO₂eq annually, which leads to 9,529,739 tCO₂eq over the 20 years of the technology's lifetime. At least one million households in northern Nigeria will be direct or indirect beneficiaries of the project. The project reduces the perceived risks of investing in the Nigerian renewable energy sector and catalyses private sector investment in the sector.
11. In case of successful implementation of planned activities, the impact of this programme is envisaged to be high from different perspectives:
 - (a) The contribution to the conditional NDC by 2030 will be around 0.02 tCO₂eq per capita which is 1.3 per cent of the planned 1.4 tCO₂eq per capita reduction (population growth to 262.2 million by 2030 is taken into consideration)⁵ from a total (all sectors) business-as-usual scenario. This share will be higher if only the energy or power sectors are considered. In addition, by 2030 the grid emission factors are likely to increase because of the high probability of a significant increase in economic activities. An increased grid emission factor is anticipated even the share of renewables in the grid will reach the planned National Renewable Energy and Energy Efficiency Policy 2015 (NREEEP 2015) target of 16 per cent, excluding large hydro power plants;
 - (b) The contribution to the national renewable targets set by NREEEP 2015 and the Nigerian Electricity Regulation Commission (NERC) for the generation of renewable energy, including solar power (table 13, page 45 of the feasibility study) for different target years is expected to be high. The on-grid and off-grid solar power target for 2020 is 1.3 GW and for 2030 it is 6.8 GW. A 400 MW power generation planned by the funding proposal will contribute up to 6 per cent of the 2030 target;
 - (c) Contribution to stabilization and strengthening, increased capacity of NESI is expected; and
 - (d) The activities have the potential to facilitate crowding-in in the Nigerian renewable energy market with utility-scale, on-grid solar power projects and the sector financiers.
12. The amount of GHG emission reduction potential considered in paragraph 11(a) above is quite impressive, but because of the high implementation risk to deliver the power to the end-user considered under investment criterion 6, efficiency and effectiveness, the impact criterion is rated as "medium".

1.2 Paradigm shift potential

Scale: High

13. According to Nigeria's Biannual Updated Report, the energy industry (1.A.1) sector ranks second with regard to the country's GHGs inventory due to its 28.2 per cent share following the agriculture, forest and other land-use sector, which has a 66.9 per cent share. Within the sector, the electricity and heat production (electricity generation) subsector is a major contributor with 80.7 per cent of total emissions for all energy industries.

⁵ Available at <<https://www.populationpyramid.net/nigeria/2030/>>.

14. There are several activities tackled by the project that could significantly change the country's power generation and supply system to low-emissions operations and increase the resiliency of the system. Among these activities the following should be highlighted:
- (a) Increase the share of non-hydro renewables, and solar power in particular, in the power generation mix;
 - (b) As far as there is currently no utility-scale solar IPP project in Nigeria, this programme is an opportunity to introduce the development of the first of such projects, which, if successfully implemented, could produce a significant demonstration effect for FGN and the energy and power market in the country;
 - (c) GCF intervention with this project into the renewable power generation process could accelerate the implementation of regulatory and capacity growth processes planned by FGN for the electricity industry system; and
 - (d) Increased access to electricity will reduce consumption of biomass fuel in cooking and hot water preparation sectors.
15. Taking into consideration Nigeria's solar resources, the project has good potential for replication not through the existing 14 IPP projects but to facilitate new utility-scale, on-grid power projects.
16. It is the opinion of the independent Technical Advisory Panel (TAP), the paradigm shift potential of this project to shift the power energy sector to a low-emissions pathway, is "high", and could be realized if this funding proposal is successfully implemented.

1.3 Sustainable development potential

Scale: Medium

17. The funding proposal will contribute to the following Sustainable Development Goals (SDG):
- (a) SDG 7 – affordable and clean energy. Focusing on universal access to energy, increased energy efficiency and the increased use of renewable energy through new economic and job opportunities and is crucial to creating more sustainable and inclusive communities and resilience to environmental issues like climate change;
 - (b) SDG 9 – industries, innovations and infrastructure. Investments in infrastructure – transport, irrigation, energy, and information and communication technology – are crucial to achieving sustainable development and empowering communities in many countries; and
 - (c) SDG 13 – climate change, and in particular a low carbon economy.
18. Among other development impacts, this solar programme and proposal will contribute to the economy by creating about 500 green jobs (per project) during the construction phase ranging from highly skilled (field technicians, operators, corporate staff, etc.) to unskilled labour work (support and services), and, additionally, about 50 jobs for long-term operations and maintenance. This will be a starting point of building the capacity of technicians and skilled experts in renewable energy technologies.
19. According to the funding proposal, most of these 14 IPP stations are planned to be constructed in the northern regions of Nigeria, where poverty is very high, access to electricity very low, the economy is not developed and the sustainable development impact could be high as a result of the installation of solar power stations in these regions. These solar stations could improve the social, economic and environmental situation in the northern regions of Nigeria. It is the opinion of TAP that in case, the most of the planned 3–5 photovoltaic (PV) stations will satisfy the required investment criteria are in the northern regions, the sustainable

development potential of the project will be high. At this stage of assessment, it can only be assumed that finally selected PV stations will be located in these poor regions and contribute to the economy of these least developed regions by creating additional jobs and providing more reliable energy to small and medium-sized enterprises. The independent TAP thinks that development of such processes depends on the willingness of the population from these regions to pay for electricity and power dispatching regime even if access to the power grid will be increased in the northern regions through planned parallel governmental programmes. Therefore, the criterion is rated as “medium”.

1.4 Needs of the recipient

Scale: High

20. In Nigeria, lack of access to reliable energy is the most pressing social and economic development issue. According to the feasibility study conducted by the accredited entity (AE), the Nigerian power supply sector has huge problems, from generation to transmission capacity, distribution and end-user regulation under the absence of a metering system. Infrastructure for electricity network covers up to 70 per cent of the population while the remaining 30 per cent remains off-grid, with clear discrepancies among regions. On the demand side, the Nigerian system operator estimates that NESI requires more than 22 GW of generation capacity to cover its needs, which nowadays is partially covered by on-grid generation and partially covered off-grid by on-site diesel/gas generators.

21. On the supply side, even though the country has 12 GW of on-grid installed capacity, only 4–5 GW was dispatched on average over the last two years, while grid technical capacity for power evacuation is around 7–8 GW. There are also significant constraints on the transmission and distribution network capacity and massive investment should be deployed in the coming years to revert this situation.

22. As mentioned in paragraph 13 above, Nigeria’s power generation subsector is the highest emitter (80.7 per cent) in the country’s energy industry sector based mainly on natural gas generation. It has a high rate of gas and power transmission losses and low utilization of the country’s abundant renewable resources, and of solar energy in particular.

23. Nigeria’s NDC outlines an ambitious programme to achieve 13 GW of renewable energy supply to rural communities currently off-grid and reach 32 million tCO₂eq emissions reduction annually by using renewable energy by 2030. Based on NREEEP 2015, NERC set a target to reach 18.2 GW renewable installations, excluding large hydro.

24. Based on information provided in the funding proposal, feasibility study, NREEEP 2015 and NDC, it is clear that successful implementation of this project in Nigeria is crucial for the country to meet the different renewables targets – and solar power production in particular – and for strengthening the energy industry in low-emissions development.

25. The operational and financial situation of the power sector is further aggravated by an unpredictable regulation, including the inadequate enforcement of contracts. As a result, the sector lacks financial viability.

26. All these national circumstances listed above demonstrate that needs of the recipient in this project is very “high”.

1.5 Country ownership

Scale: Medium

27. Based on the high potential of renewable energy, NREEEP 2015 was developed by FGN defining the strategies to realize the large potential for renewable energy sources in Nigeria. NERC set a target of generating a minimum of 2 GWh of electricity from renewables by the year 2020 and 18.2 GWh, excluding large hydro, by 2030.

28. Nigeria's NDC demonstrates the country's high interest and ownership of planned measures in the power sector. The mitigation options proposed in the NDC for the energy sector address both energy demand and supply, and are expected to deliver significant development benefits along with the mitigation of GHGs.

29. The adaptation strategy of Nigeria considered in the NDC also covers activities for high emitting sectors, which should reduce the risks identified, in particular, in the power supply sector. The adaptation strategy plans to increase the resiliency of the power supply sector, which could be done through:

- (a) The inclusion of increased protective margins in construction and placement of energy infrastructure (i.e. higher standards and specifications);
- (b) Undertaking risk assessment and reduction measures to increase resilience of the energy sector;
- (c) Strengthening existing energy infrastructure, in particular through early efforts to identify and implement all possible no-regrets actions;
- (d) Developing and diversifying secure energy backup systems to ensure both civil society and security forces have access to emergency energy supply; and
- (e) Expanding sustainable energy sources and decentralizing transmission to reduce vulnerability of energy infrastructure to climate impacts.

30. Through different programmes and regulatory acts, FGN expressed its high commitment to the renewable power generation sector, including off-grid and on-grid solar power generation. Diversification of power sources and inclusion of more sustainable options in power generation are considered by FGN as priority options. In support of this approach, in November 2015, FGN approved its FIT regulation and put the electricity distribution companies under obligation to buy at least 50 per cent of power generated from renewables and price tariffs according to the long-run marginal cost and levelized cost of energy.

31. In 2016, NBET negotiated and signed PPAs with 14 IPPs for 1.32 GW of solar PV with an estimated investment cost of over USD 1.8 billion, accepting a generation tariff of USD 115/MWh for the energy produced. As the current generation tariff is around USD 55/MWh, in March 2018, the Minister of Finance advised that the country couldn't pay the previously negotiated tariff and should instead proceed if developers accept a significantly lower generation tariff of USD 75/MWh, refusing to approve a put-call option agreement. As of August 2018, only two solar PV developers, Afrienergia Power (50 MW) and CT Cosmos (70 MW), have signed the put-call option agreement.

32. The success of the development of the renewable energy investments in Nigeria is strongly linked to the need to reset the electricity supply industry, NESI. In that sense, in 2017, FGN approved the PSRP, which aims to develop a series of carefully thought out policy, regulatory, operational, governance and financial interventions to be implemented over the next five years to restore financial viability.

33. According to the feasibility study, the PSRP, if not the only one, is a great opportunity to transform the NESI in the next five years. If correctly implemented, it will create the right environment for the deployment of renewable energy investments in Nigeria. So far, the PSRP has only been partially implemented, with some important interventions being deferred until after the general elections in 2019. This very important decision made by the country is weakened by delays in implementation process, which makes the country's ownership "medium".

1.6 Efficiency and effectiveness

Scale: Medium

34. Efficiency of this funding proposal depends on the proposed financial scheme to be successful from a GCF money management point of view and on the technical realization of the project concept to deliver the power generated from renewable energy (via grid supply) to the final consumer and ensure GHG emissions reductions.

35. At this level, assessment of the proposed financial/investment scheme of the funding proposal could be considered efficient, considering all existing and anticipated financial constraints and securing the identified financial risks.

36. The funding proposal has high potential to demonstrate an investment scheme that could be successful in Nigeria for crowding-in IPPs and private investors into the solar power industry. It was assessed that the debt service cover ratio for the considered solar projects is around 1.16x without GCF participation, which demonstrates a high investment risk for the programme; while with GCF participation, the debt service cover ratio grows to 1.38x, which could be considered as acceptable risk. To mitigate the revenue risk in terms of payment delays by the off-taker, the AE has incorporated a debt service reserve account equal to six months of debt service as project selection criteria.

37. Regarding the potential of realization of the GCF target to achieve maximum reduction of GHGs, it is the opinion of TAP that the risk is very high when considering all existing constraints. Its success fully depends on the national circumstances and effective enforcement of the regulations and programmes introduced by FGN for removing different barriers to improve the national grid supply capacity. Therefore, at this stage of assessment, the final effect/impact is assessed as “medium”.

38. The solar programme implementation structure is also designed to allocate risk appropriately through the requirement for adequate transmission and evacuation plans to be in place prior to disbursement for construction of the plant and the take or pay arrangement under the PPA. However, implementation of this requirement is not under the control of the AE.

39. The cost per tCO₂eq of reduced GHG emissions for GCF is calculated to be USD 10 and, for whole project, USD 49 for the 20 years of each solar project’s lifespan. Considering the current state of the electricity supply sector, the price does not seem very high. However, as was mentioned above, it’s the opinion of TAP that delivery of generated solar power energy to the final consumer might not be so high, which will be reflected in the final cost of GHG emissions reductions. This is one more reason for scoring the criterion as “medium”.

II. Overall remarks from the independent Technical Advisory Panel

40. The independent TAP fully recognizes the high risk of the funding proposal not to reach the planned impact of GHG emissions reductions because of significant constraints existing in the country's grid delivery process, including: grid absorption and evacuation capacity; the low technical capacity of the electricity distribution companies; a low rate of connectivity to the grid; unpredictable tariff regulations; and a lack of willingness of end-users to pay for unreliable electricity. At the same time, TAP takes into consideration that GCF intervention could have a high impact on proper and timely implementation of the planned ongoing support programmes of FGN and different donors, and that the government is dedicated to improving the electricity supply industry. Thus, TAP recommends the funding proposal for approval.

41. In order to reduce this implementation risk of delivery power to end-users, TAP recommends that the Secretariat form an agreement with the AE to perform regular analysis of the delivery of energy produced by solar IPP and, in case of issues, coordinates with the programme stakeholders to review appropriate measures that would address and help to improve the situation.

42. Also, TAP recommends that the Secretariat monitors parallel projects (FP027, FP095 and FP099) financed by GCF for African countries, including Nigeria, that will implement activities in Nigeria that are complementary to the activities planned in this funding proposal.

Reply to the Independent Technical Advisory Panel assessment findings (FP 104)

Proposal name: Nigeria Solar IPP Support Program

Accredited entity: Africa Finance Corporation (AFC)

Impact potential
Noted and no further comments.
Paradigm shift potential
Noted, we believe that post the 2019 elections, there will be a renewed commitment from the FGN on the implementation of the PSRP as well as other related programs that are focused on improving NESI. And as such, there will be increased confidence with regards to the Programs ability to execute against its intended GHG emission reduction targets.
Sustainable development potential
Noted and no further comments.
Needs of the recipient
Noted and no further comments.
Country ownership
Noted, we believe that post the 2019 elections, there will be a renewed commitment from the FGN on the implementation of the PSRP as well as other related programs that are focused on improving NESI.
Efficiency and effectiveness
Noted and no further comments.
Overall remarks from the independent Technical Advisory Panel: The recommendation from the ITAP for the AE to perform regular analysis of delivery of energy produced by the Solar IPP's and in cases of issues coordinate with the relevant power sector stakeholders is well noted.



GENDER ASSESSMENT AND ACTION PLAN

NIGERIA SOLAR INDEPENDENT POWER PLANT (IPP) SUPPORT PROGRAM

23 January 2019

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GLOSSARY

AFC	Africa Finance Corporation
AfDB	African Development Bank
FCT	Federal Capital Territory (Abuja)
GAP	Gender Action Plan
GCF	Green Climate Fund
HIV- AIDS	Human Immunodeficiency Virus-Acquired Immune Deficiency Syndrome
IFC	International Finance Corporation
IPPs	Independent Power Producers
ISS	Integrated Safeguards System
OS	Operational Safeguard
PAR	Project Appraisal Report
PBO	Program-Based Operation
UN	United Nations
WB	World Bank Group
ERGP	Economic Recovery and Growth Plan
ECREEE	Ecowas Center for Renewable Energy and Energy Efficiency
FCT	Federal Capital Territory
GDP	Gross Domestic Product
FGN	Federal Government of Nigeria
CSO	Civil Society Organization
CO ₂	Carbon Dioxide
SSA	Sub-Saharan Africa
MW	Megawatts
PPA	Power Purchase Agreement
SME	Small and Medium-Sized Enterprises
NBET	Nigerian Bulk Electricity TradingPlc
NEEDS	National Economic Empowerment and Development Strategy
FCT	Federal Capital Territory
GEO	Gender and Equal Opportunities
VAPP	Violence Against Persons Prohibition
FMWA	Federal Ministry of Women Affairs
GDI	Gender Development Index

OECD	Organization for Economic Cooperation and Development
UNESCO	United Nations Educational, Scientific and Cultural Organization
WHO	World Health Organization
UNICEF	United Nations International Children's Emergency Fund
SDG	Sustainable Development Goals

EXECUTIVE SUMMARY

In most Sub-Saharan African countries, lack of access to electricity and other forms of energy adversely affects businesses, thereby reducing standards of living and leading to a depression of the overall economy. Limited access to energy for the most basic needs of cooking, lighting, income generation, schooling and healthcare limits human development in many spheres, including but not limited to education, nutrition, food security, water and sanitation, market-based work, health and gender relations. In Nigeria, many women and men, particularly those with low incomes living in urban and peri-urban areas, are comparatively more disadvantaged in terms of their ability to access electricity and modern energy.

Nigeria has a population of over 180 million people and currently faces major energy challenges, with as many as 98 million people, approximately 55% of its citizens, lacking access to grid-connected electricity, as at the end of 2015. Even with an installed capacity of approximately 11,165 MW as at December 2017, available daily generation capacity has been erratic, ranging from 3,000 MW to 5,000 MW due to gas availability, transmission and distribution constraints. This is short of the required electricity to supply the Country's huge and growing population. This shortfall in generation capacity has led to the proliferation of private fossil-fuel based generator use by many individuals and businesses, a situation which is inefficient and polluting. The rampant use of fossil-fuel based generators has the resultant effect of increasing greenhouse gas (GHG) emissions in the country. Between the years 1990 through 2010, GHG emissions increased from 164 million tonnes (MT) CO₂ eq to 263MT CO₂ eq. Population growth, along with predicted economic growth in Nigeria is expected to drive GHG emissions to over 900 MT CO₂ eq by 2030. Furthermore, the installed energy mix in Nigeria is 26% from hydropower and 74% from oil, gas and other fossil fuels, highlighting the significant amount of room for growth of solar power within the Country's installed energy mix.

To show its commitment to increasing power generation and diversifying away from thermal and hydropower, the Federal Government of Nigeria, through the Nigerian Bulk Electricity Trading Plc ("NBET"), the electric power off-taker, signed several utility-scale solar Power Purchase Agreements (PPAs) in July 2016, to supply around 1,125 MW of power to the Nigerian national power grid. The Nigeria Solar IPP Projects (a total of 14 IPP projects, mostly located in the northern part of Nigeria) will bring significant diversification of the Country's energy mix from diesel generation sources. To date, none of these projects have been able to reach financial close.

In support of the development and implementation of the Nigeria Solar IPP Program, the Africa Finance Corporation ("AFC") and the African Development Bank ("AfDB") are together proposing to work with the Green Climate Fund ("GCF") on a Nigeria Solar Intervention Program, which will catalyze the delivery of c.400MW of renewable power, ensuring the successful financing, construction and operation of the first utility-scale solar power projects in the Country. The Program will consist of US\$300 million, to be equally provided by GCF, AfDB and AFC and will provide concessional debt alongside local financial institutions, for the construction of 3 – 5 of the Nigeria Solar IPP Projects. Each of the 14 IPP's will submit relevant documentation and those that meet the selection criteria will be shortlisted as sub-projects of the Program. The total cost of the projects to be selected is expected to be c. US\$400 million. The Program is expected to reduce or avoid 476,487t CO₂ eq on an annual basis (9,529,739 t CO₂ eq. over the life of the Program) and impact at least 1 million households in Northern Nigeria as direct or indirect beneficiaries. The Program is also expected to reduce the perceived risks of investing in the Nigerian renewable energy sector and catalyze private sector investment in the sector.

This document presents an overview of a gender assessment across Nigeria, with a focus on Northern Nigeria and the States in which the sub-projects are located. Challenges to mainstreaming gender considerations into sub-projects are identified and discussed in detail. A summary of stakeholder consultations undertaken during the preparation of this document is presented, including a list of the potential stakeholder institutions and key actions considered influential in the consideration of gender aspects within each of the focus States.

The document concludes with a Gender Action Plan (GAP), which sets out specific goals and targets related to positive gender actions, which the Program will seek to achieve within the sub-projects. It is expected that the sub-projects will develop Sub-project-specific Gender Action Plans, with specific measurable targets for the medium and long term. As part of the preparation of the Sub-project-specific GAPs, further consultations will be undertaken with all relevant stakeholders (local government, local and international Civil Society Organisations or CSOs), to ensure that gender considerations are effectively mainstreamed and considered in all sub-projects.

1 INTRODUCTION

1.1 Background

The population of Nigeria has been growing steadily with the average annual population growth rate only slowing from 3.2 % in 2006 to 2.5 % in 2014.¹ In 2015, the population of Nigeria was estimated at 183 million people, consisting of 92,387,474 males and 90,989,254 females, with sex ratios of 102 men / per 100 women². Almost 70 % of the Nigerian population is aged below 30 years.

Nigeria is a Federation comprising the Federal Government, 36 State Governments and the Federal Capital Territory (FCT or Abuja), with 774 local governments. The Nigerian government administration is divided into 3 main branches – the Executive, the Legislative and the Judiciary, which are independent organs that provide checks and balances in governance. There are several political parties in Nigeria but two are most prominent; the All Progressives Congress, that won the most recent 2015 elections with 211 seats (206 male and 5 female) and the People's Democratic Party with 142(132 male and 10 female) seats in the House of Representatives.

Nigeria released its Economic Recovery and Growth Plan (ERGP) for medium term planning for the period, 2017 – 2020. The objectives of the Plan are to restore economic growth, build a globally competitive economy and invest in Nigerians by driving social inclusion, job creation, youth empowerment and improved human capital. To achieve the objectives of the Plan, the Nigerian government prioritized 5 steps which include:

- Stabilizing the macroeconomic environment;
- Achieving agriculture and food security;
- Ensuring energy sufficiency;
- Improving transportation infrastructure and;
- Driving industrialization focusing on Small and Medium-Sized Enterprises (SMEs).

Nigeria has one of the world's highest economic growth rates, averaging 7.4%³ annually, but over 42.4% of the population currently live below the poverty line, according to the United Nations (UN). Oil exports contribute significantly to government revenues and about 15% of Gross Domestic Product (GDP), despite employing only a fraction of the population. Agriculture, however, contributes to

¹ Nigeria Country Profile, United Nations Economic Commission for Africa, 2016

² 2015 Statistics Report on women and men in Nigeria, National Bureau of Statistic, November 2016

³ According to the Nigeria economic report released in July 2014 by the World Bank

about 45% of GDP, and employs close to 90% of the rural population, who are often engaged in subsistence farming.

Mainstreaming gender considerations into project design, development and implementation is often challenging, even though women and men are affected differently by project interventions. In Nigeria, many women and men, particularly those with low incomes living in urban and peri-urban areas, are more disadvantaged than most in terms of their ability to access electricity and modern energy. The lack of access to electricity and other forms of energy adversely affects businesses, thereby reducing standards of living leading, to a depression of the overall economy. Limited access to energy for the most basic needs of cooking, lighting, income generation, schooling and health centers limits human development in many spheres, including but not limited to education, nutrition and food security, water and sanitation, market-based work, health and gender relations.

1.2 Scope and Objectives

Based on the above context and to ensure compliance with the GCF Gender Policy and Action Plan Framework, the scope and objectives of this document are to:

- Undertake a Gender Assessment including a review of the Nigerian context related to gender, with a specific emphasis in Northern Nigeria and sub-project States;
- Prepare a GAP with goals and indicative actions, which will be implemented as part of the Program to ensure that positive gender action is mainstreamed into the sub-projects.

In addition to the above, a detailed regulatory review related to gender is also undertaken. Some stakeholder consultation has been undertaken and several potential partners and CSOs, which could be engaged during the development of sub-project-specific GAPs have been identified and presented in the document.

2 REGULATORY POLICY FRAMEWORK RELATED TO GENDER

2.1 The Nigerian Context

2.1.1 The Gender Policy Framework in Nigeria

The 1999 Constitution of the Federal Republic of Nigeria prohibits discrimination on the basis of places of origin, sex, religion, status, ethnic or linguistic association. Successive governments have always demonstrated commitment to upholding this and to promoting gender equality and women's empowerment in varying degrees. To facilitate gender equality and women's empowerment, the Federal Government of Nigeria created a favorable national legal and policy framework and put in place institutional mechanisms in this regard.

Moreover, Nigeria as a member of the United Nations signed and ratified various relevant international instruments, treaties and conventions without reservation. These instruments have always emphasized that member nations put in place the necessary mechanisms needed to eliminate gender discriminations and ensure equality and human dignity to all men and women.

The Federal Government of Nigeria in 2000 adopted a National Policy on Women. This Policy was reviewed in 2006 and upgraded to become the National Gender Policy. Other key government policies with gender equality and empowerment of women frameworks include the National Economic Empowerment and Development Strategies (NEEDS) in May 2004; and the Transformation Agenda of the immediate past administration who, in developing the Vision 2020, had a 'Special Interest Group on Women' to oversee the development of policy statements that engender 'sustainable human and national development built on equitable contribution of the Nigerian women, men and children'.

2.1.2 Convention on the Elimination of All Forms of Discrimination against Women

The Federal Republic of Nigeria consists of 36 states and the Federal Capital Territory (FCT), with diverse cultural and religious practices even within each state. In many cases, including issues relating to women and children, federal laws are not applicable at state level. Attempts to domesticate the Article 18 of the Convention on the Elimination of All Forms of Discrimination against Women (the Convention) at the Federal level continues to generate public and parliamentary debates, with some legislators, religious groups and leaders arguing that the provisions will erode local customs and religious beliefs.

In order to render the Convention applicable domestically, the National Assembly has been debating the Gender and Equal Opportunities Bill (GEO Bill) since 2017.

The GEO Bill contains provisions that prohibit discrimination against women and invalidates pre-existing discriminatory laws, policies, rules, directives and customs. The Bill specifically mentions the need to protect the right of women and girls to conclude contracts, administer property and be treated equally to men in courts and tribunals. The GEO Bill also sets out provisions aimed at addressing harmful cultural practices that violate the rights of widows. If the GEO Bill is not passed into law at federal and state levels, existing laws that discriminate against women remain operative in different parts of the country. Such laws include, Section 55 of the 1990 Labour Act of Nigeria which bars women from being employed in night work, except as nurses; and Section 360 of the Criminal Code Act which makes the indecent assault of a woman a misdemeanour punishable with a two-year prison term, as opposed to the three years prison term imposed for indecently assaulting a man, which is a felony in Section 353.

2.1.3 The Violence Against Persons Prohibition Act (VAPP Act)

The Violence Against Persons Prohibition Act (VAPP Act) was passed by the National Assembly in 2015, but this Law is currently only applicable in the FCT. The Law criminalises several acts which are harmful and discriminatorily against women, such as female circumcision or genital mutilation; forcefully ejecting a person from their home; forcing a person to be financially dependent on another or economically abusing them; and harmful widowhood practices. In addition, the law broadens the narrow definition of rape as contained in Criminal and Penal codes. While some additional amendments are required to bring the Act fully in line with international standards, by criminalising these acts which are predominantly carried out against women and girls, the VAPP Act has expanded the scope of protection available to women and girls in FCT.

CSOs have continued to lobby individual states to enact the VAPP Laws or similar legislation to protect women and girls against abuse. At least 13 States now have domestic laws which protect some of the rights of women and girls. However, many of these laws have limited scope and do not guarantee women the full range of human rights that Nigeria has committed to implement by ratifying the Convention.

2.1.4 Gender Related Laws at State-level

Though the gender focal points were set up in other related ministries and government offices, coordination among ministries and offices is not sufficient and there needs to be a coordinated framework for the coherent measures. As Nigeria is a Federal Republic, each state has the authority to draft its own legislation. As a result, each State has different laws on gender issues.

It is very significant to mention that the Plateau State government gazetted its own Gender Equal Opportunity Law passed since 2015 in the State, making it the first to do so in Northern Nigeria. The Law gazetted on June 27, 2018 is to protect women from all forms of discrimination in politics, economy, traditions and social issues.

Table 2.1. Gender related laws at the State level in Nigeria

The law on prohibition of female human trafficking	• The Sharia (Sharia law) Article 239, 2000 in Zamfara State • The Edo State Criminal Code (Amendment) Law 2000
The law on abolition and prohibition of Female Genital Mutilation	• Edo state (2000), Ondo state
The law on protection of human rights of widow	• The Prohibition of Infringement of a Widow's and Widower's Fundamental Human Rights Law, No. 3, 2001
The law on prohibition of early marriage, and ensuring primary school enrolment for girls	• Ban on early marriage in Kebbi and Niger states • The state law which secures primary school enrollment for girls in Edo, Kano, Borno, Gombe, and Bauchi states

Source: Federal Ministry of Women Affairs and Youth Development (2002) (2008b)

2.2 Sharia Law in Nigeria

2.2.1 History of Sharia Law in Nigeria

The values, beliefs, ideas, arguments and discussions of Sharia Law have been a significant part of the religious and cultural context of Nigeria for centuries and date back to the revivals brought about by Usman Dan Fodio in the early 19th century. One of the significant aims of the Dan Fodio-led Hausa Kingdom, during the years of the 19th Century Holy War, was to improve the standards of living of Muslim women. After the death of Usman Dan Fodio, the practicalization of precepts based on this religious system of rule where obscured for several decades, with a return to patriarchal systems. However, there was a resurgence of the rule based on Sharia Law precepts, with the declaration by Governor Ahmed Sani Yerima of Zamfara State in 1999.

Since 1999, Sharia Law has been instituted, either partly or fully, as a main form of civil or criminal law in 12 States in Nigeria. These are presented below.

Table 2.2. List of Nigerian States that have adopted Sharia Law

1. Zamfara State	2. Kano State	3. Sokoto State	4. Katsina State
5. Bauchi State	6. Borno State	7. Jigawa State	8. Kebbi State
9. Yobe State	10. Kaduna State	11. Niger State	12. Gombe State

It is important to note that Kaduna, Niger and Gombe States have only instituted Sharia Law in parts of these States with large Muslim populations.

Although there were significant protests against the institution of Sharia as a formal system of law in Northern Nigeria (this mostly stemming from non-Muslim minorities in implementing States), given the judgement pronounced in the case of *Alkamawa V Bello* (1998), Sharia Law is now seen as a distinct and universal legal system in the Country.

2.2.2 Challenges with implementing Sharia Law in Nigeria

Although the historical foundations of Sharia Law may not have included an overt gender bias, there are currently widely acknowledged biases in the implementation of penal codes under Sharia Law. It is thought that the misapplication of Sharia Law and/or mis-pronouncement of judgements under this system is likely to be linked to the ignorance of women and possible biases stemming from the judgment and prosecution systems.

As an example, there is a social custom where widowers according to Sharia law can fall into economical poverty when the husband passes away because they traditionally have no right to their property.

2.2.3 Sharia and Gender Issues in Northern Nigeria

There is limited information available on the direct impact of Sharia Law implementation on the state of women in Northern Nigeria. Before the advent of formal Sharia Law in 1999, the situation of women in a typical town in northern Nigeria was described as follows: *"The plight of women is one of perpetual silence and immobility. The houses are built in such a way that women and children are housed in separate quarters from men"* (Women's World: Rome ISIS, 1989). This points to the fact that Sharia Law implementation may not have necessarily worsened the lot of the Northern Nigerian woman. Indeed, there are assertions that the institution of Sharia Law was used instead as a tool to gain political prominence during the resurgence of Nigeria's democratic era in 1999.

The use of Sharia Law as a political and cultural tool is an issue of worldwide debate. Gender proponents highlight one of the fundamental messages in the

Qur'an, which emphasizes the equality of all Muslim believers and does not condone oppressive acts towards women. Instead, studies have postulated that the treatment of women in Northern Nigeria is influenced more by cultural and historical factors than religious factors.

The existence of Sharia Law in Northern Nigerian States is a significant factor in the implementation of gender action plans, targets and objectives for the Nigeria Solar Program. The Program aims to recognize the important influence of religious codes and practices, such as Sharia Law, on the norms, values and systems of life of communities around the implementing projects, while recognizing the importance of gender equality and using culturally appropriate means to promote appropriate gender practices.

2.3 Nigerian Institutional Provisions and Arrangements

The national machinery for the promotion of gender issues in Nigeria is the Department of Women Affairs of Federal Ministry of Women Affairs and Social Development (FMWA), which was established in 1995. Each State in Nigeria has the State Ministry of the Women Affairs and Social Development. The objective and vision statement of FMWA is as follows: *To have a Nigerian society that guarantees equal access to social, economic and wealth creation opportunities to all, irrespective of gender; and one that places premium on protection of the child, the aged and persons with disabilities, while focusing attention on key operators in both private and public sectors on mainstreaming the concerns of these groups of people in national development process.*

Their role is to serve as the national vehicle to bring about speedy and healthy development of Nigerian women, children, the socially disadvantaged and person with disabilities; and the mainstreaming of their rights and privileges in national development process

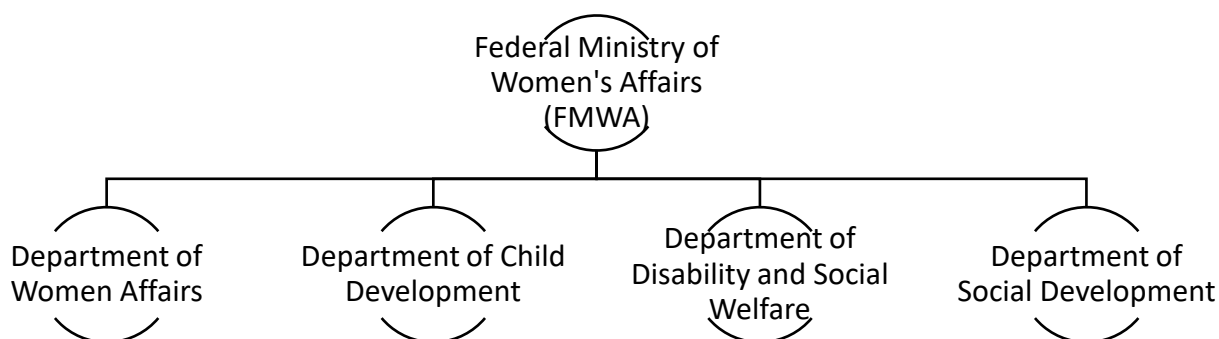


Figure 2.1. Organizational Chart of the Federal Ministry of Women Affairs

Though the gender focal points were set up in other related ministries and government offices, coordination among ministries and offices are not sufficient and there needs to be a coordinated framework for the coherent measures.

2.4 International Policies, Standards and Institutions Relating to Gender

The next section outlines the gender policies from GCF, AFC and AfDB. They are complimentary and the Program will integrate these gender policies in the 11 sub-projects.

2.4.1 Green Climate Fund

GCF⁴ has placed gender as a key element of its programming architecture, and its commitment to gender equality centres on gender-responsive climate action Programs and projects that benefit women and men.

Gender mainstreaming is central to the GCF's objectives and guiding principles, including through engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The GCF Governing Instrument states that: *"The Fund will strive to maximize the impact of its funding for adaptation and mitigation ... promoting environmental, social, economic and development co-benefits and taking a gender-sensitive approach."*

GCF's gender-responsive approach is captured in the GCF Gender Policy and Action Plan⁵, which was adopted by the Fund's governing body in 2015. Through the Gender Policy and Action Plan, GCF is committed to:

1. Gender equality and equity;
2. Inclusiveness in all activities;
3. Accountability for gender and climate change results and impacts;
4. Country ownership—through alignment with national policies and priorities, and broad stakeholder engagement, including women's organizations;
5. Gender-sensitive competencies throughout GCF's institutional framework—skills, knowledge and behaviours acquired from training and

⁴ <https://www.greenclimate.fund/how-we-work/mainstreaming-gender> [modified text]

⁵ <https://www.greenclimate.fund/documents/20182/818273/1.8 - Gender Policy and Action Plan.pdf/f47842bd-b044-4500-b7ef-099bcf9a6bbe> [Accessed 06/12/2018]

experience that enable GCF Secretariat staff members to apply a gender-lens throughout their work; and

6. Equitable resource allocation—so that women and men benefit equitably from GCF's adaptation and mitigation activities.

Per the GCF requirements, gender equality considerations should be mainstreamed into the entire project cycle to enhance the efficacy of climate change mitigation and adaptation interventions and ensure that gender co-benefits are obtained. This applies to all projects, not only those intended from the outset to center on women or to have a gender focus. Gender mainstreaming is fundamental to any project intervention and does not necessarily signify additional costs; in fact, mainstreaming gender makes climate interventions more effective and efficient.

2.4.2 Africa Finance Corporation (AFC)

AFC has a Gender Policy, which seeks to integrate gender considerations in its investment activities. AFC's approach and commitment to its gender policy as it relates to its investment activities, implies that AFC will:

- i. work towards conducting gender analysis to ensure that gender issues are addressed as part of its investment process;
- ii. encourage investment entities to educate their supply chain in managing gender equity in accordance with global best practice;
- iii. integrate gender equity assessment into the AFC Investment process;
- iv. not establish business relationships with organizations that condone, support or otherwise participate in trafficking, including labor or sexual exploitation;
- v. include sex disaggregated data in environmental and social action plans.

2.4.3 The African Development Bank (AfDB)

2.4.3.1 Gender Considerations at the AfDB

The African Development Bank's, Department of Gender, Women, and Civil Society as part of its implementation of Gender Strategy Pillar II on Women Economic Empowerment, looked at business opportunities in the infrastructure sector. In this regard, the department with other development partners has commissioned a Pre-feasibility Study on Business Opportunities for Women in a Changing Energy Value Chain in West Africa. This pre-feasibility study is part of the ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE)

project. The validation workshop took place in April 2018 which was attended by women owned business in the energy sector.

The Pre-feasibility study aimed at identifying barriers and challenges to promoting women entrepreneurship in the energy sector across West Africa, and using this information to develop a gender responsive, regional energy market development strategy that taps into the innate entrepreneurial capacity of ECOWAS women.⁶

Results from the study show two types of forces impacting business models and transformations in energy-based businesses. These are (i) external forces (i.e. changing energy generation, transmission, storage, and distribution dynamics), (ii) and internal forces (i.e. a company's strategic capabilities which include its operational, human resource and financial dynamics). Both external and internal forces are drivers of change in the energy value chain which bring about changes in market models in the locality, country or region.

The study considers that women entrepreneurs act as agents of change on the energy value chain, capitalizing on the changes that happen in the energy market to deliver benefits – in the form of better products and services to consumers ('beneficiaries of change') creating both economic and social impact. Women entrepreneurs, acting as agents of change, are required to undergo certain transformations in their business models, in order to adjust to the changing market models.

The challenges that exist for women entrepreneurs range from handling operational complexities, technological expertise, employee training and development, recruitment challenges to financial foresight. The necessity to balance personal and business responsibilities, deal with cultural stigma and lack of technical (and or operational) and financial expertise, exerts pressure and restricts the growth of women-owned business, hence there is an urgent need to address their business challenges to help them thrive and succeed in the face of West Africa's energy transition.

2.4.3.2 Approaches to Consultation, including Vulnerable Groups

In addition, the Bank (IESIA) Guidance Notes contain detailed information on the mechanism of consultation defining various aspects of consultation including consultation objectives, requirements, approach in carrying out a consultation (how to carry consultation, how shall people be consulted, who to be involved

⁶ This study evaluated the drivers of change not only through secondary data (collected from study reports and data collected from agencies) but also primary data (collected through field interviews with women entrepreneurs).

and at what stage). Marking the specific aspect on stakeholders' engagement, the (IESIA) Guidelines Notes also provides guidance on vulnerable groups' identification as per OS 1 requirements on vulnerable groups and offer differentiated measures for vulnerable groups' inclusion in development.

2.4.3.3 Vulnerable group inclusion

The AfDB ISS defines vulnerable individuals or groups as those within a project's area of influence who are particularly marginalized or disadvantaged and who might thus be more likely than others to experience adverse impacts from a project. The vulnerability can be determined by identifying the likelihood that an individual or a group faces harder conditions as the result of the implementation of a project.

Vulnerable status may stem from a group's gender, economic status, ethnicity, religion, cultural behavior, sexual orientation, language or physical and psychological health conditions. Vulnerable groups may include, among others, female-headed households, those below the poverty line, the landless, those without legal title to assets, ethnic, religious and linguistic minorities, Indigenous Peoples, those who are disabled, etc.

Vulnerable groups are more likely to be exposed to adverse impacts in large-scale projects with a large area of influence, potential cumulative impacts, and multiple affected communities, than in small-scale projects that have site-specific issues.

2.4.3.4 Objective and scope of vulnerable group identification

The objective of identifying vulnerable groups is to enable a strategic focus on the consideration of their views and specific needs during the project planning, and thereby to specifically avoid harm to them, as well as to ensure that they have the opportunity to participate in and benefit from the proposed Program. Having identified the vulnerable groups, the objective becomes to define differentiated measures for them to ensure that they are protected and that suitable benefits are adequately planned and directed to them as set out below.

2.4.3.4.1 Vulnerable Groups and Gender

Projects may have different impacts on women and men, owing to their different socio-economic roles and their varying degrees of access to and control over assets, productive resources, and employment opportunities.

Gender discrimination often limits access to the resources, opportunities, and public services necessary to improve standards of living. In addition, there may be norms, societal practices, or legal barriers that impede the full participation of people of one gender (usually women, but potentially men) in consultation,

project planning, decision-making, implementation of project activities or the sharing of benefits.

2.4.3.4.2 Other Vulnerable Groups

Those of low economic status, particularly those below the poverty line, the landless and those without legal title to assets may also lack the resources and capacity to participate in project decision-making or benefit-sharing to the same degree as those of higher economic status.

In addition, those with health conditions, those who are disabled, etc., are also groups that commonly lack the capacity, means or voice through which to avoid negative project impacts and reap project benefits.

2.4.3.5 Differentiated measures for vulnerable groups' inclusion in development

Once groups have been identified as vulnerable, the borrower or client shall propose and implement differentiated measures so that adverse impacts do not fall disproportionately upon them, and so that vulnerable groups are not disadvantaged in sharing development benefits and opportunities. Differentiated measures are required to respond to the requirements of specific types of vulnerable groups. The AfDB ISS provides guidance to differentiated measures requirements to specific types of vulnerable groups including measures responding to vulnerable gender groups, indigenous people, vulnerable groups experiencing resettlements and other vulnerable groups such as groups of low economic status, those with health conditions, those who are handicapped, etc.

Meaningful consultation is of vital importance in determining what differentiated measures are necessary for the vulnerable groups in question, as well as in seeking broad community support (BCS) from such vulnerable groups. There should be a targeted and meaningful consultation process, backed by adequate information, and carried out with each vulnerable group.

Specific, targeted consultation sessions with each vulnerable group are important because consultations with non-vulnerable groups may not always reveal the special conditions or concerns of vulnerable groups, and how these might be addressed in a differentiated and targeted manner.

Consultation around differentiated measures for vulnerable groups requires a socially and culturally sensitive approach that shall ensure that:

- The vulnerable group in question is represented in discussions, and that members of this group are given ample and appropriate opportunities and channels to express their views, concerns and aspirations in the language and manner of their choice, without external manipulation, interference, coercion, or intimidation.

- Representative bodies and civil society organizations, as well as a sufficient number of members of the vulnerable group themselves, are included in the consultation process.
- Local leaders deemed to “represent” the views of vulnerable members of the community actually have the members’ consent and understand their views and perspectives.
- Spaces for discussions are created that are perceived to be “safe” from the perspective of the vulnerable group, and that are easily accessible to them.

The consultations with each vulnerable group should primarily seek to elucidate the special conditions and concerns of the group in question, and the form that associated differentiated measures should take in order to ensure that the vulnerabilities of the group in question are not further exacerbated by the project and that the group is given the opportunity and the capacity to benefit from the project according to their views and needs.

2.4.3.6 Gender Mainstreaming and Vulnerability Assessment

The term “vulnerable groups” refers to people who, by gender identity, ethnicity, age, disability, economic disadvantage or social status may be more adversely affected by project impacts than others and who may be limited in their ability to claim or take advantage of project benefits. Vulnerable individuals and/or groups may also include people living below the poverty line, the landless, the elderly, women- and children- headed households, refugees, internally displaced people, ethnic minorities, natural resource dependent communities or other displaced persons who may not be protected by national and/or international law. It is important to identify and address these groups during the early consultation phases of the projects to avoid placing additional strains on these groups because of the project.

The empowerment of vulnerable groups, especially women and youth groups is essential for public good and more so that a number of them are farmers as well. The feasibility study has revealed that, in the two regions, women are largely involved in agricultural and related activities. Thus, adequate women engagement is a key aspect to ensuring the sensibility and sustainability of the overall project management. Therefore, demonstrated efforts should be done to encourage the participation of women in the project. Annex 5 outlines some information that should be provided in an assessment of the challenges and opportunities for gender concerns.

The primary objective of the vulnerable persons' assessment and assistance measures is to avoid the occurrence of project-induced vulnerability, and if it occurs, to mitigate this through preventive and follow-up measures.

Criteria used to assess Project-induced vulnerability include pre-Project poverty, household composition, income, food supply, housing, social support, and health.

The criteria are used to establish household vulnerability relative to local conditions. Vulnerability thus becomes locally defined as those households that are recognized to be in a difficult situation against the background of general poverty in the area.

Vulnerability should be viewed in two stages: pre-existing vulnerability and transitional hardship vulnerability. Pre-existing vulnerability includes that stage which would be present with or without Project development. Transitional hardship vulnerability occurs when those directly affected by the Project, whether predisposed or not, are unable to adjust to new conditions due to shock or stress related to Project activities.

Project measures to identify vulnerable households and individuals include:

- Participatory engagement techniques to confirm community perceptions of well-being and to identify at-risk households;
- Analysis of baseline data to identify at-risk households;
- Implementation of household monitoring surveys designed to reveal trends in social welfare (household composition, assets, sources of income, expenditures etc.);
- Self-registration at offices of households that identify themselves as vulnerable or at risk; with all such registrations leading to an evaluation of that household by the project/investor team in order to assess the households' vulnerability; and
- Regular visits to all physically displaced households and any economically displaced households identified as vulnerable during resettlement planning and implementation processes to re-assess those households' vulnerability. Such visits will occur at least once a quarter; and each visit will be recorded in the database flagging changes to indicators that are problematic.

3 GENDER ASSESSMENT

3.1 Project Locations and Overview

In July 2016, the Federal Government of Nigeria signed PPAs for 14 solar power projects located in different parts of the country, mostly in northern Nigeria with a total generation capacity of 1,175MW. The 14 projects are shown in Table 3.1.

Table 3.1. Summary of all the Solar IPP signed by the Federal Government of Nigeria

	Project / Project Company	State	Capacity (MW)
1	Pan Africa Solar	Katsina	75
2	Nigerian Solar Capital Partners	Bauchi	100
3	Afrinegia Power Limited	Nasarawa	50
4	KVK Power Limited	Sokoto	50
5	Middle Band Solar One	Kogi	100
6	Novia Scotia Power Development Company	Jigawa	80
7	CT Cosmos	Plateau	70
8	Oriental Renewable Solutions	Jigawa	50
9	Quaint Abiba Power	Kaduna	50
10	Anjeed Innova Group	Kaduna	100
11	EN Africa	Kaduna	50
12	Motir Dusable Limited	Enugu	100
13	Nova Solar 5 Farm Limited	Katsina	100
14	LR Aaron Power	Abuja	100

*Project No 14 (LR Aaron Power) is not currently being considered for inclusion under the Nigeria Solar IPP Program.

The projects are mostly located in Northern Nigeria as shown in Figure 3.1 below.

UN

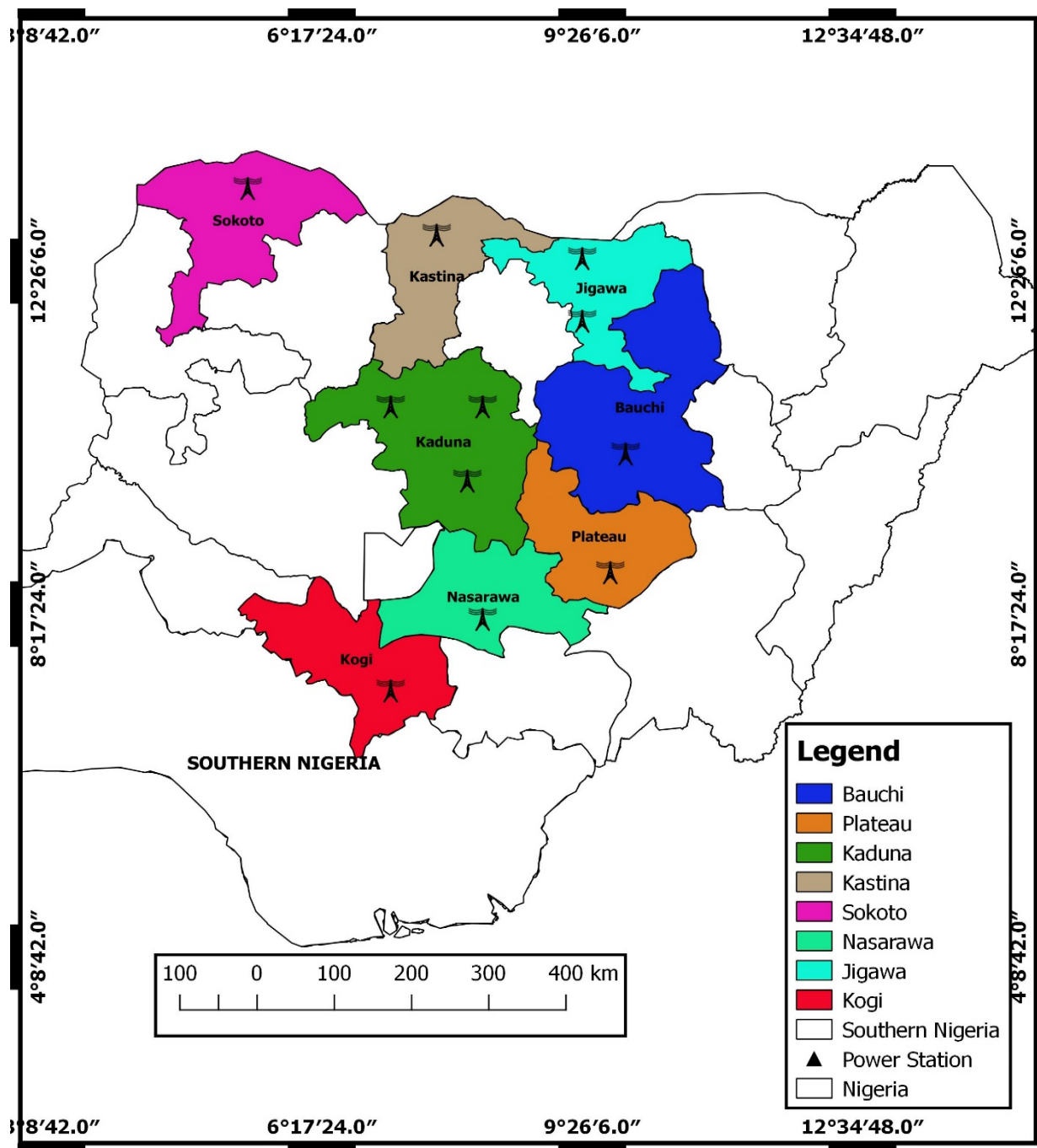


Figure 3.1. Location of the Solar IPP Subprojects in Northern Nigeria

3.2 Gender Characteristics and Key Indicators across Nigeria

The following section provides a summary of a number of key indicators related to gender across Nigeria.

3.2.1 Gender Gap Index

Nigeria is ranked 125 out of 145 on the 2015 Gender Gap Index countries with a score of 0.638. On the AfDB Gender Equality Index 2015, Nigeria is ranked 23rd out of 52 countries. The AfDB Index reflects women's status in three dimensions of equality: economic opportunity, social development and law and institutions. The ranking is based on a score of 0-100, with 100 representing perfect gender equality. Nigeria's overall score is 54.7%, it ranks 18th in economic opportunities, and an unsatisfactory 32nd in human and social development and 30th in laws and institutions.

3.2.2 Human Development Index

The 2014 female Human Development Index (HDI) value for Nigeria is 0.468 in contrast with 0.556 for males, resulting in a Gender Development Index (GDI) value of 0.841. Nigeria is ranked 152 out of 188 countries on the UNDP's Gender Inequality and GDI; the Organization for Economic Cooperation and Development's (OECD) Social Institutions and Gender Index of 0.3911 is very high (significant gender gaps in education, economic empowerment and political participation remain in Nigeria) and the World Economic Forum's Gender Gap Report. The country is ranked 39 out of 52 on the Ibrahim Index of African Governance, with a score of 44.9 (out of 100) in overall governance. This is lower than the African average (50.1) and lower than the regional average for West Africa (52.4).

3.2.3 Education

High population growth has put pressure on an already challenged education system. In 2011, according to UNESCO, children of primary school age accounted for over 16% of the population. Estimates suggest that 10.1 million are not receiving a formal education and 26% do not complete primary school. Primary school enrolment has fallen from 84.8% in 2005 to 57.4% in 2014. Secondary school net enrolment has stagnated at a low level, around 35 % (see figure 14). In 2008 only 61% of men and 41% of women could read and write. Enrolment in tertiary education was 10.4% for men and 8% for women in 2005. The enrolment in tertiary education overall was 16.7% in 2014.⁷

The literacy rate of young women between 2004 and 2014 increased from 60.4% in 2004 to 80% in 2008. In 2012, it declined to 66%, followed by a slight increase in 2014 (66.7%). There were more literate young women in the urban regions (85.3%) than in the rural areas (57.8%). The %age of literate young women in the south-

⁷ Nigeria Country Profile, United Nations Economic Commission, 2016

east (93.5%) was much higher than in the rest of the regions. The north-east (33.0%) and the north-west (35%) had the lowest %age of literate young women in 2014.⁸

3.2.4 Health

The health profile of Nigeria has improved in recent years, but it remains worrisome overall. The under-5 mortality rate has improved from more than 191 per 1,000 live births in 1990 to 89 per 1,000 live births in 2014 (see figure 11). However, it remains higher than the regional and global averages. The most frequent causes of child mortality are malaria (20%), pneumonia (17%), other diseases (14%), prematurity (12%) and diarrhea (11%). Child malnourishment is widespread. The rate of children under five with stunted growth has stagnated above 40 % since 2000. For the population, the most frequent causes of death are communicable diseases (81%), non-communicable diseases (14%) and injuries (5%). Nigeria is one of the few countries worldwide in which polio still persists, although it is limited to the north-eastern states.⁹

Fertility is at 6 children per woman, which is above the regional average (5 children) and the world average (2.5 children). Maternal mortality was 243 per 100,000 live births in 2014, which was higher than the world average (200 deaths per 100,000 live births) (see figure 12). Life expectancy at birth was 49 years in 2005 and has increased to 52.3 years in 2012 (53.4 years for women and 51.7 years for men).¹⁰

3.2.5 Political Participation

Nigeria's gender gap is widening for want of political empowerment, leadership and good governance. Women are under-represented in all political decision-making bodies and most often occupy subordinate positions, particularly at the community and household levels. The lack of women in decision-making positions may be one explanation for Nigeria's low investment in sectors that are crucial to human development outcomes, such as health and education. Although more women than men are registered to vote, women are most often excluded from decision making at all levels by male dominated patronage networks, the absence of agreed quotas and a party system that fails to nominate women candidates for electable seats. Fear of violence and restrictions on mobility may also deter women.

3.2.6 Employment

A large share of the labor force of Nigeria is employed in the primary sector: 40.1% of the male labor force and 19.9% of the female labor force are employed in

⁸ NBS, 2015.

⁹ Nigeria Country Profile, United Nations Economic Commission, 2016

¹⁰ Ibid

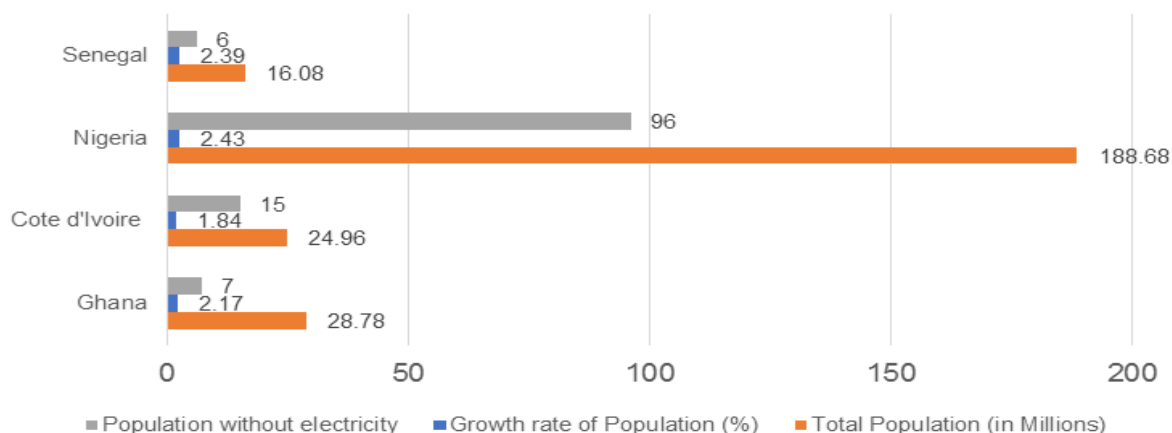
agriculture, forestry and fishing. Another important source of employment is the service sector: 24.9% of the labor force is employed in wholesale and retail trade, including repairs. Those sectors are followed by other services (7.1%), accommodation and food services (5.6%), transportation and storage (4.1%) and education (3.2%). Finally, 11% of the labor force is employed in manufacturing and 2.4% in construction.

3.2.7 Poverty

In spite of Nigeria's economic growth and rich endowment in terms of human and natural resources, social disparities abound, including poverty and gender inequality. Poverty levels are high with women particularly the rural dwellers make up the majority of the poor. Poor assets ownership has greatly contributed to this particularly in the North East (NE) where majority of the beneficiary states are located. For example, in a situation where about 70% of women are involved in agriculture, over 93% of the women do not own land in the NE (Adamawa 94.9%, Bauchi 92.5%, Gombe 97.7%, Taraba 93.5%). The story is similar with Anambra and plateau states with 90.9% and 66.5% respectively (NDHS 2013).

3.2.8 Demographic changes and growing urbanization

Demographics are drivers of present and future demands for electricity. Nigeria has the fastest growing demographics, which is about the average population growth rate for West Africa, i.e. 2.7% per year. The data from ECOWAS Center for Renewable Energy and Energy Efficiency (ECREEE) West Africa Power Pool (WAPP) projections highlights that total consumption of electricity will almost double by 2020 and triple by 2030 in these 4 selected countries combined.



Source: IMF, CIA and IEA Databases 2016

Figure 3.2. Demographics & Electricity Access

Nigeria has the highest urbanization rate with 4.3%. The rate at which the population growth and urbanization are happening, electricity distribution and waste accumulation will be some of the biggest challenges in urban planning for cities of the future, and this will open opportunities for technologies that can convert waste to electricity. During the interviews it was found that some women entrepreneurs share their interest in waste-to-energy technologies, but all of these were start-up ideas.

3.3 Gender Characteristics and Key Indicators in Sub-project States

As part of the gender assessment, a review of the literature was undertaken across the states in which the sub-projects are located, to identify specific gender issues, which may become material considerations during the design and implementation of the sub-project.

It is expected that as part of the implementation of the sub-projects, more specific reviews will be undertaken to inform the development of the sub-project-specific GAPs. The table below detail statistics in some Northern Nigerian States and in sub-project States.

Table 3.2. The gender related index in some Northern Nigerian states

State	Life Expectancy (year)	Adult Literacy (%)	Primary and Secondary enrolment rate (%)	Income per person (USD)
Average (36 states)	52	55.1	76	715
Bauchi	52	28	37.5	54
Yobe	52	38.9	33.5	72
Kano	53	26.7	46.5	157
Borno	54	18	34.0	129
Kaduna	51	48.0	73.5	266
Sokoto	53	39.7	28.8	299
Jigawa	50	50.5	23.6	150
Niger	58	60.2	47.0	675
Katsina	54	39.5	34.5	442
Kebbi	53	54.3	26.9	515
Gombe	51	69.1	31.1	38

Source: UNDP (2009) *Human Development Report Nigeria 2008-2009*, Table 3.1, and p.152.

Table 3.3. Gender Statistics in Sub-project States

State	Population 2010 Projected ¹¹		Life Expectancy		Adult Literacy		Primary & Secondary School Enrolment		Income per person (USD)	
	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male
Bauchi	2,616,508	2,714,425	52	47	28.0	49.2	37.5	59.4	54	277
Nasarawa	1,043,584	1,064,133	55	47	28.7	54.2	87.0	113.8	640	1,813
Kogi	1,850,380	1,886,193	50	46	43.7	67.3	109.1	127.9	88	209
Katsina	3,217,092	3,324,175	54	51	39.5	57.6	34.5	64.7	442	1,553
Sokoto	2,073,425	2,101,331	53	48	39.7	62.6	28.8	51.5	299	2,685
Jigawa	2,428,957	2,468,430	50	45	50.5	73.9	23.6	41.6	150	1,816
Kaduna	3,408,496	3,484,459	51	43	48.0	67.0	73.5	106.3	266	1,114
Plateau	1,790,869	1,781,360	47	43	72.2	88.3	81.7	134.3	122	267

Because each state has the authority to draft its own legislation, each state has different laws on gender issues. The project will investigate the gap between the state legislation vs the federal and international standards and propose recommendations at sub-project level.

3.3.1 Access to Energy and Gender-related Challenges

While rich in energy resources as well as being the largest oil producing country in Africa, Nigeria still has major challenges for its citizens' access to modern energy such as electricity, liquid and gaseous fuels, modern cooking options, and mechanical power. With 55.6% access¹², nearly half the population is un-electrified and most households, especially in the rural areas (only 33% access) use mainly biomass to cook. While access to modern energy services is documented as being a necessary, if not sufficient, condition for economic growth and development, Nigeria's energy sector has struggled with low investment, inadequate and ailing infrastructure, which resulted in shortages and consequent rise in costs, access and affordability to the vast majority of people,

¹¹ stmat.info/files/uploads/53129/annual_abstract_of_statistics_2011.pdf

¹² World Bank: World Development Indicators: Electricity Production, sources and Access.

in particular low income earners most of who are women, the vulnerable, society's marginalized and poor men.¹³

In Nigeria, as within the ECOWAS region, and most of Sub-Saharan African countries, there is heavy dependence on traditional biomass with 85% of people still cooking with charcoal and firewood. Biomass represents 57% of final energy consumed¹⁴. A large number of women and men, particularly those with low incomes or those living in rural areas are more disadvantaged than most in terms of their ability to access electricity and modern energy. In most of Sub-Saharan African countries, lack of access to electricity (apart from South Africa with 85.4%¹⁵ access rate) and other forms of energy adversely affects their businesses, more importantly deprives them of their right to enjoy a better standard of living with subsequent depression of the overall economy. These are the energy poor. Energy poverty – defined as the lack of adequate modern energy for the most basic needs of cooking, heating, lighting, income generation, community schools and health centres is limiting human development in many spheres, including but not limited to education, nutrition and food security, water and sanitation, market-based work, health, and gender relations.¹⁶

Figure 3.3 shows that better access to cleaner energy, cheaper (in terms of time and effort) and more efficient energy services comes with increasing level of productivity and invariably increasing income. This establishes the inverse relationship between energy poverty and gender productivity. “Energy poverty” disproportionately hurts women and children who are forced to spend up to 20 hours a week gathering biomass (wood, dung, crop waste) and drinking water, instead of going to school, learning, and bettering their situation.

Women and men experience energy poverty differently. Women, mostly the rural and urban poor, are at the bottom of the energy ladder, depicting the gendered nature of the experience of poverty with women bearing its worse consequences. More educated and affluent women can afford to use electricity or liquefied petroleum gas (LPG) for cooking. The use of inferior fuels (e.g. cow dung, twigs, paraffin, charcoal, firewood etc.), with very few alternatives for rural and urban poor women and men, more adversely affects women than men.

¹³ Country Gender Profile (draft), African Development Bank

¹⁴ Situation Analysis of Energy and Gender Issues in ECOWAS Member States 2015

¹⁵ World Bank: World Development Indicators: Electricity Production, sources and Access

¹⁶ Nigeria Country Gender Profile (draft), African Development Bank, 2016

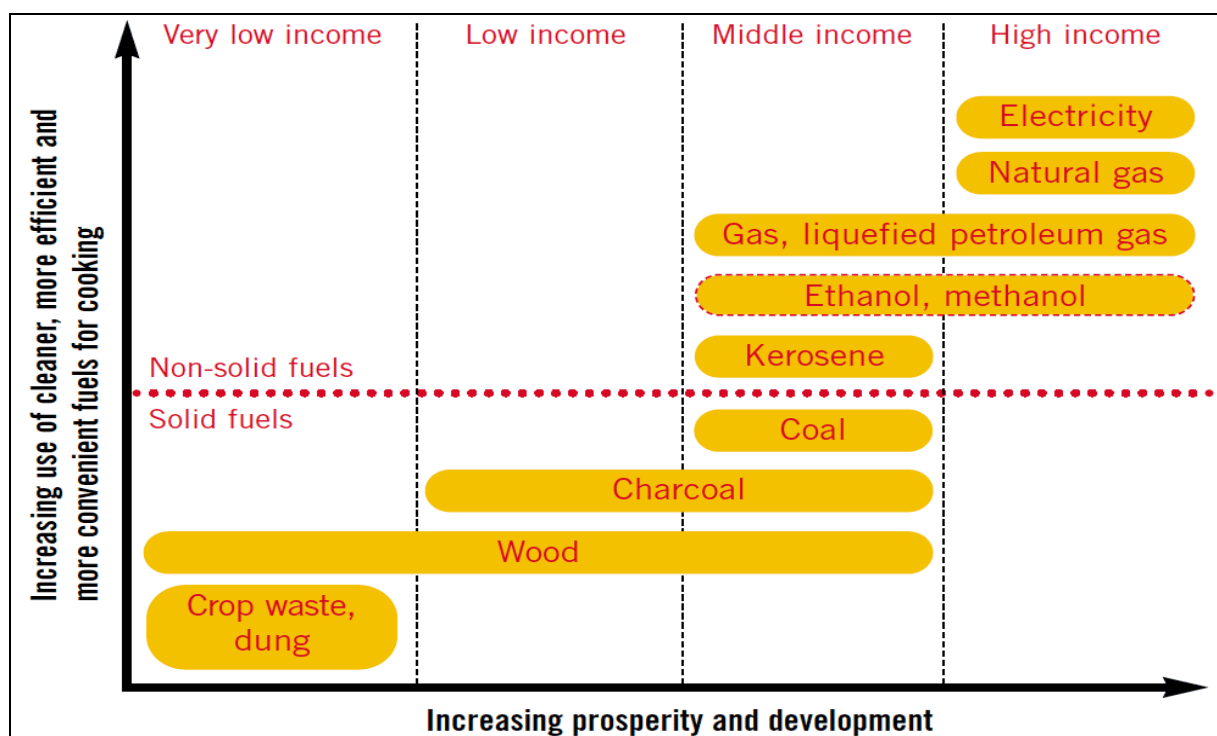


Figure 3.3. The energy ladder – Household energy and development inextricably linked

The primary source of energy for most of the rural people in Nigeria is biomass which may be sourced in some cases more than five kilometres away. This has a negative impact on women, not only in travelling long distances on foot but women, and their children, normally carry heavy loads of firewood on their heads, which can compromise physical health and wellbeing. Moreover, according to WHO reports, exposure to indoor air pollution is responsible for over 4 million deaths a year, worldwide, easily making it the most critical environmental killer humans face – more death than Malaria, HIV/AIDS, and lung cancer combined.

On energy (electrification) access: i) there is differential access by men and women to electrification, possibly because of socioeconomic status; ii) women and men have different preferences for off-grid lighting products and appliances; iv) in patriarchal societies of which Nigeria is one, men usually make buying decisions within the household with men going for 'luxury' type goods (i.e.: television) rather than domestic appliances that can help lessen the domestic chores of women. Also, men and women use electricity differently.

The mechanical power and productive uses of energy, even within the agricultural sector, is different for men and women as they have varying roles in farming and employ energy efficient technology differently. Men and women

grow different crops, partly because of culture, partly because of resource constraints and access to appropriate technology and extension services. More women are involved in manual or low-tech agro-processing than men. More men are involved in energy-intensive manufacturing than women, with better financial outcomes. However modest changes are occurring in response to changing socio-economic trends in the country.



Figure 3.4. Children studying – when there is reliable electricity Source: ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE) publications

Energy and development–gender implications: Women in electrified households are said to be more likely to find paid work outside of the home with no significant change for men. Men are reported to be better positioned to acquire and make use of mechanical power for productive use and, therefore, are more likely to increase their productivity and income. Access to electricity allows both men and women to engage in more income-generating activities. Modern cooking fuel and cooking appliances decrease exposure to harmful air pollutants for both men, women and children. In some cases, men are more at risk where they are responsible for cutting and burning wood for charcoal. There are better education outcomes with electrification, as school attendance increases for both male and female children but more for female children by releasing domestic labor constraints on women and girls. Electrification allows school children, both girls and boys, to study at night therefore increasing performance (see figure 3.5 above). Countries with better access have higher female literacy rates.

3.3.2 Women's Access to Climate Financing

"A growing emphasis on renewables is a response to both climate change and security of supply concerns". Electricity generation from renewable sources accounted for an average of 61.5% of the total final electricity consumption across the 4 selected countries¹⁷ (with Nigeria being highest at 87% and Senegal the lowest at 43%)¹⁸. There is such continuous pressure to produce more electricity from renewable energy resources than through fossil fuels.

As well, climate change impacts are adversely affecting women and girls. Lower rainfall levels, deforestation etc. requires women to intensify search for wood, fetch water at lengthier distances etc. This aspect will be examined in more detail within the scope of the 11 projects.

3.3.3 Vulnerable Groups

All 11 Solar IPP projects requires large expanse of land which are mainly in rural areas. It is a known fact that agricultural land use is predominant in rural areas of Northern Nigeria. What this means is that most of the people within the project areas will depend on agriculture and land-based resources for livelihood. Acquisition of land for this project without proper mitigation measures will expose some social groups to economic vulnerability. This might include women, widows, farmers, pastoralists and women heads of households, as well as the elderly and people with disabilities.

3.3.4 Indigenous Groups

Indigenous peoples, also known as first peoples, aboriginal peoples or native peoples, are ethnic groups who are the original settlers of a given region, in contrast to groups that have settled, occupied or colonized the area more recently. According to the World directory of minority and indigenous people in Nigeria published by the Minority Group International in May 2018, the main minority and indigenous people are Hausa / Fulani (29 %); Yoruba (21 %); Igbo (Ibo) (18 %), Ijaw (10 %), Kanuri (4 %), Ibibio-Efik (3.5 %), Tiv (2.5 %), Edo (Bini) (less than 1 %), Nupe (less than 1 %). In the last census conducted in 2006. The country did not then, and does not now, collect or analyze data disaggregated by ethnicity, religion or language. As such, there are no Indigenous Peoples in Nigeria per the AfDB standards and will therefore not be covered within the scope of the ESMF. However, it is recommended that the proponent investigate this issue further and confirm in final documentation.

The Land Use Act of 1978 which vested the ownership of land to the state and kept in trust by the governor of the state will not require an Indigenous Peoples

¹⁷ Large-scale hydroelectric power contributed the highest share of electricity generation.

¹⁸ <http://gtf.esmap.org>

Plan for the sub-projects in Northern Nigeria. This is because the process of land acquisition for projects and sub-projects are usually started with state government with the involvement of the Ministry of Land who oversees the process of traditional acquisition through the compensation process.

3.4 The United Nations and Gender Considerations in Nigeria

3.4.1 The United Nations and UN Sustainable Development Goals

The 2030 United Nations Sustainable Development Goals (SDGs) Agenda is clear: there can be no sustainable development without gender equality. *Turning promises into action: Gender equality in the 2030 agenda*, a global monitoring report by UN Women, asks: How far have we come in turning this new development consensus into results for women and girls and what is needed to bridge the remaining gaps between rhetoric and reality?

Women and girls around the world must have equal rights and opportunity and be able to live free of violence and discrimination. Women's equality and empowerment is one of the 17 Sustainable Development Goals which is also integral to all dimensions of inclusive and sustainable development. In short, all the SDGs depend on the achievement of Goal 5: Achieve gender equality and empower all women and girls. SDG Goal 7, Ensure access to affordable, reliable, sustainable and modern energy for all, is a dimension which this Program will embed within the gender domain.

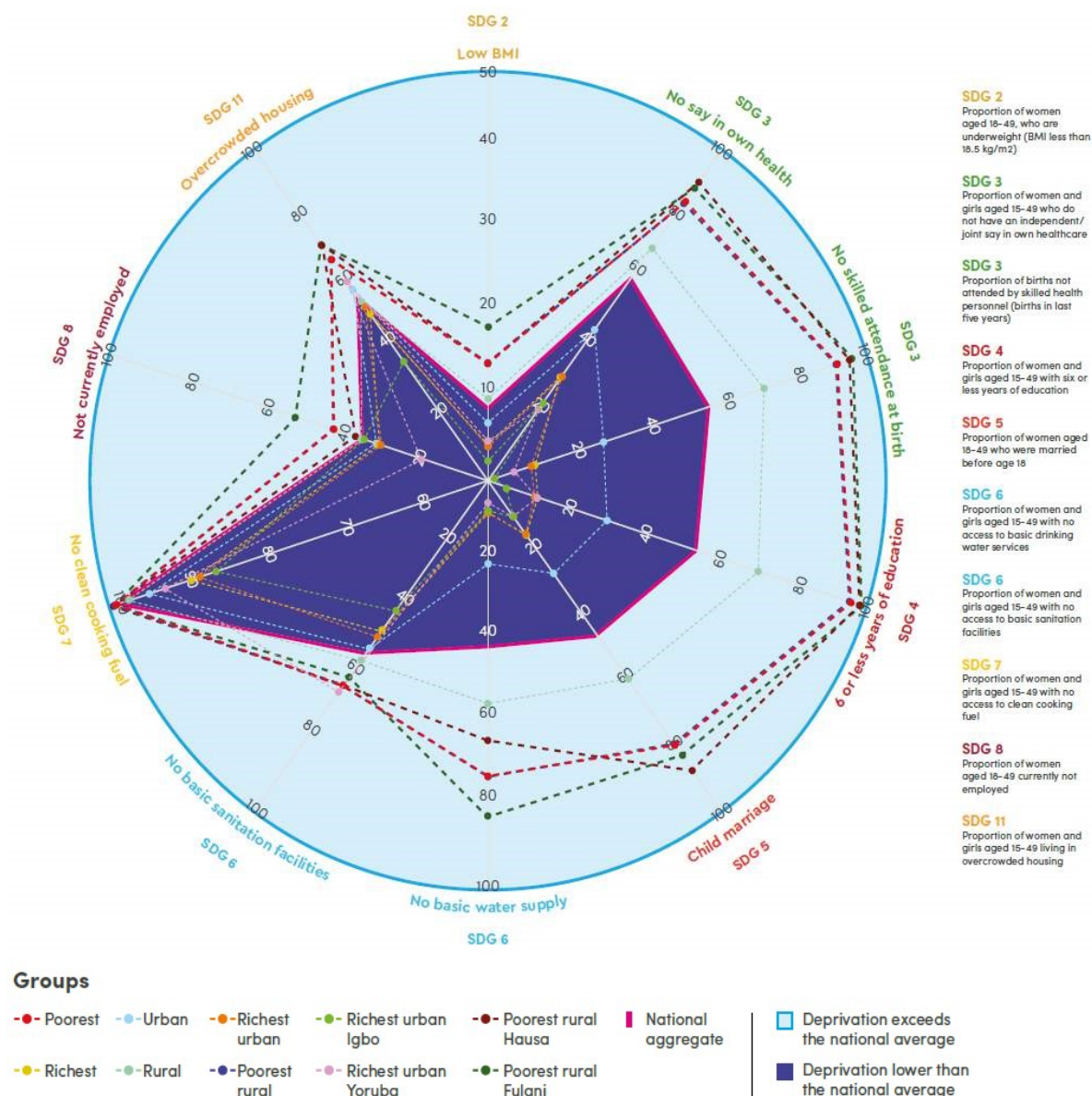
The United Nations' gender equality goal requires urgent action to eliminate the many root causes of discrimination that still curtail women's rights in private and public spheres by 2030. This is around discriminatory laws, eliminating gender-based violence, equal distribution of economic resources and sexual and reproductive rights. In the 2017 Progress Report, the UN noted that "*gender inequality persists worldwide, depriving women and girls of their basic rights and opportunities. Achieving gender equality and the empowerment of women and girls will require more vigorous efforts, including legal frameworks, to counter deeply rooted gender-based discrimination that often results from patriarchal attitudes and related social norms*"¹⁹

¹⁹ UN Economic and Social Council (2017). [Progress towards the Sustainable Development Goals: Report of the Secretary-General](#) (E/2017/66).

In the UN Report, a spotlight is put on gender-based discrimination across all 17 SDGs (Figure 2.1). Note:

- The goal-by-goal review shows that gender inequalities—deeply rooted and present across all countries, including Nigeria—are pervasive in every dimension of sustainable development. Using micro-level data, the report shows how across countries, women and girls who experience multiple and intersecting forms of discrimination are often left behind and excluded from progress.
- These groups of women and girls are excluded not only from access to education and health, but also other key measures of well-being including access to clean water, fuel and housing. Addressing these challenges requires dramatic advances in statistics, financing and policies for gender equality. The diagram below demonstrates the situation facing women in Nigeria.

The UN further reported that across countries, it is those women and girls who experience multiple and intersecting forms of discrimination who are often the furthest behind. To identify such inequalities among women and girls within countries, the report undertakes analysis across a range of development dimensions for four countries, finding that those who are deprived in one dimension are likely to experience deprivations in other dimensions as well.



Source: UN Women calculations based on microdata from the 2013 Nigeria DHS (NPC, Federal Republic of Nigeria and ICF International 2014).

Notes: Different scales are used across each of the 10 axes, each corresponding to the maximum and minimum values for each given indicator. Select groups are shown given space limitations. For full group disaggregation, see Annex Table 3.

Figure 3.5. Diagram 1: Inequalities in SDG-related outcomes for different groups of women and girls, Nigeria, 2013

For the case of Nigeria, the report finds:

- **Education.** Wealth is a driving force behind educational attainment: 13% of women and girls from the richest households' report completing six or less

years of education, while in the poorest households, 96.5% are education-poor.

- **Child marriage.** A low-income, rural woman of Hausa ethnicity is eight times as likely to be married before the age of 18 as a high-income, urban woman of Yoruba ethnicity.
- **Violence against women and girls.** 23% of women have been victims of physical or sexual violence by a previous husband and 17.3% by another relative. Igbo women and girls are the likeliest to report being victims of violence at the hands of a relative: 25.2%.
- **Clustered deprivations.** 15% of all women aged 18–49 (or 5.2 million women) are simultaneously deprived in four SDG-related dimensions. These women were not only married before the age of 18 and education poor, but they also reported no agency in healthcare decisions and claimed to be unemployed at the time of the survey.

3.4.2 UNICEF's Gender Related Actions and Impacts in Nigeria

The section below details areas in which UNICEF works actively to produce gender significant impacts in Nigeria. The work being undertaken by UNICEF reflects the typical type of gender related issues encountered in Northern Nigeria where most of the projects under consideration are located. Considering that organizations such as UNICEF work across States, their specific areas of interventions could help inform the types of interventions that could be undertaken at subproject-specific levels.

3.4.2.1 Provision of Basic Education

Nigeria still has 10.5 million out-of-school children - the world's highest number. 60% of those children are in northern Nigeria. About 60% of this number of out-of-school children are girls. Many of those who do enrol drop out early. Low perceptions of the value of education for girls and early marriages are among the reasons. Some northern states have laws requiring education of girls and prohibiting their withdrawal from school. Girls' primary school attendance has been improving, but this has not been the case for girls from the poorest households.

Increased enrolment rates have created challenges in ensuring quality education, as resources are spread more thinly. UNICEF supports efforts to

increase equitable access to quality basic education and to improve learning achievements. Special emphasis is given to girls' education. The Education Program focuses on three main areas:

- **Strengthening Education Systems** - UNICEF helps improve planning, budgeting and monitoring, and assess learning outcomes and teacher competency. It also helps build the capacity of School Based Management Committees.
- **Improved and equitable access to quality basic education** - UNICEF supports early childhood development in disadvantaged communities, and helps traditional Koranic schools adopt a broader curriculum. Cash transfers are being piloted in Sokoto and Niger states to help families afford the cost of sending girls to schools, while scholarship incentives help attract qualified female teachers to rural schools.
- **Improved quality of teaching and learning outcome** - UNICEF helps teachers improve their skills, promotes quality standards for child-friendly schooling, and supports capacity building for emergency preparedness and response.

3.4.2.2 Social Policy Research and Communication

UNICEF supports efforts to strengthen safety nets, research on the situation of women and children, and communication for development.

- UNICEF supports initiatives such as cash transfers for girls' education and in-kind support to access health services. It also promotes support for children with disabilities, as well as youth employment schemes.
- UNICEF has supported the drafting of the Gender Economic Opportunity bill as well as the Violence Against Persons Prohibition bill. It has also supported state and federal level gender reviews.
- UNICEF has participated actively in the design and drafting of a national social protection policy. It also provides technical advice for its domestication and implementation at the state level.

3.4.2.3 Water, Sanitation and Hygiene

Many schools in Nigeria lack safe, private toilets and hand-washing facilities. This affects enrolment and performance, particularly in the case of girls. The impact of water, hygiene and sanitation falls disproportionately on women and girls, the main carriers of waters. UNICEF aims to increase sustainable access to and use of improved water sources, sanitation facilities and hygiene practices. Over the

years, UNICEF contributions have helped millions of people gain access to improved water sources and sanitation however a lot of work remains to be done.

1. UNICEF is a key ally in the government's efforts to eliminate open defecation by 2025.
2. The Program promotes low-cost, community-based approaches, climate change adaptation and greater engagement of women.
3. UNICEF supports the provision of child and gender-friendly WASH in schools and primary health institutions.
4. The WASH Program is an essential component in integrated Programs for the community management of acute malnutrition and in efforts to scale up Maternal, Newborn and Child Health services.

3.4.2.4 Health

UNICEF strategies include scaling up of maternal and neonatal services in high-burden communities and conducting health weeks to provide high-impact interventions, including routine immunizations, care for pregnant women, birth registrations, promotion of handwashing and distribution of mosquito nets.

3.4.2.5 HIV and AIDS prevention and Control

- Activities dealing with HIV-AIDS are included in most of UNICEF Nigeria Programs and are focused on the elimination of mother-to-child transmission, care and treatment of children, women and adolescents living with HIV, and support for HIV prevention among adolescents.
- Interventions take place at federal and state level and include policy development, strengthening of health services and community mobilization. This includes support of community groups and volunteers to identify pregnant women in their communities and educating them about attending antenatal care (ANC) services. During ANC, pregnant women get an HIV test, and receive ARVs if they test HIV positive.
- Integration of HIV-testing of pregnant women during maternal and child health weeks: HIV positive pregnant women are referred for PMTCT services.

The projects which have been quoted, and in general the work being done by UNICEF is wholly on women and children and as such UNICEF will possess sex-disaggregated data that could be useful when preparing the sub-project specific Gender Action Plans.

Although UNICEF is used as an indicative example, other CSO interventions and projects related to gender in the subproject areas will be investigated when developing Subproject-specific Gender Action Plans. At this stage of the Program and where required, sub-projects will build on synergies obtained from the work and interventions by UNICEF and other similar organizations.

3.5 Conclusions on the Gender Assessment

The participation of women into all levels of the energy value chain will lead to more effective clean energy initiatives, unlock greater return on investments, and expand the prospects of reducing emissions. What is required at this time is the implementation of all appropriate regulatory measures to ensure an institutional framework for women to conduct business. Without this legal protection, many of the attempts to integrate or mainstream gender in the renewable energy domain will be futile. Therefore, at the sub-project levels, consultations with stakeholders would include institutional authorities (federal and state) to design and mainstream gender considerations into the sub-projects.

The women of Nigeria have the potential to unlock an incredible opportunity if given the opportunity. Women can increase growth now and for the future, give women more control over their lives and those of their children, and foster investment in skills and health of children. Jobs define much of who we are and how we live, and when Nigerian women enter work, they are likely to develop a stronger say in their own destiny. Promoting women's access to gainful employment can unleash a strong force for innovation, productivity, and economic growth. With income opportunities also comes more control over household resources, and there is evidence that women are more likely than men to invest resources in children's health and education. This is good for long-term economic growth, as well. All these changes are needed as Nigeria tries to move towards a more diversified and inclusive economic growth.

It also documented that the work ethic held by women in Nigeria tend to be stronger than those of men. Of the permanent jobs which would be created under the IPP Solar Projects, it would be worthwhile to investigate how most women can be employed for these RE projects. Furthermore, it has also been documented that for the same job, women earn less than their male counterparts. Adjusting the salary scales to be on par for women in Nigeria would increase productivity on both the local and national levels.

Women have an important share in the IPP Solar energy market as users of improved off-grid electricity. Projects seeking to expand energy access through off-grid solutions can improve women's productivity, alleviate time poverty and in turn enable women to engage in paid work or start their own businesses. Several studies have found that energy projects that have incorporated gender issues at

the project design stage have 'greater end-user acceptability, initiated more income-generating activities, and increased input and output in entrepreneurial activities' (Alstone et al., 2011).

As part of the sub-projects, specific and context specific gender assessments and consultations will be undertaken to inform the development of the sub-project-specific Gender Actions Plans, which will be aligned with the vision, set in the Funding Proposal.

4 GENDER CONSIDERATIONS IN SUB-PROJECT STATES

4.1 Gender Characteristics and Key Indicators across the Sub-project States

As shown in Section 3.1, there are currently 11 sub-projects being considered under the Nigerian Solar IPP Support Program. The 11 sub-projects are mainly located in the northern parts of Nigeria, across 8 states namely, Sokoto, Katsina, Jigawa, Bauchi, Kaduna, Plateau, Nasarawa and Kogi States.

Of the 11 sub-projects (across 8 States) under consideration, a decision on which specific sub-project to be funded under the Program is yet to be made. Substantive consultations with stakeholders as well as the identification of specific gender related data will be collected when the sub project specific Gender Action Plans are prepared. However, as part of the preparation of the current document, desktop research was undertaken to identify specific gender related indicators, issues per State, as documented in published literature and documents.

This section of the report summarises the outcomes of the desktop study, focusing on the context of key gender issues in the States where sub-projects are located.

4.1.1 Gender dis-aggregated data in the States with Sub-projects

A comparison between male and female statistics on specific indicators directly associated with women (population, life expectancy, adult literacy, education and income) is presented in Table 3.3 below.

Table 4.1 Gender dis-aggregated data in the States with sub-projects

State	Population Projected ²⁰ 2010		Life Expectancy		Adult Literacy		Primary & Secondary School Enrolment		Income per person (USD)	
	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male
Bauchi	2,616,508	2,714,425	52	47	28.0	49.2	37.5	59.4	54	277
Nasarawa	1,043,584	1,064,133	55	47	28.7	54.2	87.0	113.8	640	1,813
Kogi	1,850,380	1,886,193	50	46	43.7	67.3	109.1	127.9	88	209
Katsina	3,217,092	3,324,175	54	51	39.5	57.6	34.5	64.7	442	1,553
Sokoto	2,073,425	2,101,331	53	48	39.7	62.6	28.8	51.5	299	2,685

²⁰ stmat.info/files/uploads/53129/annual_abstract_of_statistics_2011.pdf

Jigawa	2,428,957	2,468,430	50	45	50.5	73.9	23.6	41.6	150	1,816
Kaduna	3,408,496	3,484,459	51	43	48.0	67.0	73.5	106.3	266	1,114
Plateau	1,790,869	1,781,360	47	43	72.2	88.3	81.7	134.3	122	267

²¹ Source: Country Gender profile Nigeria. JICA, 2011

4.1.2 Key Gender Issues per State

A summary of the key genders likely to be encountered in each State where the sub-projects are located is presented in the table below.

4.2 Gender Committee

To undertake gender mainstreaming activities within the scope of these projects, it is recommended to establish a Gender Committee. The Committee would be charged to monitor the gender baseline study and provide a more detailed roadmap to identify the various interventions required to ensure that this project becomes gender-mainstreamed. One of the expected results, for example, is for the sub-projects to develop a Gender Mainstreaming Policy, Sexual Harassment and Discrimination Policy and Equal Opportunity Policies. These can be developed together with the environmental and social management systems (ESMS).

The Committee can also include external members, such as a representative from the Federal Ministry of Women Affairs (FMWA) and state/ local representatives from the project states.

One of the first steps the Committee can undertake at the onset of the project is to determine recruitment strategies that have a strong emphasis on how the sub-project can overcome challenges faced by women wanting to work in IPP's hubs located in rural, remote areas.

²¹ https://www.jica.go.jp/english/our_work/thematic_issues/gender/background/pdf/e10nig.pdf

Table 4.2 Summary of key gender related issues in each of the States with Sub-projects

State	Project	Capacity	Gender Issues
Sokoto	KVK Power Limited	100MW	1. Land is mainly owned by men 2. Most rural women do not receive the same level of education as male peers. 3. Women need to build on advocacy skills and be empowered to take messages about education (especially for girls), better health services, women's policy priorities in the local government budgets²². 4. Family planning and number of children are challenges for women²³ 5. In 2017, The Sokoto State Government said it allocated over N102 million for the welfare of women, children and vulnerable groups in 2017 budget and had successfully established partnerships with several organizations (e.g. UNFPA, UNICEF, UNDP and USAID) to support gender empowerment programs²⁴. 6. Establishing women centers for continuous education has been helping increase literacy rates²⁵
Jigawa	Nova Scotia Power Development Ltd	80MW	1. Ownership of land is exclusive right of men 2. Women especially in the rural areas of Jigawa are culturally restrained from public accessibility thus making it difficult for women to participate in decision making and planning in their communities. 3. The lack of resources and power in the hands of women has significantly contributed directly to the inability of girls to attend and succeed in school. 4. Higher adult illiteracy, unemployment and poverty among women
	Oriental Renewable Solution	50MW	
Kaduna ²⁶	Anjeed Innova Group	100MW	1. Land ownership is seen as exclusive rights of Men. This indicates that compensation for land will mainly go to men as land owners while only compensation for tenants (land users) may go to the women. 2. Unemployment rate is higher among female than male 3. Adult literacy is higher among male than female 4. Primary and Secondary schools' net enrolments are higher in male than Female.
	EN Africa	50MW	
	Quaint Abiba Power Ltd	100MW	

²² <https://www.ndi.org/our-stories/rural-womens-activism-shakes-local-governance-northern-nigeria> [Accessed 12/12/2018]

²³ https://healthcommcapacity.org/wp-content/uploads/2015/08/HC3-FP-Assessment-in-Nigeria-Revised-24_Jul15.pdf [Accessed 12/12/2018]

²⁴ <https://www.pulse.ng/news/local/aminu-tambuwal-sokoto-state-govt-allocates-n102m-for-women-children-in-2017-commissioner-id6732475.html> [Accessed 12/12/2018]

²⁵ http://www.millennia2015.org/files/pubs/90/M15_2008_04_W1_4_Ngozi_Patience_Nwosu.pdf [Accessed 12/12/2018]

²⁶ http://www.sparc-nigeria.com/RC/files/1.1.27_Kaduna_State_Government_GSI_Guideline_July_2014.pdf [Accessed 12/12/2018]

Kogi	Middle Band Solar One Ltd		1. Ownership of land is exclusive right of men 2. With enhanced education, women are generally better at adopting more productive land management practices²⁷. Men are more influenced by land size. 3. Women generally have a subordinate role in society²⁸ and there is a need for more empowerment programs
Katsina	Pan African Solar	75MW	1. Factors limiting the education of the girl child (family background, religious isolation, disability, early marriage and pregnancy, gender-driven violence, cultural discrimination and attitudes against women's status and role). These apply to most northern states in Nigeria²⁹ 2. More men have access to land than women 54% to 46% male female access to land 3. Ownership of land is exclusively for men while land owned by females are held in trust by men
Plateau	CT Cosmos	70MW	1. Ownership of land is exclusive right of men 2. The Plateau State government has gazetted the Gender Equal Opportunities Law (GEO), passed since 2015 in the State, making it the first in Northern Nigeria
Nasarawa	Afrinergia Power Ltd	50MW	1. Commercial poultry production is of interest to women and can be used to empower them and their role in the community. Women generally have poorer access to resources for commercial activities³⁰ 2. Female participation is frequent in crop postharvest activities and poultry management³¹ while male participation was frequent in crop pre-harvest operations only 3. Women are discriminated upon during recruitment and employment opportunities³² 4. Female Genital Mutilation³³ and female empowerment
Bauchi	Nigeria Solar Capital Partners	100MW	1. Family planning and number of children are challenges for women³⁴ 2. Cultural practices reduce access to education for the girl child³⁵ (applicable across most States in Northern Nigeria) 3. Gender based violence in homes³⁶

²⁷ <https://www.ajol.info/index.php/as/article/view/173250> [Accessed 12/12/2018]

²⁸ http://ijhsnet.com/journals/ijhs/Vol_2_No_2_June_2014/2.pdf [Accessed 12/12/2018]

²⁹ <http://article.sapub.org/10.5923.j.jamss.20170302.03.html> [Accessed 12/12/2018]

³⁰ <http://www.lrrd.org/lrrd22/9/okoh22160.htm> [Accessed 12/12/2018]

³¹ [http://www.aessweb.com/pdf-files/2-22-2\(1\)2012-AJARD-10-16.pdf](http://www.aessweb.com/pdf-files/2-22-2(1)2012-AJARD-10-16.pdf) [Accessed 12/12/2018]

³² <https://iirdpub.org/get/IJSSMR/VOL.%203%20NO.%207%202017/IMPLICATION%20OF%20GENDER.pdf> [Accessed 12/12/2018]

³³ <https://www.iiste.org/Journals/index.php/RHSS/article/view/7918/7671> [Accessed 12/12/2018]

³⁴ https://healthcommcapacity.org/wp-content/uploads/2015/08/HC3-FP-Assessment-in-Nigeria-Revised-24_Jul15.pdf [Accessed 12/12/2018]

³⁵ <https://nau.edu/uploadedFiles/Academic/COE/About/Projects/Persistent%20Gender%20Inequality%20in%20Nigerian%20Education.pdf> [Accessed 12/12/2018]

³⁶ <https://www.vanguardngr.com/2017/12/gender-violence-bauchi-magistrates-promotes-counseling/> [Accessed 12/12/2018]

			4. Access to land and resources ³⁷ limited for women resulting in poverty
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4.3 Stakeholder Engagement and Consultation

As part of the sub-projects, broad based consultation on gender specific issues will be undertaken with a number of international and local CSOs to identify specific focus areas that will contribute towards mainstreaming positive gender action into the individual sub-projects considered for funding.

This section of the report summarizes some of the consultation undertaken during the preparation of this GAP and list some key organizations that may be consulted as part of the preparation of sub-project specific gender assessment and gender action plans.

4.4 Relevant Stakeholders

A selection of Non-Governmental Organisations (NGOs) and CSOs operating in the northern part of the Nigeria and within the states where the sub-projects are located is presented in Table 4.1. To note, this list is not exhaustive and more detailed consultation will be undertaken at sub-project level to identify additional NGOs and CSOs working on gender related interventions in each State and sub-project area. Developing synergies with existing organizations in the project area will assist in defining specific goals and targets for the implementation of the Gender Action Plans for each sub-project considered under the Program.

Table 4.1. Potential NGOs and CSOs to be engaged with at the State level

State	Project	Capacity	NGOs and CSOs		
			Local	International	National
Sokoto	KVK Power Limited	100MW	Save the Child Initiative	Action Aid Nigeria	Development Initiative for Community enhancement (Mandate covers purely women and girl child Programs). Based in Abuja but covers the whole country.
			Life Helpers Initiative (LHI), a Sokoto-based civil society organization (CSO) known for health, education and socio-economic interventions	"Save Children" UNICEF DFID	
					Centre for Community Excellence (Mandate covers community development and

³⁷[https://www.researchgate.net/publication/305765984 WOMEN PARTICIPATION IN HOUSING DELIVERY BAUCHI METROPOLIS IN PERSPECTIVE](https://www.researchgate.net/publication/305765984_WOMEN_PARTICIPATION_IN_HOUSING_DELIVERY_BAUCHI_METROPOLIS_IN_PERSPECTIVE) [Accessed 12/12/2018]

State	Project	Capacity	NGOs and CSOs		
			Local	International	National
Jigawa	Nova Scotia Power Development Ltd	80MW			gender issues. Based in Zamfara state and covers the entire Northern Nigeria. Women Environmental Program (WEP) the mandate is to address gender injustices on issues relating to the environment and economic and social rights of women and young people. Empower to Empower Initiative. Based in Abuja but also covering Northern Nigeria
	Oriental Renewable Solution	50MW			
Kaduna	Anjeed Innova Group	100MW	I Care Women and Youth Initiative		
	EN Africa	50MW	Hope for Community and Children Initiative		
	Quaint Abiba Power Ltd	100MW	TBC		
Kogi	Middle Band Solar One Ltd	100MW	Participation Initiative for Behavioral Change in Development (PIBCID)		
Plateau	CT Cosmos	70MW	Vulnerable and Abused Intervention Initiative		
Nasarawa	Afrinergia Power Ltd	50MW	Hope Rising for the less Privileged and Vulnerable Foundation		
			Nasarawa State Ministry of Women Affairs, Association of Smallholder Agro-Producers in Nigeria (ASSAPIN)		
Bauchi	Nigeria Solar Capital Partners	100MW	Mother and child Care Enhancement Foundation (McCEF)		
			Bauchi State Network for CSOs (BASNEC) Fahimta Women and Youth Development Initiative (FAWOYDI)		

Some of the gender related issues covered by these NGOs and CSOs include:

1. **Low school enrolment for females, early marriage:** Women are not considered much in employment, but the situation is different in Sokoto state where affirmative action is being enforced.
2. **Social seclusion with restriction on women to freely express themselves:** The concept of (*Baa Shiga*) restriction of access into compounds because of the women and girls. Men are not restricted in social life, but women are.
3. **Restriction to adequate health care:** Some men insist that Female Doctors only can attend to their wives or that they must be present during the consultations. This is argued from a cultural background.
4. **Restricted access to women in terms of decision making:** On issues like family planning and sexual practices.

It is expected that all of these organizations, as well as others that have not yet been identified, will be engaged with during the development of sub-project specific GAPs, to ensure effective gender mainstreaming and monitoring into the sub-projects.

4.5 Stakeholder Engagement Activities during the preparation of the ESMF

4.5.1 Stakeholders Consulted

As part of the preparation of the Environmental and Social Management Framework (ESMF), stakeholder engagement activities were undertaken with a number of stakeholders at Federal, Regional and local level. Consultation was also undertaken with several CSOs. The CSOs were selected based on their core areas of interest in renewable energy and positive climate action in Northern Nigeria.

As part of the stakeholder engagement process, consultation was undertaken with the entities set out below.

Table 4.2. Specific details of stakeholder engagements undertaken

S/N	Entity	Date	Person Contacted	Designation
1	Federal Ministry of Environment (FMEnv)	04/12/2018	Mr. J.A. Alonge	Director EA Department
		03/12/2018	Mr. Abass Suleiman	Deputy Director EIA
		05/12/2018	Mrs. Odetoro	Deputy Director
2	Jigawa State Ministry of Environment	03/12/2018	Alhaji Aliyu Hadejia	Assistant Director

S/N	Entity	Date	Person Contacted	Designation
3	Dutse Local Council, Jigawa State	03/12/2018	Ayuba Ahmed	Director of Works
4	Invest Jigawa	04/12/2018	Mrs. Jamila Farouk	Director
5	Kogi State Ministry of Environment	04/12/2018	Mr. Samson Benu	Chief Environmental Officer
6	Lokoja Local Council	04/12/2018	Mohammed Bochi Lawal	Director Agriculture & Environment
7	Kogi Investment and Properties Ltd		Mrs. Obadaki	Ag. Managing Director
8	Kaduna Environmental Protection Agency	04/12/2018	Dr Yusuf Rigasa	Director General
9	Nigerian Electricity Management Services Agency	04/12/2018	Mr. John Obajimi	Health and Safety Officer
10	Transmission Company of Nigeria	05/12/2018	Mrs. C.I.B Sako	Environmental and Social Specialist
11	Renewable & Rural Power Access Department, Federal Ministry of Power, Works and Housing	05/12/2018	Engr. Ajeigbe A. Jacob	Assistant Director
12	Empower to Empower	05/12/2108	Ms. Joy Ini	Program Coordinator

Specific consultation was not undertaken with Project Affected Peoples (PAPs) at project level because this process will be carried in subsequent phases for the sub-projects. In addition, avoiding consultations with PAPs from any one site of the potential sub-projects will eliminate bias.

A summary of the outcomes of the engagements undertaken with each of the stakeholders including dates and communication means is presented Table 4.3 below.

Table 4.3. Summary of consultations undertaken during the development of the ESMF related to gender

S/N	Entity	Date	Person Contacted	Means of Communication	Discussion Objectives on Gender related Issues of Interest
1	Federal Ministry of Environment (FMEnv)	04/12/2018	Mr. J.A. Alonge.	Face-to-Face meeting	- Land acquisition for the project, issues of compensation and livelihood restoration, the need to engage the PAPs from project design and site selection. - Allowing PAPs to participate in determination of mitigation for social issues through consultations. - For awareness across board. - Implementation of Extended Producers Responsibility and Individual Producer responsibilities. - The role of regulators at National, State and Local Levels should be considered and capacity building for monitoring and support should be considered for agencies regulating the projects.
		03/12/2018	Mr. Abass Suleiman	Face-to-Face meeting	
		05/12/2018	Mrs. Odetoro	Face-to-Face meeting	
2	Jigawa State Ministry of Environment	03/12/2018	Alhaji Aliyu Hadejia	Telephone Conversation	Main issue has to do with land take and proper identification of land owners as well as transparent system of compensation
3	Dutse Local Council, Jigawa State	03/12/2018	Ayuba Ahmed	Telephone Conversation	Compensation for land taken for the project
4	Invest Jigawa	04/12/2018	Mrs. Jamila Farouk	Telephone Conversation	Involvement of traditional institution and adequate consultation with PAPs at project design state to avoid frictions over land take for solar projects
5	Kogi State Ministry of Environment	04/12/2018	Mr. Samson Benu	Telephone Conversation	Issues surrounding land acquisition for the projects should be handled by project developers completely before government allocation and granting of title
6	Lokoja Local Council	04/12/2018	Mohammed Bochi Lawal	Telephone Conversation	Issues surrounding the involvement of local council in compensation for land as well as involvement of local government in E&S monitoring because they are the closest tier of government to the projects
7	Kogi Investment and Properties Ltd	04/12/2018	Mrs. Obadaki	Telephone conversation	Loss of farmland should be properly compensated and alternatives provided

S/N	Entity	Date	Person Contacted	Means of Communication	Discussion Objectives on Gender related Issues of Interest
8	Kaduna Environmental Protection Agency	04/12/2018	Dr Yusuf Rigasa	Telephone Conversation	Environmental concerns like pollution should be considered in the choice of materials most especially the PV modules as well as transformer oil.
9	Nigerian Electricity Management Services Agency	04/12/2018	Mr. John Obajimi	Face to Face Meeting	- Considerations of Health and Safety in project design and implementation as well as in choice of electrical installation. - Ensuring that safety standard codes are adhered to.
	Transmission Company of Nigeria	05/12/2018	Mrs C.I.B Sako	Face to Face Meeting	Adherence to Environmental issues of concerns as documented in the SESA that is being produced by TCN
10	Renewable & Rural Power Access Department, Federal Ministry of Power, Works and Housing	05/12/2018	Engr. Ajeigbe A. Jacob	Face to Face Meeting	Conduct of Environmental Impact Assessment for the projects should be made compulsory.
11	Empower to Empower	05/12/2108	Ms. Joy Ini	Telephone Conversation	Women should be considered in project implementation and CSR should from projects focus on women development

Once sub-projects to be considered for the Program have been selected, AFC as the Accredited Entity shall ensure that stakeholder consultation undertaken as part of project-level gender assessments will include local communities and CSOs and are used to identify energy needs and priorities of women and men that the projects can address.

5 GENDER ACTION PLAN

5.1 Overall Objective of the GAP

Based on the Gender Assessment undertaken in the previous chapter, a GAP has been developed, which will be implemented as part of the Nigeria Solar IPP Support Program. It is expected that for all selected sub-projects, a specific Sub-project Gender Action Plan will be prepared, with measurable metrics to be used throughout the life of the project.

Table 5.1 outlines potential targeted initiatives which the project proponents will be encouraged undertake throughout the life of the solar IPP projects, to achieve gender objectives for the Program. This is an indicative list and will be refined once on-the-ground information and facts are collected, especially about project acceptance by local population and economic viability.

It should be noted that the targets and goals in the table below are overarching and aspirational goals. As part of the Program, more project-level consultation will be undertaken with relevant stakeholders, so that specific targets can be set for each subproject eventually selected for the Program. The specific indicators and targets will be detailed in Subproject-specific Gender Action Plans. As part of the implementation of the selected subprojects, the AFC will ensure that subproject-specific GAPs contain sex-disaggregated targets that are rationalized by baseline information collected before the beginning of each project. At this stage, the final cost estimates for the implementation of the specific aspirational goals have not been stated, as the subprojects are yet to be specified and costs in the table below are indicative.

While women's empowerment, a focus on women's empowerment in renewable energy developments can further enhance the effectiveness and sustainability of these investments.

Finally, it will be vital to gather specific gender data relevant to subprojects, to design context specific strategies, policies and Programs that are able to effectively address gender issues.

5.2 Responsibilities for Implementing the GAP

AFC, as the Accredited Entity, will have overall ownership of the program level GAP and will ensure that it is implemented at sub-project level. This will include ensuring that the sub-projects selected for funding under the program develop subproject specific GAPs with baseline data and specific performance metrics.

Project-level GAPs will be implemented by sub-project Sponsors, who will be directly responsible for achieving project-level gender-based objectives. Sub-

project sponsors will be required to submit reports of project-level GAP implementation and achievement of objectives, to the Accredited Entity.

At the Program level, the Accredited Entity will be responsible for collating project-level GAP implementation information and will monitor and report (to GCF), on the implementation of the Program-level GAP and achievement of indicators and targets. This will be done through the following reports:

- i. Evaluation Reports
- ii. Annual Performance Reports
- iii. Semi-Annual Performance Reports

5.3 Gender Action Plan

Table 5.1. Proposed Gender Action Plan for the Program (below)

	Expected Program Output	Gender-related Issue	Justification	Activities	Indicator	Target	Responsible Organization	Indicative Cost
1	Increased number of households, and individuals (males and females) with improved access to low-emission energy sources;	Access to Energy	It is assumed that the Program, by generating relatively cleaner solar power, will increase access to low-emission energy sources	Surveys to determine baseline numbers for access to high emission energy	Number of households and individuals (disaggregated into male and female) with improved access to low-emission energy sources	A minimum increase of 15% of households (in each sub-project area) with increased access to low energy sources.	IPP sub-project developer (SPV)	Approx. 20,000USD per sub-project
2	Increased Number of Jobs created for women during Construction and Operation phase of sub-projects	Direct Job Creation	Equal employment opportunities should be offered to both men and women, during the construction and operational phases of sub-projects	- Require Sub-Project Sponsors to set minimum targets for employing women in construction and operation. - Collaborate with targeted CSOs on various education projects and initiatives.	Percentage of women employed in each sub-project	200 permanent jobs to be created in the Operation and Maintenance phase Alternatively, a minimum of 5% of all construction and operation jobs to be provided to women	IPP sub-project developer (SPV)	No additional cost

		Indirect Job Creation and Entrepreneurship Development	There is a limited number of direct jobs which can be created by sub-projects. It is expected that indirect job and employment opportunities will be created through sub-projects by increased access to energy.	Monitor statistics for women-operated businesses over a determined period, through surveys.	Number of new women-operated businesses in determined project area	50 new businesses per sub-project	IPP sub-project developer (SPV)	15,000 USD per sub-project (for conduct of surveys)
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